The Dual Labour Market and the Motherhood Employment Penalty in Japan: Evidence from Household Panel Data

Beatriz Gietner*

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Abstract

Japan has among the most generous parental leave statutes in the OECD, yet one of the widest motherhood employment gaps. Using an event-study design applied to 662 first births in the JHPS and KHPS panels (2004–2022), I show that the penalty is strongly stratified by labour-market tier. Employment falls 13.2 percentage points at childbirth, with a clean near-lead null at $t = -2$. Among women employed before birth, non-regular contract status predicts non-employment at $t = 0$ with an odds ratio of 7.50 ($p < 0.001$), while the binary indicator for employers with fewer than 100 employees is positive but imprecise. Post-birth "recovery" is largely an extensive-margin illusion - returning mothers shift from full-time to part-time work, and conditional hours, earnings, and implied hourly wages remain durably below pre-birth levels. Consistent with this re-entry bottleneck, use of government training subsidies is rare and access barriers (awareness, eligibility, and uncertainty) fall disproportionately on mothers who lose employment-insurance-linked eligibility after childbirth exit. These patterns are consistent with weaker effective access to job-continuity protections in non-regular and sub-100-employer segments; statutory entitlements alone appear insufficient where labour-market dualism channels mothers into employment tiers with limited institutional support.

Keywords: motherhood penalty, child penalty, dual labour market, Japan, event study, non-regular employment

JEL Codes: J13, J16, J21, J31, J41

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1 Introduction

Japan presents a puzzle. Over the past three decades, successive governments have enacted increasingly generous work-family policies: parental leave now covers the vast majority of employees, childcare capacity has expanded, and equal employment opportunity legislation has been in place since 1986 (OECD, 2024a; Ministry of Health, Labour and Welfare, 2019). Yet Japanese women's labour force participation continues to exhibit the characteristic M-curve - a sharp decline during prime childbearing years followed by partial recovery - that has defined the country's gender employment gap for half a century (Ochiai, 1997; Yu, 2009). If the policy infrastructure exists, why are exits still so concentrated in some parts of the labour market?

A large literature studies the labour-market consequences of childbirth for women. Recent work by Kleven et al. (2019a,b) popularized the child-penalty framework and documented its importance across OECD countries. The framework identifies when employment changes occur but says less about how the penalty is distributed across labour-market positions (see also Angelov et al., 2016; Adda et al., 2017; Waldfogel, 1998). In institutionally homogeneous labour markets like Denmark's, that distinction may matter less. In Japan, where over 50 percent of employed women hold non-regular contracts with fundamentally different protections than regular workers (OECD, 2024a), the composition of the penalty cannot be inferred from aggregate event-study plots.

Despite extensive descriptive work on Japan's dual labour market and gender inequality (Brinton, 1993; Yamaguchi, 2019; Dumauli, 2019; Houseman and Osawa, 2003), existing studies have not shown within one panel how the regular/non-regular divide and firm size stratify childbirth exit, translate into large absolute-risk differences at the childbirth margin, and shape the composition of post-birth recovery. This chapter does that.

Isolating the role of labour market structure from individual preferences is difficult. Women who hold non-regular jobs may differ from regular workers in unobserved dimensions such as career commitment, family planning intentions, and risk tolerance that independently predict exit. I do not treat the stratification results as causal effects of contract type or firm size. The design here is descriptive: it emphasizes the magnitude and composition of employment change, compares quit-flagged and not-quit-flagged women on observable pre-birth characteristics, and uses placebo, reweighting, balanced-panel, and missingness checks to test whether the main patterns are driven by obvious support or composition failures. For the lower-power pre-birth mechanism models, I use Firth-corrected logit models and bootstrap inference to assess sensitivity to the low event count.

Using an event study of 662 first births observed in the JHPS and KHPS from 2004 to 2022, I document that Japan's motherhood employment penalty is strongly stratified by labour market tier. By labour-market tier I mean the combination of contract type, employer size, and the associated effective access to job continuity through motherhood. The estimates are descriptive event-time trajectories in the analytic panel and should be interpreted with explicit support and composition constraints. Within this sample, exits are concentrated among women in jobs with weaker continuity through motherhood: non-regular contracts, small employers, and weaker effective leave access.

This chapter reframes the motherhood penalty as an institutionally patterned trajectory shaped by where in the labour market a mother sat before childbirth. The headline employment break at $t = 0$ is sharp, and the timing also fits a broader family-formation pattern. The composition of pre-birth exits shifts around the marriage-to-birth transition, and the aggregate employment break appears at childbirth itself.

The employment rate at $t = -1$ (the year before first birth) is 50.3 percent. Relative to this baseline, the event-study estimates show no evidence of a near pre-birth decline at $t = -2$ (coefficient -0.012, $p = 0.70$), followed by a 13.2-percentage-point drop at $t = 0$. In back-of-the-envelope terms, that implies about 87 of the 662 mothers in the sample moving from employment to non-employment in the birth year. Measured against the women who were working at baseline, this is roughly one in four. Employment moves back toward its pre-birth level by $t = +3$, but the recovery is misleading. Among women who were full-time at $t = -1$, fewer than half (45.7 percent) remain full-time at $t = +1$, and by $t = +5$ only 42.2 percent do. Recovery in employment does not imply recovery in job quality, and the distinction is one of the chapter's central qualitative findings.

Among women who were employed at $t = -1$ and are observed again at $t = 0$ (N = 330; 124 exits), non-regular status at $t = -1$ is a very strong predictor of non-employment at childbirth (OR = 7.48, $p < 0.001$). The indicator for employers with fewer than 100 employees is positive but imprecisely estimated. The model-implied absolute risk of exit ranges from 14.6 percent for regular workers in large firms or government employment to 66.5 percent for non-regular workers in firms with fewer than 100 employees; the corresponding raw cell rates are 13.8 and 64.8 percent. This cell contrast is best read as a model-implied risk gradient driven primarily by contract type. Firm size contributes directionally but with less precision. The pre-birth exit models at $t = -2$ provide supporting evidence on early risk stratification, with stable directional results despite the low event count. Their contribution is predictive and descriptive, with no claim of causal identification.

Among women who remain employed, the intensive margin shows persistent losses. The estimates are conditional-on-employment event-study coefficients, and the reference period is $t = -1$. The mean weekly hours among employed mothers at $t = -1$ are 34.1. The coefficient at $t = +5$ is -10.1, meaning that employed mothers work about 10 fewer hours per week than in the pre-birth reference year. In the positive-earnings sample used for the earnings event study, mean annual earnings at $t = -1$ are 264.7 man-yen (approximately 2.65 million yen). The coefficient at $t = +5$ is -96.2 man-yen:

employed mothers continue to lose about 962,000 yen per year at $t = +5$ relative to the pre-birth reference period, with no clear recovery within the observation window. I interpret these estimates as conditional intensive-margin losses and not as full-population effects. The pattern is directionally consistent with recent administrative evidence from local tax records (Fukai and Kondo, 2025). The regular/non-regular gradient also closely parallels the heterogeneity documented by Kikuchi (2026) using the larger JPSED panel.

The father-side adjustment remains limited throughout the window. In the partner-linked analytic sample, only 1 of 661 partners reports taking childcare leave (0.15%). Mothers also provide roughly five times more childcare than fathers throughout the post-birth window.

The chapter's contributions are calibrated to the strength of the underlying evidence. The strongest result is descriptive: employment drops sharply at childbirth, while the near pre-birth lead is essentially flat. This birth-year break is stable across cohort splits, survey sources, IPW reweighting, balanced-panel restrictions, and a matched childless placebo. The composition of post-birth recovery is the next layer of evidence. Aggregate employment moves back toward its pre-birth level, but the recovery masks a persistent shift toward lower-quality work: full-time attachment, hours, and earnings do not return to baseline within five years. Labour-market tier, and especially non-regular contract status, strongly predicts exit at the childbirth margin, with an odds ratio of 7.48 in the main risk set. The interpretive layer comes last. Taken together, these patterns, alongside near-zero paternal adjustment, are consistent with unequal effective access to job continuity across contract types and firm sizes, although that institutional interpretation is not separately identified by the panel design.

This evidence hierarchy structures the chapter as a whole: the abstract, results, and conclusion use stronger language for stronger evidence and explicitly mark more tentative claims as interpretive. Table 1 summarises this hierarchy near the outset, and Appendix D.14 provides a consolidated robustness table for the core employment coefficients ($t = -2$, $t = 0$, $t = +5$) together with a denominator map linking each headline statistic to its underlying risk set. The table ranks claims by evidential strength, not by substantive importance. The institutional reading is the chapter's interpretive payoff, disciplined by the stronger descriptive and predictive results that it draws on.

Relative to designs built around administrative earnings alone, this chapter offers a clearer view of where the penalty concentrates, how large the childbirth-margin risk gradient becomes across labour-market tiers, and why apparent employment recovery can mask durable post-birth downgrade (Kleven et al., 2019b,a; Angelov et al., 2016; Bertrand et al., 2010; OECD, 2024a; Yamaguchi, 2019).

The remainder of the chapter proceeds as follows. Section 2 describes the institutional context of Japan's dual labour market and gendered work culture. Section 3 presents the data and empirical strategy. Section 4 documents the aggregate birth-year break and the

Table 1: Evidence hierarchy used throughout the chapter

Level	Core evidence	Interpretation rule
High	Employment break at $t = 0$ with near lead null; stable across major sensitivity checks	Treated as the chapter's strongest descriptive fact
Moderate	Childbirth-margin risk stratification by labour-market tier; post-birth compositional downgrade	Treated as robust association evidence, not causal mechanism identification
Contextual / Interpretive	Institutional channels (care logistics, fiscal thresholds, workplace practices)	Used to discipline interpretation, not as separately identified effects

Notes: "Contextual / Interpretive" refers to the interpretive layer above the identified descriptive patterns, not to weak underlying descriptive evidence. Detailed evidence-to-claim mapping and full diagnostics are reported in Section 8 and Appendix D.

composition of pre-birth exits. Section 5 presents predictive risk stratification by labour-market tier and reports heterogeneity by job type and firm size. Section 6 examines the intensive margin and the transformation of the penalty over time. Section 7 presents father-side descriptive evidence. Section 8 reports robustness checks. Section 9 discusses policy implications, external validity, limitations, and concludes.

2 Institutional context: Japan's dual labour market and the gendered household

Several features of the institutional environment shape why Japanese women exit employment at motherhood: a labour market divided into structurally distinct tiers, a fiscal architecture that creates sharp earnings thresholds for secondary earners, a household division of labour that concentrates caregiving on mothers, regional and marriage-related variation in employment norms, and a work-family policy regime that aligns more closely with regular employment than with non-regular trajectories. This section describes them in turn. A compact reference summary of these institutional features appears in Appendix E.

2.1 The regular/non-regular divide

Japan's labour market is organised around a distinction that has no direct equivalent in most Western economies. Regular employees (*seishain*) hold open-ended contracts with a single employer, receive seniority-based wages, firm-specific training, and strong de facto protections against dismissal. Non-regular employees (*hi-seiki*) hold fixed-term, part-time (*pāto*), or dispatch contracts with limited tenure, flat wage profiles, and few of the implicit guarantees that define regular employment (Brinton, 1993, 2001; Yamaguchi, 2019; Houseman and Osawa, 2003; Kambayashi and Kato, 2017). As of 2020, approximately 37 percent of the total workforce and over 50 percent of employed women held non-regular positions (OECD, 2024a).

The contract category "part-time" (*pāto*) is a legal employment classification and does not necessarily imply fewer working hours. In the analytic panel at $t = -1$, 36 percent of non-regular workers work full-time hours (35+ per week) and 12 percent of regular workers work part-time hours. Throughout the chapter, "non-regular" refers to contract type and not to hours.

This distinction is important for the motherhood penalty because it determines access to the institutional accommodations that allow continued employment through and after childbirth. Regular employees at large firms benefit from internal labour markets: their positions are held during leave, their return is expected, and seniority continues. Japan's Child Care and Family Care Leave Act formally extends parental leave eligibility to all employees who have worked for their employer for at least one year, with generous replacement rates (Ministry of Health, Labour and Welfare, 2019). On paper, non-regular workers are covered. In practice, this formal coverage is often fragile. For fixed-term workers, eligibility depends not only on tenure but also on whether the contract is expected to continue beyond the leave horizon, and MHLW guidelines explicitly treat non-renewal or reduced renewal frequency around leave as forms of disadvantageous treatment (Ministry of Health, Labour and Welfare, 2019, 2004). Smaller employers also face tighter staffing margins and fewer replacement possibilities when workers take extended leave (OECD, 2024a; Yamaguchi, 2019). As a result, legal entitlement (de jure access) and usable entitlement (de facto access) can diverge, especially for women in non-regular and smaller-firm employment.

Firm size reinforces this divide. Large employers can usually redistribute tasks, maintain staffing buffers, and preserve positions during leave spells. Small and medium-sized firms, by contrast, often operate with thin margins and limited replacement capacity, so even short absences are harder to accommodate. This creates a practical gradient in continuity: the larger and more formalized the employer, the more feasible leave-taking and post-leave return become. Section 5 returns to this gradient in the childbirth-margin stratification results.

2.2 Fiscal thresholds for secondary earners

Beyond contract-based segmentation, Japan's tax and social-insurance system creates sharp earnings thresholds for secondary earners that shape married women's earnings independently of motherhood. The well-known 1.03 million yen tax threshold sits alongside additional thresholds around 1.06 and 1.30 million yen linked to mandatory social-insurance enrollment (Nagase, 2012; OECD, 2024a). These thresholds appear to be a broader structural feature constraining married women's earnings, not a motherhood-specific phenomenon. In the comparison sample used for the earnings distributions, the cumulative shares below these thresholds are at least as high among childless married employed women as among returning mothers: among childless married employed women, 49.7% earn at or below 103 man-yen and 60.0% at or below 130 man-yen, compared with 47.8% and 55.9% respectively for returning mothers at $t = +3$ to $t = +5$ (see Section 6 and Figure 11). The pattern in this chapter is consistent with motherhood shifting women from full-time positions with earnings above the cutoffs toward reduced-hours jobs, often on non-regular contracts, where these thresholds are more likely to bind.

2.3 The corporate warrior and the gendered household

Japanese men work among the longest hours in the OECD. The figure of the *salaryman* or corporate warrior remains a defining feature of Japanese corporate life despite recent policy efforts to reduce overtime (Nemoto, 2016; North, 2009; OECD, 2024a). The 2018 Work Style Reform Act introduced overtime caps, but enforcement remains uneven: men substitute paid overtime for unpaid overtime, leaving total hours largely unchanged (Burdin et al., 2024). The culture of long hours has deep organisational roots that the reform has not displaced (Kuroda and Yamamoto, 2013).

Long hours among husbands map onto a binding constraint on household time allocation. When one partner regularly works 50 or more hours per week, the household's effective scope for redistributing care tasks after childbirth is narrow. The gap is filled by the other partner, by paid care, or by some combination of the two. In Japan, where formal care for children under three is rationed and gendered norms continue to shape expectations about who provides care (Nemoto, 2016), the redistribution defaults to the mother. The result is a household-level division of labour that can be hard to renegotiate even when individual preferences would point elsewhere.

The time-use data in this sample reflect that asymmetry. The wife-to-husband childcare ratio is roughly 5-to-1, in line with official time-use statistics and prior work on Japanese work-family norms (Statistics Bureau of Japan, 2021; Nemoto, 2016). Section 7 returns to this asymmetry with direct father-side descriptive evidence on childcare time and leave take-up.

2.4 Marriage, motherhood, and regional norms

A further institutional dimension is the continuing role of marriage as a transition point in women's employment. In rural Japan and smaller cities, traditional household norms can create pressure to exit employment at marriage, independently of childbirth (Ochiai, 1997; Yu, 2009; Raymo and Iwasawa, 2005). In metropolitan areas, this norm is weaker, while the motherhood penalty appears more closely associated with labour-market segmentation and corporate work culture. This geographic heterogeneity suggests that aggregation can mask different local compositions of marriage-related and childbirth-related exits. For this chapter, the point is interpretive. Some exits that appear to be tied to childbirth may also reflect regional variation in marriage-related withdrawal norms. Section 5.7 returns to this issue directly in a metropolitan versus non-metropolitan split.

2.5 Policy perspectives

Japan's work-family policy regime has expanded steadily since the 1990s, with the explicit aim of reducing employment discontinuity around childbirth. The Child Care Leave Act introduced paid leave with income replacement for eligible workers, with extensions up to two years when childcare placement is unavailable (Ministry of Health, Labour and Welfare, 2019). In parallel, the state expanded childcare capacity and even encouraged firms to increase women's workforce participation (OECD, 2024a; Nishitateno and Shikata, 2017; Yamaguchi, 2019). The reform evidence is mixed: increasing cash benefits alone appears to have had limited effects on maternal job continuity (Asai, 2015), while childcare expansion improved maternal employment where care availability increased (Nishitateno and Shikata, 2017). The central question for this chapter is why policy expansion has not produced comparable continuity across labour-market tiers.

In practice, policies do not operate uniformly across labour-market tiers. They are most effective for women already in regular employment. Income-replacement leave requires employment-insurance eligibility, which depends on stable contracts and minimum hours. Childcare expansion can support return-to-work only when a job match still exists to return to. For women whose fixed-term contracts lapse around childbirth, the binding issue is employment continuity, with childcare access addressing only part of that constraint. The policy architecture aligns more closely with *seishain* careers than with non-regular trajectories, a point that becomes more important as non-regular employment expands (Genda, 2005; Piotrowski et al., 2015; OECD, 2024a).

A practical issue is timing and affordability of care by child age. In Japan, care for ages 0–2 is largely provided through nursery-type settings (*hoikuen*/certified centres), while age 3 onward expands into a broader early-education menu (*yochien* plus integrated centres), with substantial policy support from age 3–5 (Children and Families Agency, Government of Japan, 2024b,a). This age-structured transition is important for the dynamics of maternal employment. The first years are typically the tightest care-constraint period, and the feasibility of sustained employment often improves only once children move into the 3+ system. Cost also remains relevant despite subsidies, because out-of-pocket burdens vary by household income, municipality, care type, and ancillary fees (OECD, 2025b, 2024a). This age-structured transition helps interpret the later recovery pattern in Section 6, especially around the medium-run horizons where employment begins to move back toward its pre-birth level. The institutional environment facing mothers changes over the child life-cycle, including care availability, cost, and the feasibility of sustained employment.

3 Data and empirical strategy

3.1 Data sources

Surveys. I draw on two complementary household panel surveys conducted by the Panel Data Research Center at Keio University. These panels are well suited to the chapter's substantive focus because, unlike administrative earnings records, they observe the job characteristics that matter for the argument here, including contract type, firm size, and work hours.[1] The Keio Household Panel Survey (KHPS) has been fielded annually since 2004, sampling households nationwide with a focus on economic behaviour, employment, and family structure. The Japan Household Panel Survey (JHPS) began in 2009 with a broader scope and a larger initial sample. Both surveys collect detailed individual-level information on employment status, work hours, earnings, job characteristics (including contract type and firm size), commute time, household composition, reasons for job changes, and other wave-specific modules. The surveys use a common questionnaire structure that facilitates the pooling of observations across both panels.

Access. The access to JHPS/KHPS microdata is provided under approved institutional conditions via the Panel Data Research Center at Keio University.[2] The replication materials therefore include only code and derived outputs, but not raw respondent-level microdata, in accordance with the data-use agreement and confidentiality requirements.

Harmonisation across surveys. The official JHPS/KHPS construction program harmonizes variable numbers across surveys through dedicated integration scripts and a published correspondence file. I follow that harmonization framework and implement additional variable-level checks for all core outcomes and mechanisms used in this chapter (employment, hours, earnings, contract type, and firm size). I also report robustness with a dataset indicator (KHPS vs JHPS), which leaves the key near-lead and childbirth-margin estimates substantively unchanged, even though the long-horizon $t = +5$ coefficient differs more across the two surveys.

Unweighted models. I estimate unweighted models. No single harmonised longitudinal survey weight exists for the pooled JHPS/KHPS panel used here, so the coefficients are interpreted as sample-average trajectories within the analytic panel of married first-

[1] KHPS was launched in 2004 and JHPS in 2009. The surveys were initially managed separately, and their questionnaires were unified from the 2014 survey onward in the integrated JHPS/KHPS framework. Both are nationwide panel surveys built from stratified sampling designs, and the official release includes harmonisation documentation (including variable correspondence files and integration scripts) as well as sample-weight materials.

[2] Panel Data Research Center, Keio University: `https://www.pdrc.keio.ac.jp/`. Access to JHPS/KHPS microdata is governed by the Center's application and confidentiality procedures.

birth women observed around childbirth. Their scope is limited to that analytic panel and does not extend to nationally weighted population moments.

Temporal coverage. I use all available waves - KHPS 2004–2022 and JHPS 2009–2022 - providing the temporal coverage needed to observe women several years before and after first birth, which is essential for the event-study design.

Institutional context. Section 2 describes the institutional environment in detail. Three features are particularly relevant for the empirical design. The regular/non-regular employment divide determines access to leave and job continuity. Firm-size variation in human-resource capacity determines whether leave can be practically exercised. Spousal tax and social-insurance thresholds shape women's labour supply incentives on re-entry. The variable definitions below operationalize these institutional features.

3.2 Sample construction

My unit of analysis is the married woman observed around the time of her first birth. I identify first births from the fertility modules of both surveys. Both KHPS and JHPS record household events (including births) that occurred between February of the previous year and January of the survey year, so the "birth year" in the data corresponds to the survey wave in which the birth was reported. I define event time k as the difference between the survey year and the birth year, so $t = 0$ is the wave in which a birth is first recorded, $t = -1$ is the prior wave (which may overlap with late pregnancy), and $t = +1$ is the first full post-birth wave.

To construct the event-study sample, I restrict the event window to $k \in [-5, +5]$. A woman enters the sample at her first observed wave within the event window and remains until she drops out of the survey or the window ends. This produces an unbalanced panel in which sample sizes increase toward $t = -1$ (N = 656) and then decline at longer post-birth horizons due to panel attrition and the end of available waves. The sample sizes at the extremes - N = 186 at $t = -5$ and N = 454 at $t = +5$ - reflect unbalanced support and differential observability across event time. Support and attrition patterns are quantified explicitly in Sections 4 and 8.

Pooling the two surveys yields 1,184 identified first births (622 from KHPS, 562 from JHPS). After restricting to women with at least one observation in the $[-5, +5]$ window and requiring at least one pre-birth and one post-birth observation, the core regression sample includes 662 women contributing 5,212 person-year observations. This is the sample I use for all event-study regressions, summary statistics, and mechanism analyses, unless otherwise noted.

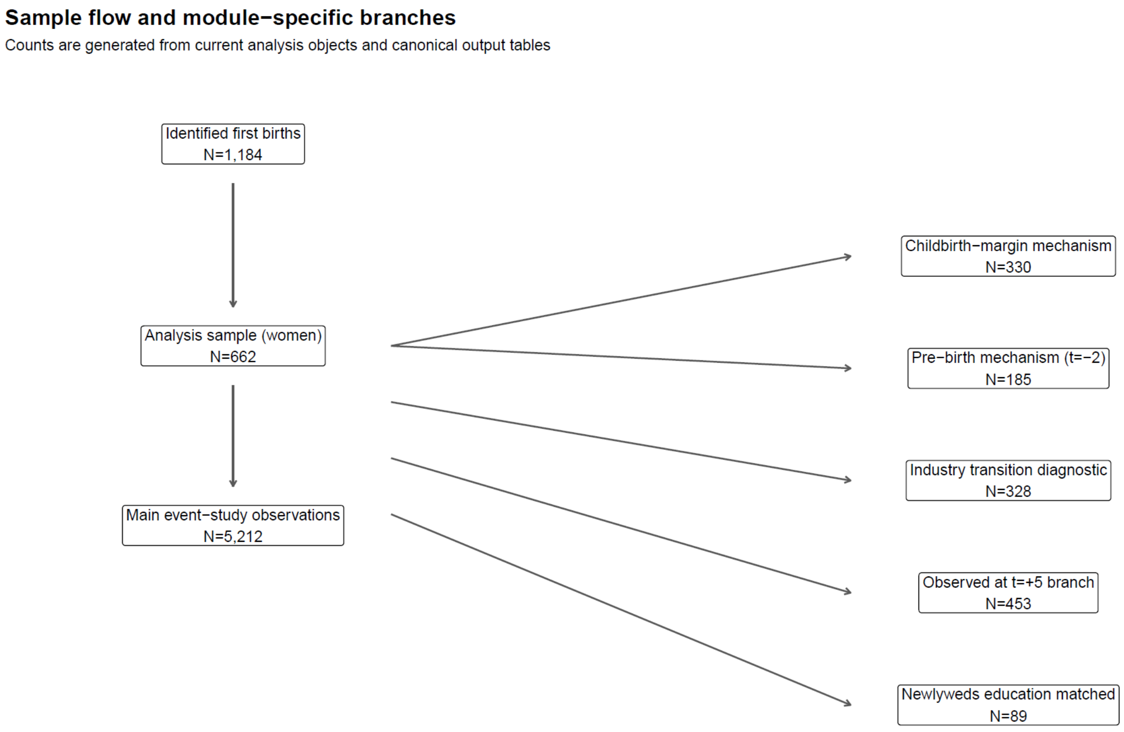

Figure 1: Sample flow from identified first births to module-specific analysis subsamples. The core event-study sample contains 662 women (5,212 person-year observations). Branches report the subsamples used for the childbirth-margin mechanism model, pre-birth mechanism model, industry transition diagnostic, long-horizon attrition/IPW checks, and newlyweds-education appendix diagnostic.

Table 2: Sample flow across analysis modules.

Step	**Module**	N
1	Identified first births	1,184
2	Analysis sample (event-study women)	662
3	Main event-study observations	5,212
3a	Childbirth-margin mechanism sample (employed at $t = -1$, observed at $t = 0$)	330
3b	Pre-birth mechanism complete-case sample ($t = -2$)	185
3c	Industry transition diagnostic sample	328
3d	Observed at $t = +5$ (attrition/IPW branch)	453
3e	Matched to newlyweds education	89

Notes: Rows 3a-3e are module-specific branches and are not sequentially nested within one another. They are reported to show which sample size applies to each analysis block.

3.3 Variable definitions

Relative to administrative or dedicated labour-force surveys, these panels measure labour-force status less cleanly. The questionnaires do not provide a single harmonized unemployment indicator that can be used consistently across waves and surveys. I define employment continuity using reported weekly work hours and interpret the main outcome as employment versus non-employment. This is a binary distinction; the data do not support a finer separation between unemployment and non-participation across all waves.

Employment status. The primary outcome is a binary indicator equal to one if the woman reports any positive weekly work hours in the survey wave. It is coded as zero when reported weekly hours are zero, and as missing when the survey response does not permit employment status to be determined consistently. This outcome is best interpreted as employment versus non-employment. It does not distinguish unemployment from non-participation, because the harmonized JHPS/KHPS questionnaires do not measure that distinction consistently across all waves. I also decompose employment into full-time (35+ hours per week) and part-time (positive hours below 35) to track the composition of employment over the event window.

Hours and earnings. Weekly hours are measured as reported usual weekly work hours among employed women. Annual earnings are reported in units of 10,000 yen (man-yen) and refer to the respondent's own labour income. Both variables are conditional on employment: the intensive-margin event studies estimate changes among working women, but not the unconditional population. At $t = -1$, mean weekly hours among workers are 34.1 (SD 16.5). Mean annual earnings are 268.9 man-yen (SD 206.9) in the descriptive baseline, while the earnings event study in Section 6 uses the positive-earnings employed sample, for which the corresponding $t = -1$ mean is 264.7 man-yen.

Contract type. I classify employed women as regular (*seishain*) or non-regular (*hi-seiki*) based on the employment status question in each survey. Non-regular employment includes part-time (*pāto*), fixed-term, temporary, and dispatch contracts. The classification follows the survey's own coding and is harmonized across KHPS and JHPS using a crosswalk of variable numbers (see Appendix D.3 for variable mapping details). Contract type and working hours are distinct dimensions with substantial overlap: at $t = -1$, 36 percent of non-regular workers work full-time hours (35+ per week) and 12 percent of regular workers work part-time hours. Throughout this chapter, "full-time" and "part-time" refer to hours worked (above or below 35 per week), while "regular" and "non-regular" refer to the employment contract type. At $t = -1$, 30.9 percent of women are employed full-time and 19.4 percent part-time (by hours); among employed women, approximately 38 percent hold non-regular contracts.

Commute time and husband's overwork. Commute time is measured as reported one-way travel time to work in minutes at the relevant survey wave. The husband's overwork indicator equals one when the husband reports working 50 or more hours per week, and zero otherwise. I include these variables in the mechanism regressions as proxies for job frictions and household time constraints around the childbirth transition.

Firm size. Employer size is measured using the survey's firm-size categories. For descriptive tabulations, I define "small firm" as fewer than 30 employees, "medium" as 30–99, and "large" as 100 or more, with government employees classified separately. The firm-size by contract-type crosstab at $t = -2$ shows 198 women with complete job-characteristic data, distributed across the categories shown in Table 5. For the parsimonious logit specifications, I collapse firm size to a binary indicator for employers with fewer than 100 employees versus 100 or more employees or government employment. The regression sample is smaller (N = 185) because the mechanism models additionally require non-missing values for commute time and the husband's overwork indicator.

Quitter classification: rule and counts. I classify women using the survey quit-status mapping at $t = -2$ (quit-flagged versus not quit-flagged), and I treat proxy transition-based definitions as robustness checks. Because the survey's quit-reason module records job-exit events and does not provide a direct currently-unemployed category at this margin, classification follows reported exit events, while contemporaneous labour-force status plays only a secondary role. Among the 662 women in the regression sample, 29 (4.4 percent) are quit-flagged at $t = -2$ (flagged as having exited employment between $t = -2$ and $t = -1$), 262 (39.6 percent) are not quit-flagged at $t = -2$, and the remainder are unclassified in the quit-module mapping at this margin.

Quitter classification: scope. Two qualifications are important for interpreting these labels. Event-study support at $t = -2$ is larger (N = 453), so the quit-flagged / not-quit-flagged labels apply only to women observed at that event time and do not cover the full analytic sample. Not-quit-flagged status is also defined by the absence of a quit event, not by continuous employment, so some women in that group are non-employed at $t = -1$.

Quitter classification: descriptive comparison. Quit-flagged women's mean pre-birth income is 148.4 man-yen versus 302.7 man-yen for the not-quit-flagged group; their employment rate is 55.2 percent versus 61.0 percent. The women who exit earliest are, on average, those in lower-paying positions.

Household structure. Because co-resident grandparents can relax childcare constraints and potentially moderate the employment penalty, I classify households at $t = -1$ into three types based on the household roster: nuclear (no co-residing parents of either spouse, 89.6 percent), patrilocal (co-residing with the husband's parents, 5.8 percent), and matrilocal (co-residing with the wife's parents, 4.3 percent). Patrilocal and matrilocal residence are identified by matching parent identifiers in the household roster to either the husband or the wife, with adjustments for whether the survey respondent is the wife herself (Type A) or her spouse is the respondent and the wife is recorded as the spouse (Type B).

3.4 Summary statistics

Table 3 reports summary statistics at $t = -1$ for the full sample, the quit-flagged group, and the not-quit-flagged group. Of the 662 women in the analytic sample, 656 are observed at $t = -1$; among those women, the employment rate is 50.3 percent, with 30.9 percent in full-time and 19.4 percent in part-time work. Among employed women, mean weekly hours are 34.1 (median 40.0), and mean annual earnings are 268.9 man-yen. The gap between the two pre-birth comparison groups is considerable: quit-flagged women earn roughly half as much as the not-quit-flagged group (148.4 versus 302.7 man-yen). These patterns motivate the mechanism analysis in Section 5, which asks whether the job characteristics that differ across these groups - especially contract type and firm size - predict exit in a multivariate framework.

Table 3: Summary statistics for the analytic sample of married first-birth mothers around $t = -1$ (662 mothers overall; 656 observed at $t = -1$). Employment shares are reported for all mothers observed at $t = -1$; weekly hours and annual earnings are reported among employed mothers only. Pre-birth comparison groups are defined using the quit-status mapping at $t = -2$ and are therefore not exhaustive categories of the full sample.

	Full Sample (N = 662)	Quit-flagged (N = 29)	Not quit-flagged (N = 262)
Employed (%)	50.3	55.2	61.0
Full-time (%)	30.9	31.0	36.7
Part-time (%)	19.4	24.1	24.3
Hours/week (mean)	34.1	32.5	33.4
SD	16.5	17.5	16.3
Income, man-yen (mean)	268.9	148.4	302.7
SD	206.9	95.2	242.2

Notes: Quit-flagged / not-quit-flagged labels are defined using quit-status mapping at $t = -2$ (not by contemporaneous $t = -1$ employment status). The remaining women are unclassified in the quit-module mapping at this margin; event-study support at $t = -2$ is larger than the classified subset. The not-quit-flagged group includes some women who are non-employed at $t = -1$ but did not record a quit event between $t = -2$ and $t = -1$.

3.5 Empirical Strategy

3.5.1 Event study specification

I estimate an event study of employment, hours, and earnings around first birth:

$$Y_{it} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[\text{event\_time}_{it} = k] + \gamma_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the outcome for woman i in year t, the indicators $\mathbf{1}[\text{event\_time}_{it} = k]$ capture each event year $k \in \{-5, \ldots, -2, 0, \ldots, +5\}$ with $k = -1$ as the omitted reference period, and γ_t are calendar-year fixed effects that absorb common macroeconomic shocks, including labour-market conditions and policy changes. The coefficients β_k trace the evolution of the outcome relative to the year before first birth.

I estimate three versions of this equation: a binary employment indicator for the full sample (extensive margin), weekly hours conditional on employment (intensive margin), and annual earnings conditional on employment. Heteroskedasticity-robust standard errors are reported throughout.

Estimand. The primary estimand is a descriptive event-time profile for married first-birth women observed in the JHPS/KHPS analytic panel. Under this specification, each β_k combines within-person change and composition change from entry, exit, and differential observation across event time. The coefficients should be read as sample-average trajectories around first birth, and not as causal treatment effects.

Data limitations. The JHPS/KHPS do not record several variables that would be needed for causal channel identification. Employer-level leave eligibility or enforcement records, contract renewal decisions, and HR accommodation behaviour are not observed. Exact childbirth date relative to survey interview is also unavailable, as is fine-grained childcare availability or slot constraints at prefecture or municipality level. A harmonized education measure for the full analytic sample is also not available; education can only be recovered for a restricted subset through the supplementary Newlyweds module, which is not part of the main questionnaire in every wave and does not cover the full analytic sample. Only 13.4% of the analytic sample can be matched to this module; Appendix D.4 reports the coverage diagnostic and matching details. These gaps limit the chapter's ability to isolate specific channels and motivate the descriptive and predictive framing used throughout.

Validation diagnostics. The descriptive event-time profile is assessed with three complementary diagnostics that are reported centrally in the chapter: a near-lead check at $t = -2$ in the full sample, a trimmed pre-period specification that removes far leads, and a balanced pre-support specification restricted to women observed across all pre-birth event times. I also report a JHPS/KHPS split check to verify that the key timing moments are not driven by pooling alone. Due to timing noise around the survey birth-report window, I choose to treat the $[-2, +2]$ window as the primary interpretation window for the birth-year break and present $[-5, +5]$ estimates as extended descriptive context.

Specification choice. The main specification uses calendar-year fixed effects without individual fixed effects. The year-FE coefficients β_k are interpreted as descriptive sample-average trajectories relative to $t = -1$, with calendar-year shocks absorbed but no claim that the remaining variation identifies within-person causal effects. I do not emphasize individual-FE versions because this short and unbalanced event-time panel leaves those specifications near-collinear and substantively uninformative in this setting.

3.5.2 Full-time/part-time decomposition

To understand whether the employment recovery masks a compositional shift, I supplement the event-study estimates with transition matrices that track changes in hours-based employment status. In this part of the analysis, full-time and part-time are defined by hours worked, while contract status is captured separately by the regular/non-regular classification. I track the employment status at $t = +1$, $t = +3$, and $t = +5$ of women who are employed full-time at $t = -1$. I report both the full-sample transitions and the transitions restricted to women with no second birth by the horizon date. The transition matrices are descriptive tabulations. They do not condition on covariates or adjust for attrition, and they document the composition of post-birth employment without claim to causal interpretation.

I also examine subsequent trajectories among women who are part-time at $t = +1$. Specifically, I track the shares who move to full-time work, remain part-time, or exit employment by $t = +5$. This complements the event-study estimates by showing whether part-time re-entry functions as a stepping stone back to full-time work or instead persists as a longer-run downgrade.

3.5.3 Risk-stratification regressions

To assess which job characteristics predict pre-birth exit, I estimate logistic regressions of the form:

$$\Pr(\text{quitter}_i = 1) = \Lambda(\mathbf{X}_i' \boldsymbol{\delta}), \tag{2}$$

where $\Lambda(\cdot)$ is the logistic CDF and $\mathbf{X}_i$ includes non-regular employment status, an indicator for employers with fewer than 100 employees, commute time, and the husband's overwork indicator. I estimate a sequence of nested specifications to show how each predictor contributes independently.

The dependent variable is quit-flagged status on the pre-birth margin at $t = -2$; see Section 3.3 for the coding details. In practice, this indicates whether a woman is flagged as having exited employment between the $t = -2$ and $t = -1$ survey waves. The regression sample is restricted to the 185 women in the complete-case mechanism sample with non-missing values on the included predictors, because the mechanism regressions require complete covariate information, including commute time and husband's overwork. This is smaller than both the 291 women classified in the quit-module mapping and the 198 women in the firm-size by contract-type crosstab.

Among the 185 women in the regression sample, 13 are quit-flagged (7.0 percent). Of the 29 women quit-flagged at the $t = -2$ margin overall, 13 remain in the complete-case regression sample. The event count is low, but the coefficients are stable across standard logit, Firth correction, bootstrap, and jackknife diagnostics. I treat these estimates as low-powered but coefficient-stable evidence on early risk stratification, with a supporting role rather than a central one in the chapter's argument. The full nested specifications are reported in Appendix D.8.

The low event count - 13 exits against 4 or 5 predictors - raises legitimate concerns about separation and small-sample bias. I address these in two ways. Firth's penalized likelihood estimator (Firth, 1993) reduces finite-sample bias in rare-event logit settings. Bootstrap confidence intervals from 2,000 resampled datasets provide nonparametric inference that does not rely on asymptotic normality. I report both corrections alongside the standard estimates in Section 5.2, while Appendix D.8 reports the nested mechanism specifications and jackknife diagnostics.

3.5.4 Interpretation and timing

Births are reported for events occurring between February of the previous year and January of the survey year. Accordingly, $t = 0$ corresponds to the survey year in which a birth was reported, and $t = -1$ is the prior survey year. The event-study estimates capture discrete changes around the reported birth window; they do not pin down exact conception or delivery dates.

The event-study coefficients are descriptive trajectories relative to $t = -1$, as established earlier. The pre-period coefficients at far leads ($t = -5$ to $t = -3$) are negative and jointly significant ($p < 0.001$), reflecting rising employment as young women enter the labour market; it is not evidence of pre-birth anticipation. The critical near-lead diagnostic is at $t = -2$: the coefficient is -0.012 (SE 0.031, $p = 0.70$), confirming that employment is stable in the year before first birth.

This near-lead result is stable across robustness checks. In the trimmed pre-period specification, $p = 0.72$. In the balanced pre-support sample observed at every pre-birth event time, the coefficient remains modest in magnitude with $p = 0.078$. In the separate balanced short-window panel observed continuously from $t = -2$ to $t = +1$, the coefficient is -0.058 with $p = 0.065$. Outcomes are measured at survey timing, so $t = 0$ is the birth-report wave, not a full post-birth calendar year. I interpret the results as descriptive changes in level and composition around birth, especially over the $[-2, +2]$ window.

3.5.5 Second-birth censoring

A substantial share of women in the sample have a second birth within the event window: 19.2 percent by $t = +2$, 33.8 percent by $t = +3$, and 44.1 percent by $t = +5$.[3] Subsequent births complicate the interpretation of medium-run coefficients because the "post-first-birth" trajectory includes responses to the second birth.

I address this in two ways. The main event study is reported on the full sample, with late-period coefficients (particularly $t = +3$ through $t = +5$) flagged as reflecting a mixture of first- and second-birth responses. A second specification excludes observations after a second birth; this yields a $t = +5$ coefficient of -0.003 (versus -0.028 in the full sample), which suggests that part of the muted recovery reflects subsequent fertility. I keep the full-sample specification as the main design because it describes the realised employment trajectory following first birth in the analytic panel, including the fact that some women proceed to a second birth within the observed window. The transition matrices are also reported separately for the "no second birth by horizon" subsample.

3.6 Interpretation, identification, and scope

This section clarifies the estimand and the scope of interpretation.

Estimand. As established in Section 3.5.1, the main event-study specification reports descriptive trajectories in an unbalanced panel, not within-person causal treatment effects. Individual-fixed-effects versions are shown in the appendix as sensitivity checks. The near lead at $t = -2$ and the birth-year drop at $t = 0$ are stable in sign and magnitude across variants.

Event-time measurement. The event-time measurement follows the survey reporting window (February-January). This implies that $t = -1$ may include late-pregnancy or early-childcare overlap for some women. I therefore interpret the key timing contrast as a near-lead check at $t = -2$ and a discrete break at $t = 0$. The earlier pre-birth event times are less clean as pre-treatment periods.

[3] These shares are conditional on being observed by the stated horizon; the $t = +5$ value of 44.1 percent is conditional on women observed at $t = +5$ (N = 454), not on the full analytic sample of 662 mothers. Section 8.4 provides the full conditioning-set clarification and comparison with full-sample shares.

Mechanism scope. The mechanism regressions target pre-birth exit risk. The regressors are job characteristics measured while employed at $t = -2$, and the dependent variable indicates whether a woman is flagged as having exited employment between $t = -2$ and $t = -1$. These models are designed to explain who exits early in the pre-birth margin; they are not designed to identify childcare-margin effects at $t = 0$.

Pre-birth versus childbirth margins. The $t = -2$ and $t = 0$ margins reflect different risk sets. The former is a pre-birth family-formation margin, while the latter is predominantly a childbirth and childcare margin. Subgroup coefficients at birth should not be expected to mechanically mirror the pre-birth mechanism logits.

Conditional estimates. The intensive-margin results for hours and earnings are conditional on employment and therefore combine within-person adjustment with composition effects in who remains employed. I interpret these coefficients as evidence on the composition and quality of post-birth work.

Bundled features. More broadly, these margins are not independent in Japan's labour market structure. Contract type, hours, and employment continuity are tightly bundled features of the same employment tier. I treat non-regular status as a marker of labour-market position and interpret post-birth hours and earnings as downstream expressions of that position. In this design, those outcomes do not separate cleanly into distinct channels.

Subgroup heterogeneity. Finally, the subgroup coefficients in the heterogeneity tables are estimated in separate subgroup event studies. They are useful for ranking exposure across groups, but they are not intended as a decomposition identity and need not aggregate arithmetically to the pooled coefficient.

4 Results: aggregate break and pre-birth exit composition

The child-penalty literature often interprets pre-birth employment declines as anticipatory adjustments: women who expect to bear children reduce their labour supply in the years before birth (Kleven et al., 2019b; Kuziemko et al., 2018). In this sample, the aggregate near lead at $t = -2$ is flat; what differs pre-birth is the composition of exits among the minority who leave before birth.

4.1 Aggregate employment trajectory

I focus first on the primary outcome: the binary employment indicator, which equals one if a woman reports positive weekly hours. The employment rate at $t = -1$ is 50.3 percent (N = 656). Against this baseline, Figure 2 reports the event-study estimates.

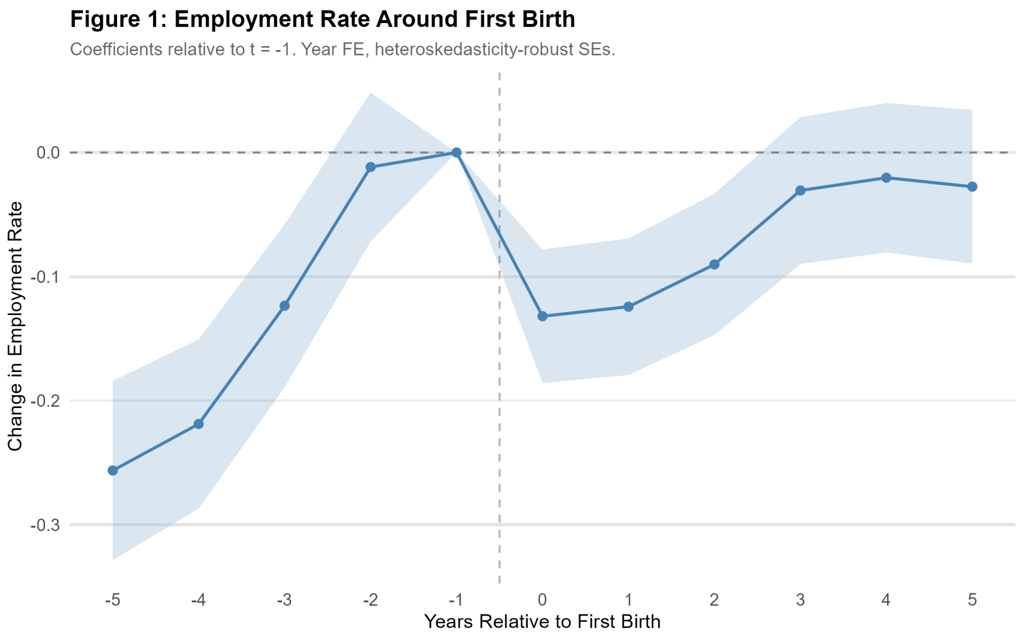

Figure 2: Event-study estimates of maternal employment around first birth (year fixed effects; heteroskedasticity-robust standard errors). Each point is the coefficient $\hat{\beta}_k$ relative to $t = -1$; bars are 95% confidence intervals. The baseline employment rate at $t = -1$ is 50.3%. Because $t = -1$ may overlap late pregnancy for a subset of women under the survey timing window, the primary interpretation window is $[-2, +2]$. Support is unbalanced, with N ranging from 186 ($t = -5$) to 656 ($t = -1$). Inference is unchanged under woman-level clustered standard errors (Appendix D.12).

The trajectory shows a clear pattern: the near lead at $t = -2$ is close to zero (-0.012, $p = 0.70$), followed by a sharp drop of 13.2 percentage points at $t = 0$ ($p < 0.001$). To put this in perspective, about one in four women who were working at $t = -1$ move from employment to non-employment in the birth year.

Employment then shows an apparent recovery, with coefficients of -0.124 at $t = +1$, -0.090 at $t = +2$, and -0.031 at $t = +3$. By $t = +5$ the coefficient is -0.028 (SE 0.032), statistically indistinguishable from zero in the pooled specification; this late-horizon point is panel-sensitive and should be interpreted with caution. Because $t = -1$ can overlap late pregnancy for a subset of respondents, the baseline may already embed partial pre-birth adjustment; if so, the estimated $t = 0$ drop is conservative relative to a fully pre-pregnancy reference.

The extended pre-birth trajectory ($t = -5$ to $t = -3$) shows large negative coefficients (-0.256, -0.219, -0.124). This pattern reflects lower employment earlier in the window, not a decline immediately before birth. The lower employment rates at earlier event times likely reflect a combination of life-cycle differences among women in their twenties and compositional selection: fewer women are observable at $t = -5$ (N = 186) than at $t = -1$ (N = 656), and those observable early are a selected subsample. The key diagnostic result is that at the near lead, where support has risen to N = 453, employment is flat relative to $t = -1$.

The support profile is materially unbalanced (N = 186 at $t = -5$, 662 at $t = 0$, 454 at $t = +5$), and observability at $t = +5$ is related to baseline employment. The IPW sensitivity that targets observability at $t = +5$ leaves the near-lead and at-birth break intact, while making the late-horizon coefficient more negative and marginally significant ($t = +5$: -0.057, $p = 0.076$). I treat late-horizon recovery as composition-sensitive. In what follows, the focus is on job-quality downgrade as the broader substantive interpretation, while the IPW result is most relevant for the long-horizon employment margin itself.

The recovery masks a substantial compositional shift. At baseline ($t = -1$), full-time workers comprise 30.9 percent and part-time workers 19.4 percent of the sample. After birth, the composition shifts sharply: among women who held full-time positions at $t = -1$, only 45.7 percent remain full-time at $t = +1$, while 33.7 percent exit employment and 20.6 percent shift to part-time. About one in three women who held full-time jobs before birth had exited employment entirely by $t = +1$, and fewer than half remained in full-time work. By $t = +5$, only 42.2 percent of those initially full-time remain so. Section 6 documents this compositional transformation in more detail.

4.2 Quit reasons: marriage versus childcare

Table 4 provides the key descriptive evidence on why women exit. Among exits at $t = -2$ (N = 29), 41.4 percent cite marriage as the reason and 6.9 percent cite childcare. The pattern reverses sharply at $t = 0$ (N = 113 exits recorded in the quit-reason module): 85.0 percent cite childcare, and only 2.7 percent cite marriage. The residual "Other/Missing" category is sizeable (51.7 percent at $t = -2$, 12.4 percent at $t = 0$), so these figures should be interpreted as suggestive.

This pattern is difficult to square with a pure anticipation interpretation. If women were leaving at $t = -2$ primarily in preparation for childbirth, childcare should appear more prominently among the stated reasons. The women who leave two years before first birth most often cite marriage - an event that typically precedes first birth by one to two years and that carries its own normative expectations about women's labour-force participation in Japan (Ochiai, 1997; Yu, 2009).

The distinction is important analytically because it suggests that the pre-birth exit margin and the birth-year break reflect different institutional pressures and may respond to different policy margins. Childcare expansion alone is unlikely to prevent marriage-related exits, just as changing marriage norms alone would not address childcare-related exits at birth. A single pooled "child penalty" estimate can obscure meaningful heterogeneity in timing and mechanism.

Table 4: Quit reasons by event time among women classified as exits in the quit-reason module. Percentages are within event-time exit groups, with missing reason responses shown explicitly. At $t = -2$, marriage is the leading reported reason (41.4%); at $t = 0$, childcare is dominant (85.0%). Missingness is high at $t = -2$ (51.7%) and lower at $t = 0$ (12.4%). The $t = 0$ count of 113 is a subset of the broader childbirth-margin non-employment transitions because the quit-reason module only captures exits with a recorded quit event.

	$t = -2$ (N = 29)		$t = 0$ (N = 113)	
Reason	N	%	N	%
Marriage	12	41.4	3	2.7
Childcare	2	6.9	96	85.0
Other/Missing	15	51.7	14	12.4

4.3 Reentry penalty and sequencing

Although employment rates recover after the birth year, the composition of work changes in ways that imply a reentry penalty. Conditional on remaining employed, mothers work roughly 10 fewer hours per week by $t = +5$ (coefficient -10.1, $p < 0.001$), and among employed mothers with positive earnings, annual earnings remain substantially below pre-birth levels (coefficient -96.2 man-yen at $t = +5$, $p < 0.001$). This pattern is consistent with reentry into non-regular or lower-hour jobs and does not look like a return to pre-birth career tracks.

In the Japanese dual labour market, contract tier and firm-level protections are difficult to recover once lost. Exit around birth can move mothers onto a lower-quality track with weaker wage progression, which shows up as persistent earnings shortfalls even as employment rates do rebound. The mechanism the chapter develops in subsequent sections takes this institutional structure as the relevant frame: aggregate recovery measured at the extensive margin can mask a durable repositioning across labour-market tiers.

The long-run trajectory also reflects subsequent births. Censoring at second birth attenuates the late-period coefficients, which indicates that additional fertility contributes to the muted recovery.

With the timing pattern established, I turn next to the question of who exits and which structural features of the labour market predict departure at the pre-birth and childbirth margins.

5 Risk stratification by labour-market tier

The centrepiece risk-stratification result is at the childbirth margin ($t = 0$), where the sample is largest (N = 330; 124 exits) and inference is sharpest. Before presenting that analysis, I examine the pre-birth margin ($t = -2$) for sequencing context: in this lower-power sample (N = 185; 13 events), the aggregate profile is flat but the composition of early exits is informative - marriage is the most common stated reason, while at $t = 0$ childcare dominates. The pre-birth models are retained as supporting evidence for the risk-stratification gradient, not as the primary mechanism identification. The empirical question is straightforward: does exit risk cluster by labour-market position? In the data, it does at both margins. Risk varies sharply by contract tier and firm size, which is consistent with segmentation in employment continuity.

5.1 Descriptive patterns: job type and firm size

I begin with bivariate comparisons from the firm-size by job-type crosstab. Among employed women at $t = -2$ who hold non-regular contracts, 12.2 percent exit employment (11 of 90). Among regular employees, 1.9 percent do (2 of 108). The interaction of job type and firm size is also informative. Among non-regular workers at small firms (fewer than 30 employees), the exit rate reaches 20.0 percent (6 of 30). Among regular workers at large firms (100+ employees), the rate is 0.0 percent (0 of 63). These raw cell rates are consistent with pre-birth exit risk being concentrated in the least protected labour-market cells, though the individual cell sizes are small and the multivariate evidence below should carry more weight.

The pattern is a first descriptive expression of the chapter's institutional reading. In some jobs, family-formation transitions can plausibly be absorbed through temporary absence and return pathways. In others, those pathways appear much weaker, and exit before birth aligns more closely with marriage-related family planning than with continued employment.

These bivariate contrasts could still reflect confounding across correlated job and household characteristics. I estimate nested logistic specifications to assess whether con-

Table 5: Pre-birth exit rates at $t = -2$ by labour-market cell (contract type × firm size), based on women with complete job characteristics (N = 198). Non-regular workers in small firms show the highest exit rate (20.0%), while regular workers in large firms show 0.0% in this sample.

Firm Size	Contract Type	N	N quit	% quit
Small (<30)	Non-Regular	30	6	20.0
Small (<30)	Regular	15	1	6.7
Medium (30–99)	Non-Regular	6	1	16.7
Medium (30–99)	Regular	19	1	5.3
Large (100+)	Non-Regular	48	3	6.2
Large (100+)	Regular	63	0	0.0
Government	Non-Regular	6	1	16.7
Government	Regular	11	0	0.0

tract tier and firm size continue to predict pre-birth exit risk once additional controls are added. Because this margin contains only 13 exit events, I treat the multivariate results as coefficient-stable supporting evidence. The section places primary weight elsewhere, and the full nested specifications are reported in Appendix D.8.

The pattern is directionally stable across those specifications. Non-regular status remains strongly associated with higher pre-birth exit risk, and employment in firms with fewer than 100 employees is also positively associated with exit, while commute time and the husband's overwork do not add much predictive power once the woman's labour-market position is included. Contract type at $t = -2$ is not merely a last-minute anticipatory switch: the pre-pregnancy stability diagnostic in Appendix D.13 shows that job type is persistent over $t = -5$ to $t = -3$. Given the low-event sample, the exact magnitudes should be read cautiously. The main takeaway is the ranking: early exit risk is concentrated in labour-market segments with weaker continuity around family formation.

5.2 Small-sample robustness

The pre-birth mechanism estimates are based on only 13 exit events in the complete-case sample, so precision is limited even though the coefficient ranking is stable across specifications. To assess whether the core pattern is driven by small-sample bias, I apply Firth's penalized likelihood estimator and construct bootstrap confidence intervals from 2,000 resampled datasets. These corrections are reported for the parsimonious two-predictor specification (Model 2); with only 13 events, the saturated model is too parameter-rich for reliable penalized estimation.

The corrected estimates preserve the qualitative ranking. The Firth estimates are attenuated relative to the standard logit, as expected, but both predictors remain statistically significant: non-regular employment (OR = 5.32, $p = 0.009$) and employment in firms with fewer than 100 employees (OR = 3.84, $p = 0.021$). The bootstrap intervals likewise exclude 1 for both predictors: non-regular $[1.68, 19.66]$ ($p = 0.008$) and the sub-100-employer indicator $[1.10, 18.34]$ ($p = 0.038$). The intervals are wide, which reflects the low event count, but the direction of the relationship is stable. I treat this as supportive directional evidence with moderate precision and keep it secondary to the chapter's main argument.

Table 6: Small-sample robustness for the parsimonious pre-birth mechanism model (Model 2), with exit at $t = -2$ as the outcome: standard logit, Firth penalized likelihood, and nonparametric bootstrap (2,000 resamples).

	OR	p	95% CI	Method
Non-regular employment				
Standard logit	6.41	0.019	–	MLE
Firth	5.32	0.009	–	Penalized ML
Bootstrap	–	0.008	[1.68, 19.66]	2,000 resamples
Firm size below 100 employees				
Standard logit	4.14	0.025	–	MLE
Firth	3.84	0.021	–	Penalized ML
Bootstrap	–	0.038	[1.10, 18.34]	2,000 resamples

5.3 Risk stratification at the childbirth margin ($t = 0$): absolute risks

Before presenting the childbirth-margin model, Table 7 compares pre-birth characteristics of regular and non-regular workers at $t = -1$. The two groups differ substantially: regular workers average 41.3 hours per week and 390 man-yen in annual earnings, while non-regular workers average 28.0 hours and 132 man-yen. In the 294-woman childbirth-margin subsample with observed contract type, exit rates at $t = 0$ are 18.3 percent for regular and 61.6 percent for non-regular workers. These differences confirm that the regular/non-regular distinction captures a substantive divide in labour-market position before childbirth, not merely a labelling difference.

Table 7: Pre-birth characteristics by contract type at $t = -1$ (employed women).

	Regular	Non-Regular
N	169	125
Mean weekly hours	41.3	28.0
Full-time (%)	88.2	36.0
Mean annual earnings (man-yen)	390.2	132.1
Median annual earnings (man-yen)	363.5	101.0
Exit rate at $t = 0$ (%)	18.3	61.6

Notes: The sample is restricted to women employed at $t = -1$ with non-missing contract-type classification. Regular = codes 1–3 (seishain, executive, family business). Non-Regular = codes 4–7 (part-time, arbeit, dispatched, contract). Full-time is defined as ≥ 35 hours per week. Exit rate = share not employed at $t = 0$ among women in this same non-missing-contract subsample who are observed at both $t = -1$ and $t = 0$.

Because the main exit spike occurs at childbirth, the primary risk-stratification model targets non-employment at $t = 0$ among women employed at $t = -1$ (N = 330; 124 exits, 37.6 percent). In this childbirth-margin sample, non-regular status at $t = -1$ is strongly associated with exit (OR = 7.48, $p < 0.001$), while the coefficient on the indicator for firms with fewer than 100 employees is positive but imprecisely estimated (OR = 1.55, $p = 0.126$).

Small-sample corrections leave the qualitative ranking unchanged: the non-regular effect remains strong, while the sub-100-employer coefficient is directionally positive but less precise. This weaker firm-size coefficient at $t = 0$ than at $t = -2$ is consistent with the selection pattern discussed in Section 5.6, where small-firm non-regular workers are disproportionately selected out before birth.

The childbirth-margin gradient is the empirical expression of the de jure/de facto gap described in Section 2.1: formal leave coverage may exist on paper, but the employment relationship does not consistently survive to make that coverage operational.

Within this same risk set ($N = 330$), the $t = -1$ composition is 149 full-time regular women (45.2%), 45 full-time non-regular (13.6%), 20 part-time regular (6.1%), and 80 part-time non-regular (24.2%); contract type is missing for 36 women (10.9%), of whom 9 are full-time and 27 are part-time. Here, full-time and part-time refer to hours worked, while regular and non-regular refer to contract status. This missingness is systematic in observables: missing contract type is concentrated among lower-hours and lower-income women, that is, profiles closer to marginal or non-regular attachment (Appendix D.9).

The absolute-risk table reports these values for transparency. In the 282-woman subsample with both contract type and firm size observed, raw exit rates are 60.8 percent for non-regular workers versus 16.7 percent for regular workers. By profile, model-implied probabilities range from 14.6 percent for regular workers in large firms or government employment to 66.5 percent for non-regular workers in firms with fewer than 100 employees, closely tracking the raw cell means.

The table-level rates use the 282 women with non-missing contract type and firm size at $t = -1$; by contrast, the childbirth-margin logit is estimated on all 330 women using missing indicators. In this risk set, 36 women are missing contract type, 16 are missing firm size, and 48 are missing at least one of the two covariates. Taken together, these results indicate that contract tier is the strongest observed predictor of childbirth-margin exit in this sample.

Table 8: Absolute-risk view of childbirth-margin exit: raw rates and model-implied probabilities at $t = 0$.

Profile / group	N	Exits	Exit rate
By contract type: Non-Regular	120	73	0.608
By contract type: Regular	162	27	0.167
By firm size: 100+ / government	175	53	0.303
By firm size: Below 100 employees	107	47	0.439
By contract × firm: Non-Regular × 100+ / government	66	38	0.576
By contract × firm: Non-Regular × Below 100	54	35	0.648
By contract × firm: Regular × 100+ / government	109	15	0.138
By contract × firm: Regular × Below 100	53	12	0.226
Profile			Predicted Pr(exit at $t = 0$)
Regular × 100+ / government			0.146
Regular × Below 100			0.209
Non-Regular × 100+ / government			0.562
Non-Regular × Below 100			0.665

Notes: Raw cell rates are computed on the 282 women in the childbirth-margin risk set with both contract type and firm size observed at $t = -1$. The model-implied probabilities come from the corresponding interaction-based childbirth-margin logit and closely track the raw cell means.

A missing-data sensitivity check shows that the non-regular gradient is highly stable across covariate treatments: OR = 7.50 (complete-case), 7.48 (missing-indicator), and 7.39 (high-information sample), all with $p < 0.001$ (Appendix D.9, Table 38). This indicates that the core childbirth-margin stratification is robust to the missing-data checks. Coefficients on the sub-100-employer indicator remain positive but less stable across specifications, so I interpret firm-size effects as suggestive and treat them with more caution. A balance diagnostic (Appendix D.9, Table 39) confirms that contract-type missingness is concentrated among lower-hours and lower-income women, consistent with ambiguous classification in marginal employment, but the non-regular OR is invariant to missingness treatment. Under this pattern, complete-case estimates are more likely to under-represent the highest-risk non-regular segment than to mechanically inflate the OR.

5.4 Selection diagnostics

A natural concern is that women in non-regular jobs may differ in ways that independently predict exit (cf. Budig and England, 2001; Correll et al., 2007). I address this concern through several descriptive comparisons and a leave-coding sensitivity check.

Pre-birth labour-force attachment. Quit-flagged women (N = 29) and not-quit-flagged women (N = 262) have similar pre-birth employment rates: 55.2 percent versus 61.0 percent at $t = -1$ ($p > 0.3$). On this margin, women who exit do not appear less attached to the labour force than those who remain.

Pre-birth income and quit reasons. Quit-flagged women earn 148.4 man-yen versus 302.7 man-yen for the not-quit-flagged group. This income measure is observed at the $t = -1$ survey wave, which follows the quit interval, so it may already reflect post-exit adjustment or reduced hours. Even so, the pattern is informative. In a simple labour-supply framework, lower earnings could reflect voluntary sorting into exit or structural vulnerability. In these data, the multivariate results point more clearly to contract type and firm size, while earnings explain less of the pattern. The quit-reason evidence at $t = 0$ shows that 85 percent of exits cite childcare, with voluntary preference cited much less often. Taken together, these patterns suggest that low income marks jobs that are more vulnerable to the motherhood transition. The evidence points away from a purely preference-based exit margin.

Sample composition. A supplementary diagnostic compares the 13 complete-case quit-flagged women to the 16 quit-flagged women excluded because of missing covariates. Both groups are concentrated in firms with fewer than 100 employees (69.2 percent among included cases and 100 percent among excluded cases), while contract type is missing for excluded cases by construction. This limits precision but does not suggest that the complete-case mechanism sample is selecting a qualitatively different group of exits.

Leave take-up. I also examine childcare-leave take-up directly. In the partner-linked analytic sample, only 1 of 661 partners reports taking childcare leave (0.15 percent). For mothers, leave take-up is higher but still limited: in the broader leave-analysis sample, 250 of 1,183 women (21.1 percent) ever report taking childcare leave. This percentage should not be interpreted as unconstrained choice. The leave item is observed only when women remain employed and report temporary absence from work; women who exit employment are coded as non-employed and therefore do not enter the leave-response risk set. Observed leave take-up thus reflects both policy availability and whether an employment relationship survives through childbirth. For many women, formal leave provisions are no longer operational once the employment relationship has ended.

Leave-coding sensitivity at $t = 0$. To assess potential misclassification at childbirth, I run a sensitivity check that reclassifies women coded as non-working at $t = 0$ as employed if they report leave take-up near birth. This affects 22 women. The $t = 0$ event-study coefficient attenuates from -0.1319 (SE 0.0275) to -0.0988 (SE 0.0276), a $+3.31$ percentage-point shift, while remaining strongly negative in both cases ($p < 0.001$). This suggests that leave-related coding contributes to measured exit levels but does not overturn the birth-year break (Appendix D.9, Table 40).

Summary. Taken together, these checks suggest that labour-market position is the more visible observed predictor of exit risk in these data. Unobserved selection into labour-market tiers may still be important, but the main gradient extends beyond the observable differences examined here.

5.5 The time-budget constraint

The risk models indicate which jobs are most exposed to exit. A separate question is why women in those jobs leave. In the logistic models, the husband's overwork indicator does not add much explanatory power once the wife's job type is included (Model 4: OR = 1.21, $p = 0.749$). I do not interpret this null as strong evidence against household constraints. In this sample, husbands' hours are uniformly high, which limits the variable's ability to explain cross-sectional differences in exit risk once labour-market position is already accounted for.

The descriptive comparisons at $t = -1$ are consistent with that interpretation. Husbands of quit-flagged women work an average of 49.9 hours per week versus 46.9 hours for husbands in the not-quit-flagged group ($p = 0.30$, N = 271), and the difference is not statistically precise. Commute time runs in the opposite direction: quit-flagged women have a mean one-way commute of 21.4 minutes versus 30.2 minutes for the not-quit-flagged group ($p = 0.11$, N = 172). The shorter commutes among exiters likely capture job quality alongside household burden, since the exiting group is disproportionately drawn from non-regular local jobs. Commute time in this sample is associated with the kind of local, lower-attachment work most vulnerable to exit.

The post-birth time-use data make the household constraint more visible. At $t = +1$, mothers average 82.7 hours per week of childcare, while fathers contribute 13.2 hours (Figure 14). The wife-to-husband ratio at this event time is 6.3-to-1; averaged across the broader post-birth window, the imbalance is closer to the roughly 5-to-1 ratio reported elsewhere in the chapter. Husbands in the sample work a mean of 47 hours per week, and 52 percent work 50 or more hours. Once commuting time is added, the typical husband has limited remaining time available to absorb a substantial share of infant care.

For mothers, the constraint is more binding still. At 82.7 hours per week of childcare - nearly 12 hours per day, seven days per week - combining infant care with employment is difficult without access to alternative care, employment continuity, or reduced-hours work. In the sample, 89.6 percent of households are nuclear, childcare leave is taken by only 21.1 percent of mothers, and work-from-home is available in only 10.1 percent of jobs. For a non-regular worker at a small firm with no co-residing grandparents and no work-from-home option, the logistics of combining employment with infant care are difficult to reconcile without reduced-hours work, alternative care, or continued job protection.

In this sample, the motherhood penalty appears plausibly constrained by labour-market structure. A purely preference-based reading fits the evidence less well. The women who exit are not necessarily choosing leisure over work; in these descriptive data, they appear to face a time-budget constraint in which infant-care requirements can exceed the hours that remain available once the husband's work schedule and limited employer flexibility are taken into account.

5.6 Heterogeneity at the birth transition

I next examine heterogeneity in the employment penalty across labour-market positions at $t = 0$. Figures 3–5 report the descriptive employment change, normalized to $t = -1$, by subgroup. The gradient is steep. Non-regular workers at large firms experience the largest penalty (-12.1 percentage points, N = 33), followed by all non-regular workers (-9.8 pp, N = 70). Regular workers - whether at large or small firms - experience penalties of only -1.6 to -1.9 pp. Small-firm workers as a group show essentially no aggregate penalty ($+0.4$ pp, N = 115), though this masks heterogeneity by contract type within small firms (-4.5 pp for non-regular, N = 35; -1.9 pp for regular, N = 54; the remaining 26 women have missing contract-type data). Because these coefficients come from separate subgroup event studies, they do not aggregate arithmetically to the pooled small-firm coefficient.

For these heterogeneity event studies, subgroup definitions are based on pre-birth job information at $t = -2$. Some women in each subgroup are already no longer employed by $t = -1$, so subgroup baselines fall below 100 percent employment.

The largest penalty falls on non-regular workers at large firms, which is consistent with a composition effect. Non-regular workers at small firms show a smaller penalty. The pre-birth mechanism evidence shows that small-firm non-regular workers are disproportionately selected out before the birth year. The $t = 0$ heterogeneity is estimated on a more selected small-firm risk set, which compresses the observed at-birth penalty in that cell. Large firms retain more women in employment up to the birth transition, while small-firm non-regular workers appear more likely to have already exited or shifted onto lower-hour trajectories before birth. The main pattern is that contract type matters more than firm size. The penalty is -9.8 pp for all non-regular workers and -1.6 pp for all regular workers, a difference about six times larger. This heterogeneity corroborates the childbirth-margin logistic results in Section 5.3 and provides descriptive evidence that the dual labour market stratifies the motherhood penalty.

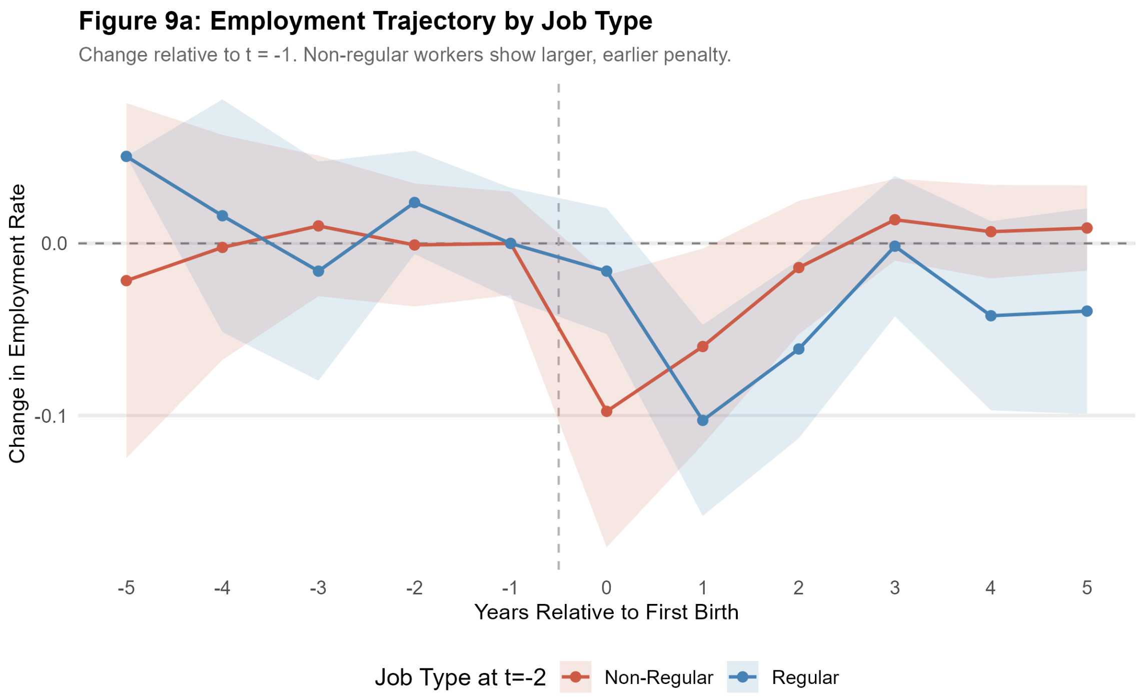

Figure 3: Employment event-study heterogeneity by contract type (year fixed effects; robust standard errors). Regular and non-regular refer to contract status, not to hours-based full-time or part-time categories. Coefficients are percentage-point changes relative to $t = -1$. At $t = 0$, non-regular workers show a -9.8 pp decline versus -1.6 pp for regular workers.

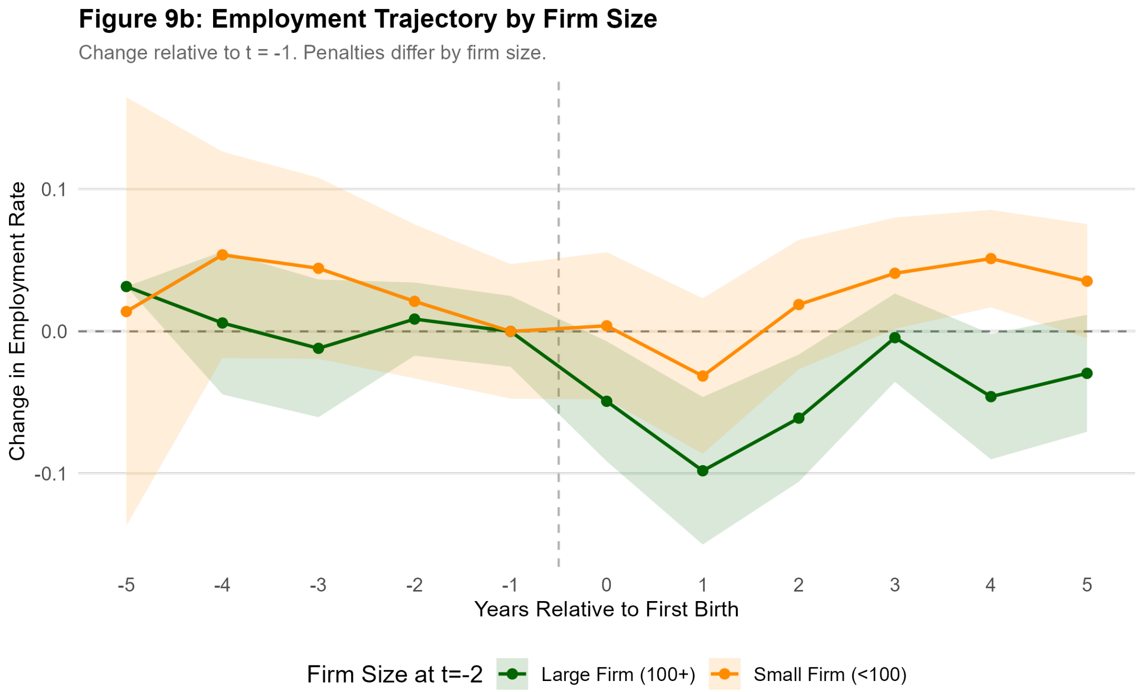

Figure 4: Employment event-study heterogeneity by firm size (year fixed effects; robust standard errors). Coefficients are percentage-point changes relative to $t = -1$. At $t = 0$, large-firm workers show a -4.9 pp decline versus $+0.4$ pp for small firms.

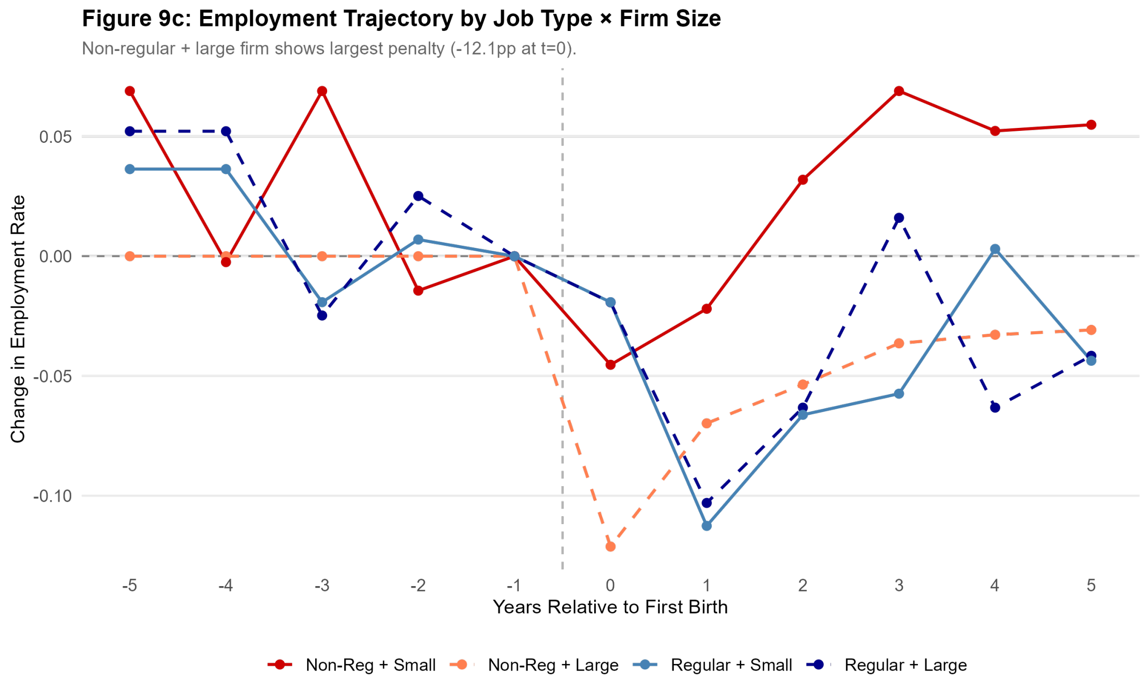

Figure 5: Employment event-study heterogeneity by contract type $\times$ firm size (year fixed effects; robust standard errors). Coefficients are percentage-point changes relative to $t = -1$. The largest at-birth decline appears for non-regular workers in large firms (-12.1 pp at $t = 0$).

5.7 Geographic non-variation

The heterogeneity results above show that the penalty varies by contract type and firm size. A natural question is whether it also varies by geography - whether the penalty is concentrated in metropolitan areas with intense corporate work cultures, or in rural areas with stronger traditional norms. I split the sample by metropolitan status (Kanto and Kinki regions, which contain Tokyo and Osaka, versus all other regions) and estimate separate event studies with the same year-FE specification.[4]

The result is a null. At $t = 0$, the employment penalty is -13.6 percentage points in metropolitan areas (N = 355 women, 2,793 obs, $p < 0.001$) and -12.9 percentage points in non-metropolitan areas (N = 307 women, 2,419 obs, $p = 0.002$). The difference of 0.7 percentage points is economically negligible. Pre-trends are clean in both subsamples: the $t = -2$ coefficient is $+0.031$ ($p = 0.46$) in metropolitan areas and -0.063 ($p = 0.16$) in non-metropolitan areas. Recovery paths are also similar, with both groups reaching coefficients near zero by $t = +3$.

A secondary split by city size (21 designated major cities versus other areas) shows a suggestive difference: the at-birth penalty is -9.9 pp in major cities and -14.8 pp

[4]In the JHPS/KHPS researcher-use files, prefecture identifiers are not provided; available geographic variables are region and city size. I use a metropolitan split (Kanto/Kinki vs. other regions) as the finest comparable geographic partition available in this study.

elsewhere. This comparison is complicated by differential pre-trends, since the major-city $t = -2$ coefficient is $+0.097$ ($p = 0.07$), so the contrast should be read with caution.

The Kanto/Kinki versus other-regions split shows no meaningful difference in the at-birth penalty, and that null is itself informative. At the same time, this comparison is coarse and cannot capture within-region heterogeneity in wages, childcare access, commuting constraints, or local labour demand. The results are consistent with the dual labour market operating nationwide, including beyond Tokyo and Osaka, while still leaving open finer-grained geographic heterogeneity in the composition of underlying channels.

Regional disparities in labour-market conditions are substantial in Japan: prefectural indicators show persistent dispersion in female employment and earnings levels (Statistics Bureau of Japan, 2025; Ministry of Health, Labour and Welfare, 2025). Childcare constraints also vary geographically, as reflected in differences in childcare capacity and waiting-child indicators across jurisdictions (Children and Families Agency, Government of Japan, 2024c,a). However, because childcare fees are locally administered and means-tested, there is no single harmonised prefecture-level fee series directly comparable to the labour outcomes used here (OECD, 2025b). Appendix Table 23 reports the corresponding all-prefecture dispersion statistics for childcare infrastructure and utilisation and provides geographic context for the metropolitan split in the main analysis.

5.8 Household structure heterogeneity

To assess whether the main pattern is driven more by household composition or by labour-market position, I examine subgroup trajectories by co-resident grandparent status and by patrilocal versus matrilocal household arrangements. These plots are descriptive and complement the main event-study estimates.

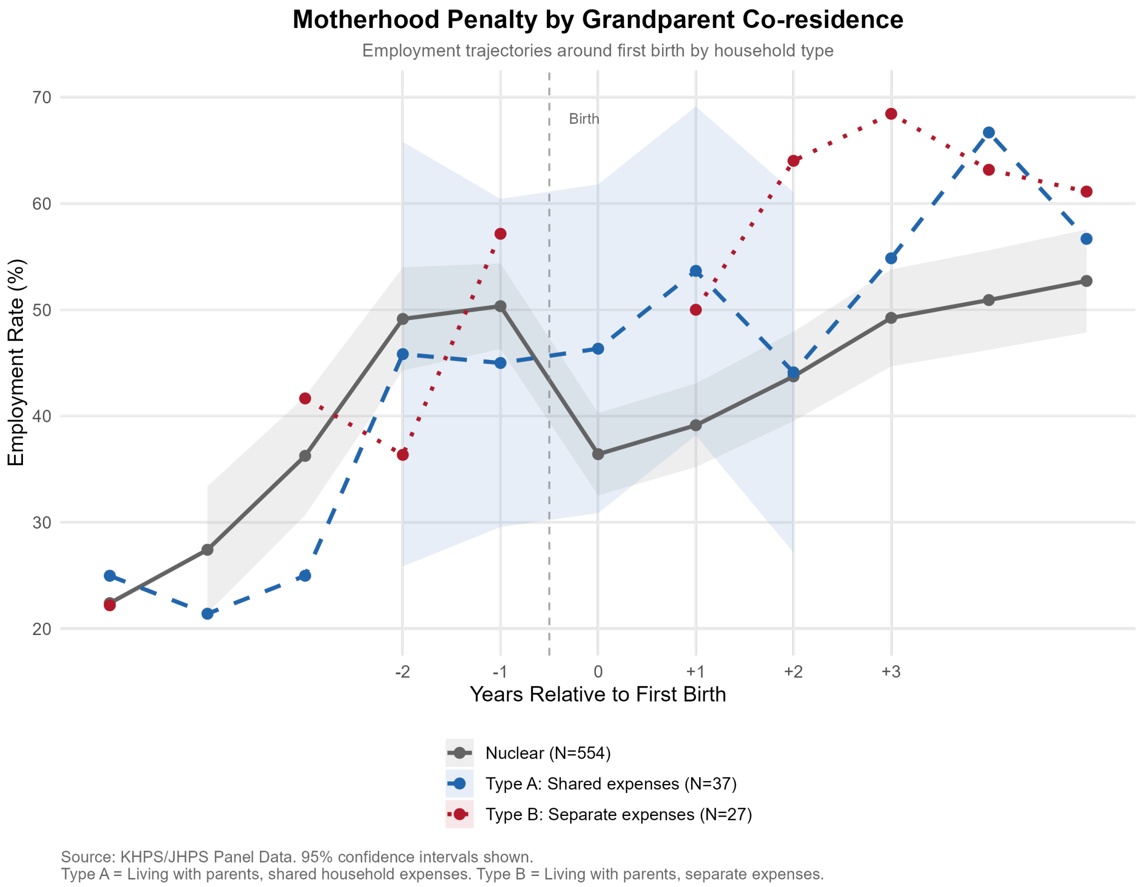

Figure 6: Additional heterogeneity diagnostic by co-resident grandparent status. The event-time coefficients are relative to $t = -1$ in the same year-FE framework. Both groups show a birth-year employment decline, with magnitude differences but no sign reversal.

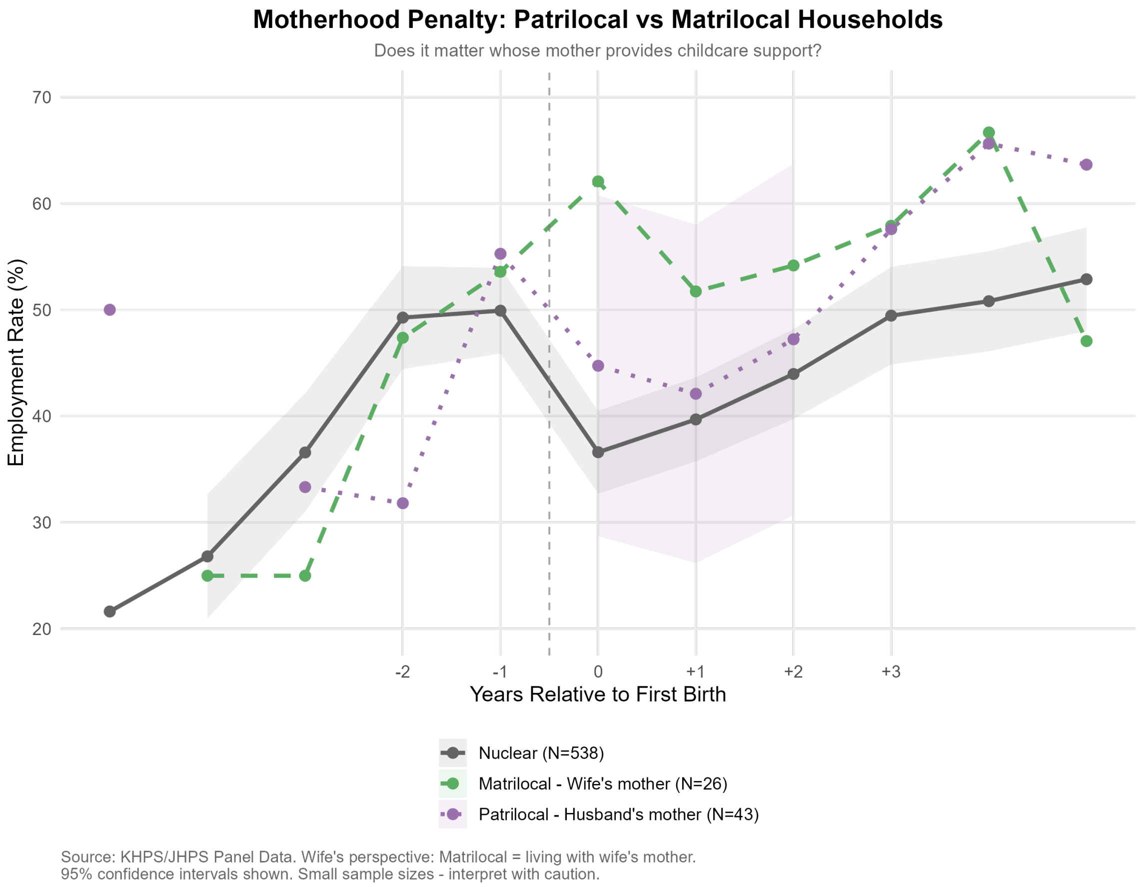

Figure 7: Additional heterogeneity diagnostic by patrilocal versus matrilocal household structure. The event-time coefficients are relative to $t = -1$ in the same year-FE framework. Both household structures exhibit the same qualitative birth-related employment decline.

Co-resident grandparents are associated with some attenuation in the birth-year decline, but not with a qualitative reversal of the pattern. Likewise, patrilocal and matrilocal households both exhibit a negative birth-related employment shift. In this sample, household structure shapes the magnitude of the birth-year decline; the dominant gradient continues to track the labour-market tier reported above.

6 The penalty's transformation: from exit to downgrade

The preceding sections focused on who leaves employment and why. But the motherhood penalty does not end with the exit decision. Women who remain employed - or who return to work after a period of absence - face a second penalty that operates through reduced hours, lower-quality re-entry, and earnings losses. This section documents the intensive margin of the penalty and shows that it comes to dominate the total penalty over time.

Among employed women at $t = -1$, mean weekly hours are 34.1 (SD 16.5). For earnings, the descriptive baseline mean is 268.9 man-yen (SD 206.9), while the positive-earnings sample used in the earnings event study has a $t = -1$ mean of 264.7 man-yen. These are the baselines against which the intensive-margin coefficients should be read.

6.1 Employment recovery and its limits

Figure 2 shows that employment partially recovers after the sharp decline at $t = 0$. The event-study coefficient narrows from -0.132 at $t = 0$ to -0.031 at $t = +3$ and -0.028 at $t = +5$. Women do return to work, but the aggregate series is misleading because the composition of employment changes substantially.

This timing is also consistent with Japan's childcare-system transitions. The years closest to birth coincide with the most constrained care period, especially ages 0-2, whereas care options broaden from age 3 onward, when enrolment pathways and subsidy coverage are typically more favourable (Children and Families Agency, Government of Japan, 2024b,a; OECD, 2025b). The observed recovery profile aligns with a staged institutional environment and does not compress into a single post-birth adjustment.

6.2 The collapse of full-time employment

Before first birth, 30.9 percent of all women in the sample are employed full-time. Here, full-time and part-time refer to hours worked and are distinct from regular versus non-regular contract status. Using the transition matrix, I track the employment status of women who held full-time positions at $t = -1$. At $t = +1$, only 45.7 percent remain full-time; 33.7 percent have exited employment and 20.6 percent have shifted to part-time. At $t = +3$, full-time persistence improves to 53.3 percent, but by $t = +5$ it falls back to 42.2 percent, a pattern partly explained by second births. When observations after second birth are censored, full-time persistence at $t = +5$ rises to 53.8 percent, which indicates that subsequent fertility explains part, but not all, of the late-horizon non-recovery.

Among women who shift to part-time at $t = +1$, relatively few transition back to full-time work. By $t = +5$, only 23.3 percent of women who were part-time at $t = +1$

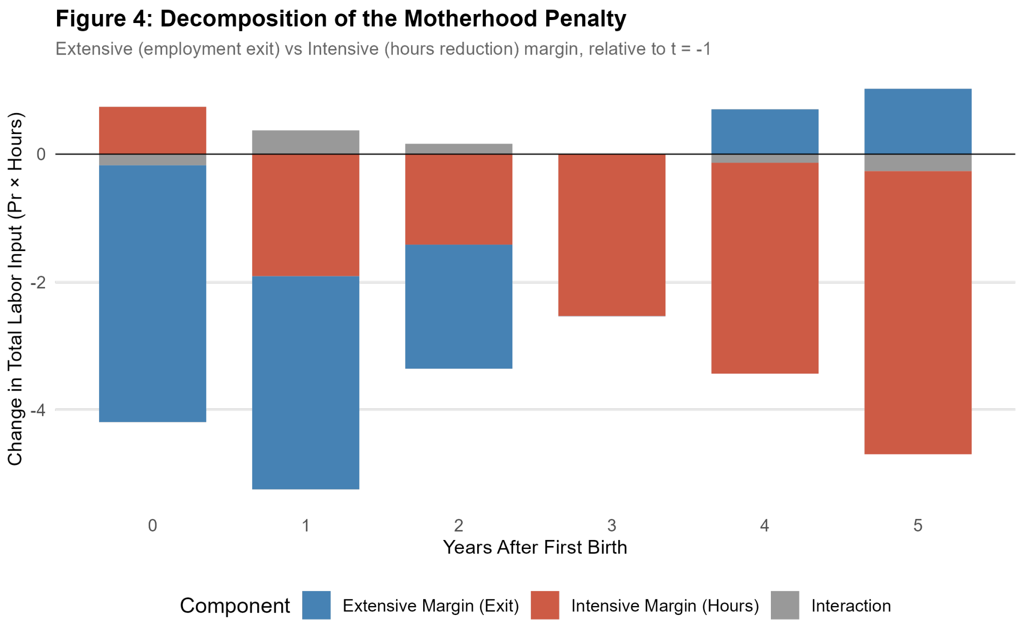

Figure 8: Decomposition of employment status into full-time and part-time shares around first birth. Full-time is defined as 35+ hours/week and part-time as employed below 35 hours/week; these are hours categories, not contract-type categories. Relative to $t = -1$, the full-time share falls sharply at birth and remains depressed, while the part-time share rises and accounts for most aggregate employment recovery (baseline at $t = -1$: 30.9% full-time, 19.4% part-time).

have moved to full-time, 18.6 percent have exited entirely, and the remainder are still part-time. This pattern is consistent with reduced-hours re-entry operating less as a temporary bridge back to full-time work and more as a persistent downgrade in job quality.

6.3 Hours and earnings conditional on working

The shift from full-time to part-time is reflected in the intensive-margin outcomes. Conditional on being employed, the event-study coefficient for weekly hours at $t = +5$ is -10.1 hours ($p < 0.001$), a 30 percent decline relative to the baseline mean of 34.1. The decline is already visible at $t = 0$, deepens further by $t = +1$ (coefficient -4.4 hours), and continues through $t = +5$. The pattern points to ongoing adjustment across the post-birth period.

Conditional earnings follow a parallel trajectory. The event-study coefficient at $t = 0$ is -49.9 man-yen ($p = 0.002$), deepening to -121.4 at $t = +1$ and remaining at -96.2 at $t = +5$ (both $p < 0.001$). Relative to the baseline mean of 264.7 man-yen in the positive-earnings employed sample, the $t = +5$ coefficient represents a 36 percent decline. The earnings loss is larger in proportional terms than the hours loss, which is consistent with post-birth employment shifting toward lower-paying and lower-quality jobs. The pattern does not look like a simple move to fewer hours on the same wage schedule.

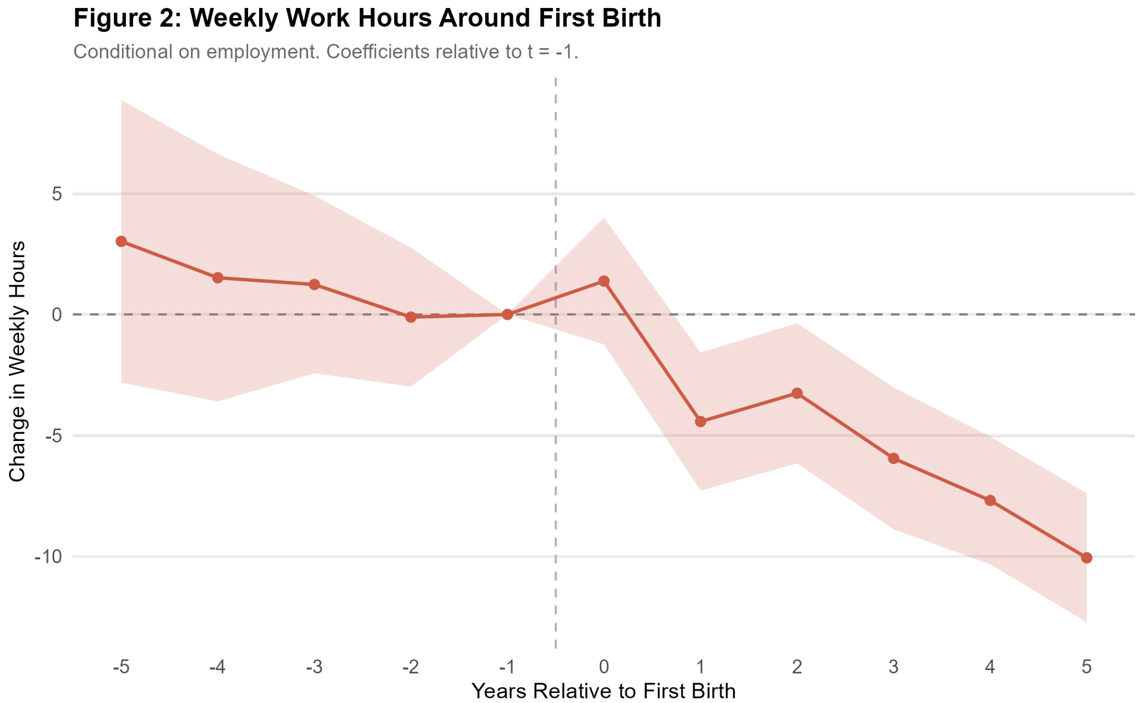

Figure 9: Intensive-margin event study for weekly hours among employed mothers only (year fixed effects; robust standard errors). Coefficients are hour changes relative to $t = -1$; by $t = +5$, weekly hours are lower by 10.1 from a baseline mean of 34.1. Bars show 95% confidence intervals.

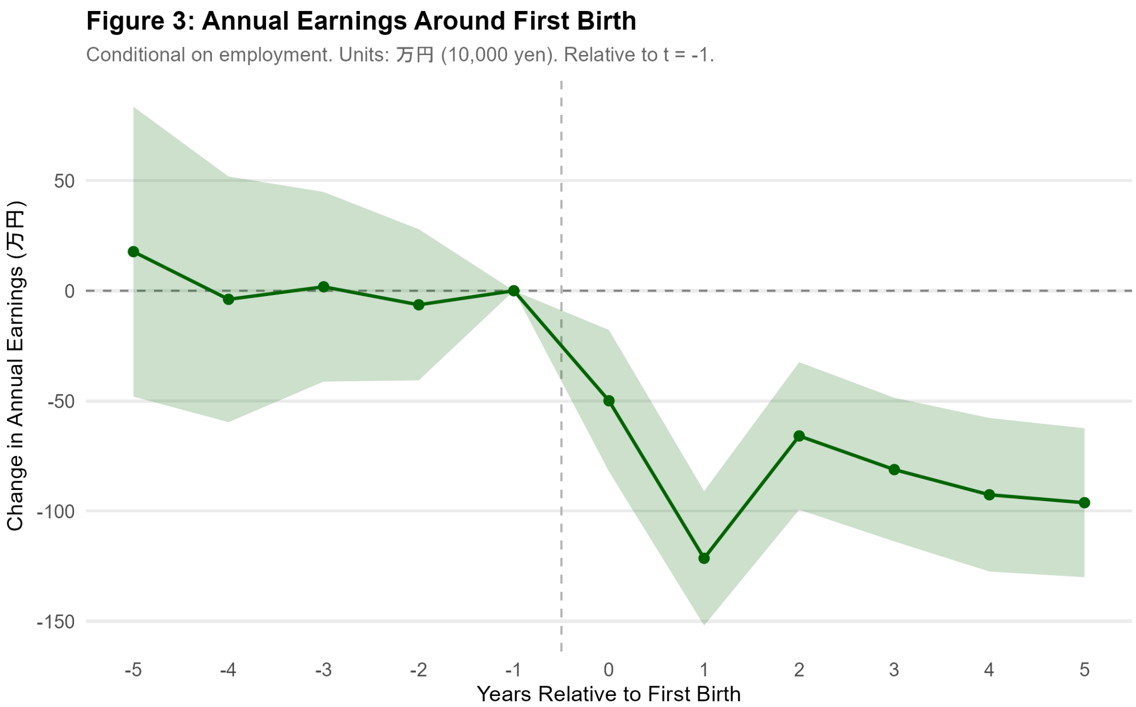

Figure 10: Intensive-margin event study for annual earnings among employed mothers only (year fixed effects; robust standard errors). Coefficients are changes in man-yen (10,000 yen units) relative to $t = -1$; by $t = +5$, earnings are lower by 96.2 man-yen from a positive-earnings event-study baseline mean of 264.7. Bars show 95% confidence intervals.

To assess whether these intensive-margin patterns are only a by-product of conditioning on employment, I re-estimate hours and earnings on the full sample by assigning zero outcomes to non-employed observations. The post-birth declines remain large and statistically strong. Relative to $t = -1$, unconditional hours are lower by 3.9 hours at $t = 0$, 6.1 hours at $t = +1$, and 6.0 hours at $t = +5$ (all $p < 0.001$ with woman-clustered SEs). Unconditional earnings are lower by 55.9 man-yen at $t = 0$, 86.8 man-yen at $t = +1$, and 52.2 man-yen at $t = +5$ (all $p < 0.001$). The near lead at $t = -2$ remains close to zero for both outcomes. This supports the interpretation that the post-birth deterioration is broader than a conditioning-on-employment artefact (Appendix D.7, Table 29).

Directionally, the most plausible selection in this setting is positive selection among post-birth employed women: those with stronger attachment and better pre-birth jobs are more likely to remain employed. If so, the conditional intensive-margin declines likely understate the full-sample deterioration associated with childbirth. I interpret the conditional coefficients as conservative in magnitude and use the unconditional outcomes as a complementary check.

Table 9 reports the earnings event-study coefficients alongside the number of women observed and employed at each event time, which makes the changing sample composition transparent.

Table 9: Annual earnings event study with sample composition.

t	N	Employed	Emp.%	Mean earn.	Coef.	SE	p
-5	186	42	22.6	254.7	17.7	33.6	0.597
-4	241	64	26.6	249.5	-3.9	28.4	0.891
-3	316	113	35.8	262.7	1.8	22.0	0.935
-2	453	219	48.3	256.8	-6.4	17.5	0.716
-1	656	330	50.3	264.7	0.0	–	–
0	662	255	38.5	217.8	-49.9	16.4	0.002
1	654	265	40.5	140.0	-121.4	15.6	<0.001
2	594	265	44.6	210.6	-65.9	17.1	<0.001
3	509	256	50.3	196.8	-81.2	16.6	<0.001
4	487	255	52.4	191.6	-92.6	17.8	<0.001
5	454	242	53.3	190.6	-96.2	17.3	<0.001

Notes: Earnings are in man-yen (10,000 yen). Coefficients come from the year-FE event study, conditional on employment and relative to $t = -1$, with heteroskedasticity-robust standard errors. N is the total number of women observed at each event time; Employed is the number with positive hours; Mean earn. is mean annual earnings among employed women.

The absence of recovery is important. In countries where the child penalty is driven primarily by career interruptions, earnings often recover as women rebuild human capital over time (Kleven et al., 2019b; Goldin, 2014; Bertrand et al., 2010; Blau and Kahn, 2017). In Japan, the lack of recovery suggests that the interruption is more than temporary. Women appear to re-enter a different tier of the labour market, with limited return to their previous career tracks.

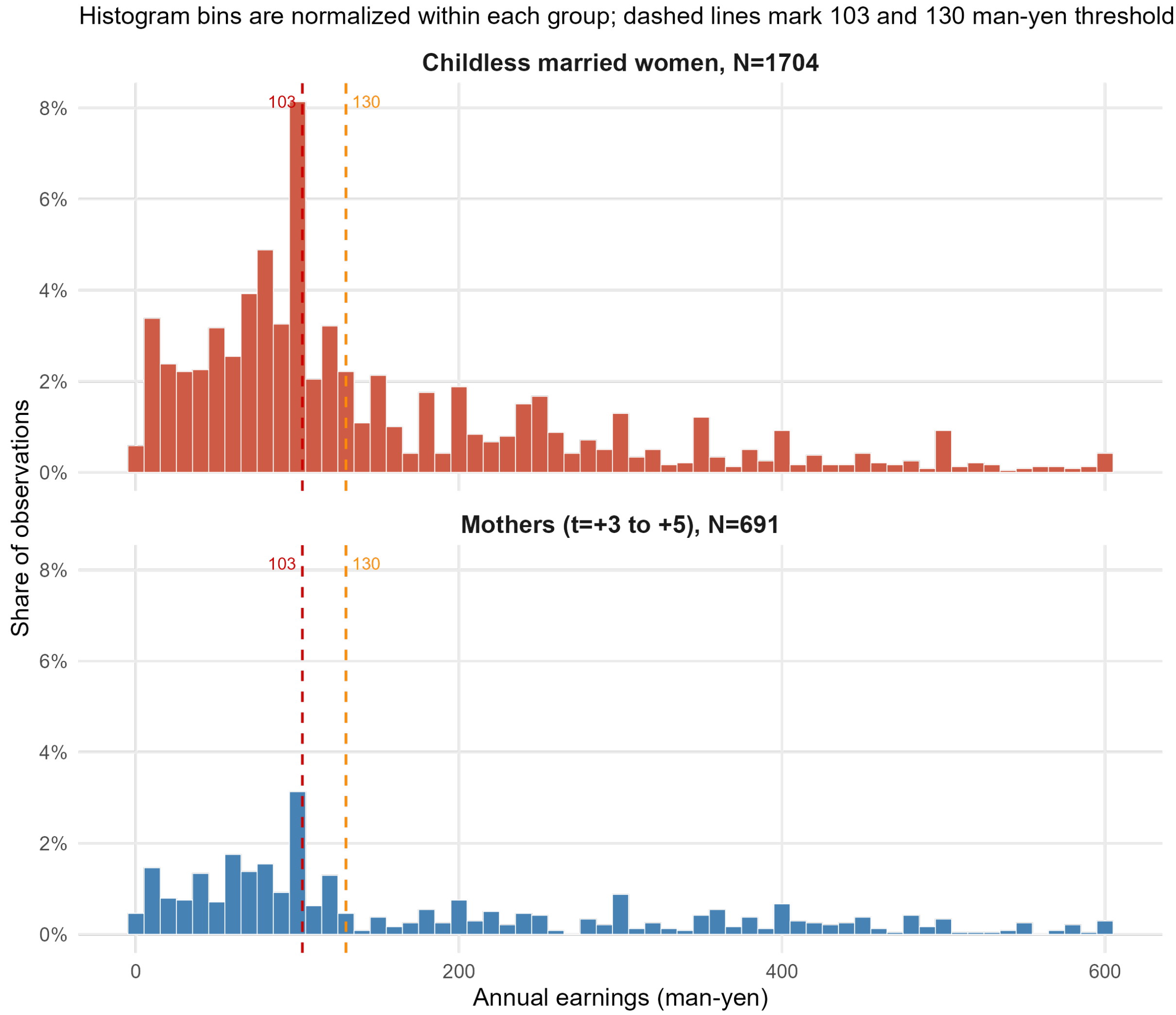

Figure 11: Normalized earnings distributions for employed observations among mothers at $t = +3$ to $t = +5$ and currently married childless women. N denotes observation counts within each panel, and the y-axis reports the share of observations within that panel. Dashed lines mark the 103 and 130 man-yen thresholds; the cumulative shares below these cutoffs are similar across groups, even though the single-bin spike at 103 is sharper for the childless sample. The figure supports the interpretation that threshold bunching extends beyond mothers and likely reflects a broader secondary-earner pattern among married women.

Figure 11 compares normalized earnings distributions for employed mothers in the post-birth return period ($t = +3$ to $t = +5$) and employed childless married women. In the pooled mothers' sample, 47.8% of observations lie at or below 103 man-yen and 55.9% at or below 130 man-yen (median 115 man-yen). Among childless married employed women, the corresponding shares are very similar or slightly higher (49.7% and 60.0%, median 105 man-yen). I interpret this comparison cautiously: the childless group is a descriptive benchmark, not a causal counterfactual. Still, the similarity of the two distributions suggests that concentration around these thresholds is not unique to mothers and likely reflects a broader structural feature of earnings choices among married secondary earners. Motherhood appears to increase exposure to reduced-hours jobs in which these pre-existing thresholds are more likely to bind.

6.4 The shifting composition of the penalty

The preceding results - the collapse of full-time employment, the persistent hours decline, and earnings losses that exceed the hours losses - reveal a penalty that changes form over time. At $t = +1$, the penalty is primarily extensive: the event-study employment coefficient is -0.124, which means that much of the overall hours loss reflects women who are not working at all. By $t = +5$, the employment coefficient has narrowed to -0.028 and is statistically indistinguishable from zero, but the hours coefficient has deepened to -10.1 and the earnings coefficient remains at -96.2 man-yen. The penalty is still present, but it has shifted from the extensive margin to the intensive margin. The apparent recovery in employment should not be read as full labour-market recovery, especially because the long horizon is panel-sensitive while the intensive margin remains persistently negative.

Figure 12 makes this shift visible using an accounting decomposition of average hours. Writing mean hours as the product of the employment rate and mean hours conditional on employment, the figure separates the change relative to $t = -1$ into an extensive component (employment decline), an intensive component (lower hours among employed women), and an interaction term. At $t = +1$, when the employment penalty is deepest, the extensive component explains most of the hours loss. By $t = +5$, the intensive component dominates: more of the remaining hours loss comes from women who are working fewer hours than before birth.

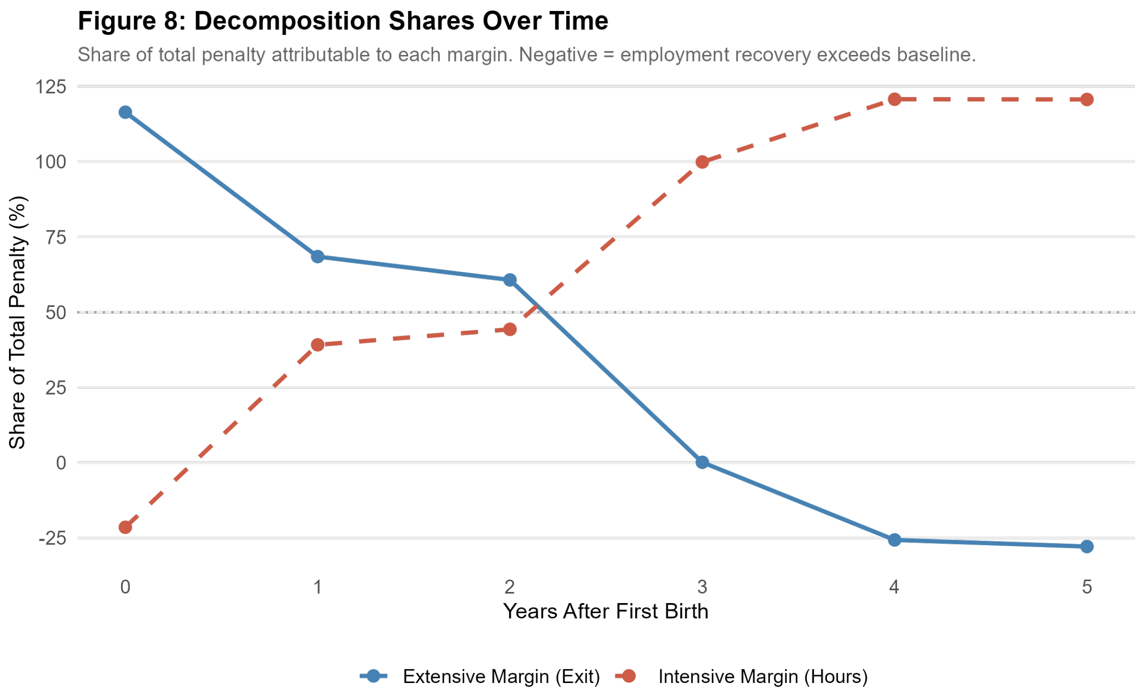

Figure 12: Accounting decomposition of the change in average hours into extensive (employment) and intensive (hours within employment) margins by horizon. Shares are computed from the identity linking mean hours to the employment rate and conditional hours, and can fall outside $[0, 100]$ when one component offsets the other. The extensive margin dominates immediately after birth, while the intensive margin dominates later horizons.

The decomposition and transition evidence are the core visual diagnostics in this section: they show that the penalty moves from immediate exit to persistent job-quality downgrade. As a separate descriptive slice, Figure 13 reports the implied hourly wage among employed mothers with positive earnings and positive hours, computed directly as annual earnings divided by annualized weekly hours. The mean implied wage declines from 1,857 yen per hour at $t = -1$ to 1,726 yen per hour at $t = +5$, a drop of 7.1 percent. This series should not be read as a mechanical transformation of the event-study coefficients above, because it is a ratio constructed on its own conditioning sample. It is best interpreted as an additional descriptive indicator of post-birth job quality. On that narrower reading, the decline is directionally consistent with the broader pattern in the data: post-birth employment appears to shift not only toward fewer hours, but also toward lower-paying and lower-quality job matches.

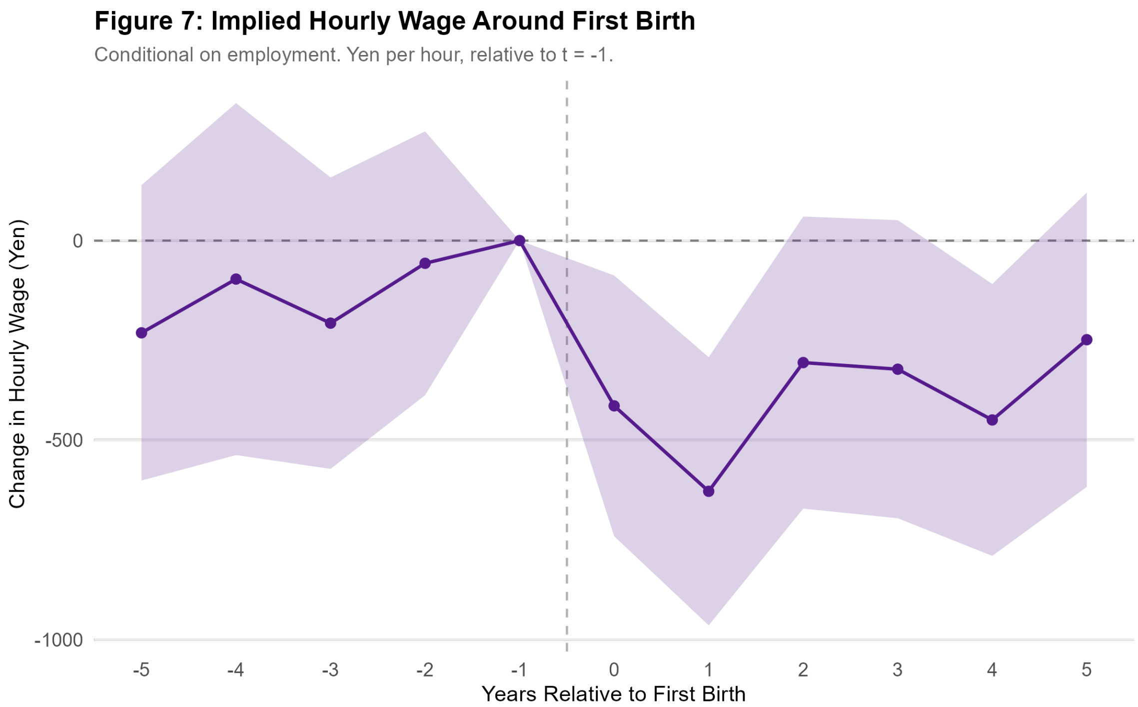

Figure 13: Implied hourly wage among employed mothers, computed as annual earnings divided by annualized hours (weekly hours $\times 52$). Mean implied wage declines from 1,857 yen/hour at $t = -1$ to 1,726 yen/hour at $t = +5$ (-7.1%), which is consistent with lower-quality post-birth job matches.

6.5 The return hazard

Among women who exit employment at or around birth, how quickly do they return? A descriptive cumulative first-return series provides a direct summary and complements the transition matrix in Section 6.2. The at-risk set consists of women not working at $t = +1$ who are still observed at each later horizon, so it narrows from 98 women at $t = +2$ to 54 at $t = +3$, 45 at $t = +4$, and 38 at $t = +5$. Within this descriptive series, 30.6 percent of the at-risk group have returned to employment by $t = +2$. Cumulative first return reaches 45.4 percent by $t = +3$ and 56.5 percent by $t = +5$. No further first returns are observed between $t = +4$ and $t = +5$ in the remaining at-risk sample.

When the sample is restricted to women with no second birth by the horizon, the pattern is nearly identical: cumulative return reaches 55.9 percent by $t = +5$. This similarity suggests that the late-horizon plateau is not explained solely by subsequent fertility. A longer absence may still be associated with persistent barriers, including loss of firm-specific capital, weaker labour-market networks, and stigma associated with long employment gaps, but this descriptive series should not be read as a formal hazard estimate.

6.6 Workplace systems and the limits of accommodation

The data include a module on workplace systems at the employer level. Among employed mothers in the sample, 60.3 percent report that their employer offers reduced working hours, and 65.2 percent report half-day or hourly leave options. These figures indicate that some degree of flexibility is nominally available. However, only 10.1 percent report work-from-home availability, and 49.1 percent report a rehiring or re-employment system related to childcare leave. The survey item does not distinguish reinstatement after statutory leave from re-employment after a prior separation, so I interpret it broadly. A further 62.3 percent report that their employer offers a pathway from non-regular to regular employment.

The transition evidence in Section 6.2 qualifies these availability shares. Relatively few women move from part-time hours back to full-time work, even when they remain attached to the labour market, which suggests that formal upward pathways are not widely realised in practice. This does not directly identify contract-status conversion, but it does indicate that upward mobility in job quality is limited along one important observable dimension. More broadly, the results in this chapter are consistent with a gap between formal availability and realised continuity. The practical value of workplace systems appears to depend on contract status, employer capacity, and whether the employment relationship survives through childbirth.

6.7 The non-regular re-entry trap

Taken together, the findings in this section are consistent with a self-reinforcing re-entry trap. Women in non-regular employment are the most likely to exit around motherhood, and post-birth employment shifts strongly toward reduced-hours work among those who remain employed or return. Among employed mothers, hours, earnings, and implied hourly wages remain below their pre-birth levels throughout the observed window. At the same time, transitions back to full-time work appear limited. The overall pattern is consistent with a durable shift toward lower-quality post-birth employment in a segmented labour market.

Because this design is not an individual-level causal re-entry model, I do not claim that every exiting woman re-enters through the same pathway. The claim is narrower: in the aggregate, post-birth re-entry appears to occur disproportionately through lower-quality positions, and these positions do not seem to function as temporary bridges back to pre-birth employment trajectories.

6.8 Skills training and the Training Grants system

An auxiliary descriptive question is whether childbirth is also associated with lower investment in job-related skills. Career interruptions can depreciate accumulated skills and generate wage losses that persist beyond the interruption itself (Mincer and Ofek, 1982; Mincer and Polachek, 1974). In Japan, where non-regular re-entry positions offer limited employer-provided training (OECD, 2021, 2022), self-directed skill investment may be one channel through which post-birth labour-market disadvantage persists.

Both the KHPS and JHPS ask respondents and spouses whether they have taken voluntary actions - such as attending school, lectures, or self-study - to improve work-related skills, and whether they have used the government's Training and Education Benefits System (*kyōiku kunren kyūfu seido*). This system, administered through Hello Work (*haroōwaku*) offices, subsidises 20-70 percent of approved training-course costs for workers enrolled in employment insurance (Ministry of Health, Labour and Welfare, 2024). A 2018 reform extended the eligibility window from 4 to 20 years for individuals who left employment due to pregnancy, childbirth, or childcare.

Table 10 reports the share of mothers reporting any skills-training activity by event time. Before birth, about one-third of mothers report active skill investment in a given year (roughly 27–39 percent across $t = -5$ to $t = -1$). This rate drops sharply to 8.6 percent in the birth-report wave ($t = 0$) and recovers only partially, reaching about 13–16 percent by $t = +2$ through $t = +4$ and 12.8 percent at $t = +5$. Whether this decline reflects reduced time availability, weaker employer support for training, lower expected returns to skill investment during interruption, or simple loss of labour-market attachment is not separately identifiable from these data.

Use of the Training and Education Benefits System is negligible in this sample: at each event time, fewer than 3 percent of mothers with valid responses report current or recent use. The more informative pattern lies in why usage is so low. Table 11 decomposes awareness and self-reported eligibility at selected event times. Roughly half of mothers have never heard of the programme, and this share changes little around childbirth (47 percent at $t = -2$, 53 percent at $t = 0$, and 45 percent at $t = +5$). Among aware non-users, most report being either ineligible or uncertain about their eligibility. Before birth ($t = -1$), 24 percent of aware non-users report that they are eligible, 38 percent report that they are not, and 38 percent do not know. By $t = +5$, the ineligible share rises to 46 percent and the eligible share falls to 19 percent, which is consistent with the loss of employment-insurance connection after labour-market exit. The 2018 reform extended the eligibility window to 20 years for mothers who left employment due to pregnancy or childcare (Ministry of Health, Labour and Welfare, 2024), but most observations in this sample predate that reform, and whether it has improved access in practice remains an open question.

The pattern points to a three-layer access barrier: invisibility (roughly half of mothers are unaware of the programme), ineligibility (38–46 percent of aware non-users report not qualifying), and uncertainty (roughly 35–40 percent do not know whether they qualify). This interpretation is consistent with OECD assessments that Japan's adult-learning system relies heavily on employer-provided training, offers limited flexibility, and underserves non-regular workers and career returners (OECD, 2021).

I treat this training evidence as auxiliary descriptive context. It supports the chapter's interpretation but does not stand as a separate headline contribution. Even on that narrower reading, the near-zero use of the training-grants system, despite its explicit design as a maternal reskilling channel, suggests that formal eligibility alone is insufficient when time constraints, information gaps, loss of employment-insurance connection, and weak rewards to additional qualifications in non-regular jobs remain in place (Asai, 2015; OECD, 2022). The training gap is best read as a plausible compounding channel for the persistent intensive-margin losses documented above, not as a separately identified mechanism.

Table 10: Skills Training Participation and Training Grants Awareness by Event Time

Event	Q2.1: Skills Training (%)			Q2.4: Training Grants (%)		
time	N_{valid}	Any (1+2)	Current (1)	N_{valid}	Used (1+2)	Aware (3)
−5	45	26.7	15.6	37	2.7	54.1
−4	70	28.6	8.6	63	0.0	46.0
−3	134	38.8	13.4	107	2.8	52.3
−2	253	35.2	14.2	185	2.2	51.4
−1	475	31.2	13.1	309	2.3	50.8
0	568	8.6	2.6	391	1.0	45.8
+1	594	10.6	4.4	387	1.3	49.6
+2	580	13.1	5.3	363	0.6	49.6
+3	493	15.6	4.3	292	0.3	53.4
+4	472	13.6	6.1	234	1.3	52.1
+5	437	12.8	6.2	177	0.0	54.8

Notes: The sample is the main analytic panel of 662 mothers around first birth. Q2.1 asks whether the wife took voluntary actions to improve job-related skills in the past year (1 = presently taking action; 2 = took action; 3 = did not). "Any" reports codes 1+2 as a share of valid responses (codes 1–3). Q2.4 asks about use of the Training and Education Benefits System (*kyōiku kunren kyūfu seido*): 1 = used and received benefits; 2 = using and will receive; 3 = aware but not used; 4 = unaware. "Used" reports codes 1+2, and "Aware" reports code 3; both are calculated as shares of valid responses (codes 1–4). Coverage varies by event time because of unbalanced panel support and item non-response. The 2018 reform extended eligibility to 20 years for individuals who left employment because of pregnancy, childbirth, or childcare (Ministry of Health, Labour and Welfare, 2024).

Table 11: Training Grants Access Barriers: Awareness, Eligibility, and Usage by Event Time

	All respondents (%)			Among aware non-users (%)		
Event time	Used	Aware, not used	Unaware	Eligible	Not eligible	Don't know
-2	2.2	51.4	46.5	–	–	–
-1	2.3	50.8	46.9	24.2	37.6	37.6
0	1.0	45.8	53.2	26.8	38.5	34.6
$+1$	1.3	49.6	49.1	19.8	35.4	44.3
$+3$	0.3	53.4	46.2	17.9	42.3	39.1
$+5$	0.0	54.8	45.2	18.6	46.4	35.1

Notes: The sample is the main analytic panel of 662 mothers around first birth. The left panel reports shares of all valid Q2.4 responses: "Used" = codes 1+2 (used and received or are receiving benefits); "Aware, not used" = code 3; "Unaware" = code 4. The right panel conditions on aware non-users (code 3) and reports self-assessed eligibility: "Eligible" = reports qualifying; "Not eligible" = reports not qualifying; "Don't know" = uncertain about eligibility. Eligibility for the Training and Education Benefits System (*kyōiku kunren kyūfu seido*) requires enrolment in employment insurance (*koyō hoken*). Women who exit employment at childbirth lose this connection, which is consistent with the rising "Not eligible" share at later event times. The 2018 reform extended the eligibility window to 20 years for individuals who left employment because of pregnancy, childbirth, or childcare (Ministry of Health, Labour and Welfare, 2024). The eligibility breakdown at $t = -2$ is omitted because of small cell sizes. Coverage varies by event time because of unbalanced panel support and item non-response.

7 Limited paternal adjustment

Parenthood creates a household shock with two potential margins of adjustment. Either partner can absorb childcare time through lower market work, and the realised allocation reflects wages, job security, workplace flexibility, and social norms within the household. Evidence from Scandinavian settings shows modest but measurable paternal adjustment, and stronger paternal adjustment is associated with smaller maternal penalties (Kleven et al., 2019b; Andresen and Nix, 2022). In Japan, I find no detectable systematic paternal adjustment in this sample window, so post-birth labour-supply responses remain concentrated among mothers.

7.1 Childcare time asymmetry

Interpretation of pre-birth childcare responses requires care because of survey timing. Births recorded in wave t can occur between February of the previous year and January of the survey year, so some observations coded as $t = -1$ already overlap early childcare; empirically, 43% of women at $t = -1$ report daily childcare. I therefore interpret childcare time primarily for $t \geq 0$, where event-time classification is cleaner. From $t = 0$ onward, the asymmetry is large and persistent. At $t = 0$, mothers report 115.5 weekly childcare hours versus 45.4 for fathers (about 2.5-to-1). By $t = +1$, the ratio widens to 6.3-to-1 (82.7 vs. 13.2), and by $t = +5$ it remains 5.3-to-1 (54.4 vs. 10.4). The unusually narrow ratio at $t = 0$ likely reflects the broad survey reporting window: the birth-year wave can include months before and immediately after birth, when paternal involvement is temporarily higher and the post-birth division of care has not yet fully settled. The key point for labour-market interpretation is persistence across the observed post-birth window. The caregiving burden remains heavily concentrated on mothers throughout that period.

7.2 Leave take-up

In the broader leave-analysis sample (N = 1,183 first-birth-linked women), only 3 husbands (0.3%) report taking childcare leave. In the partner-linked analytic subsample used for the event studies (N = 661), the corresponding figure is 1 of 661 (0.15%); the small difference reflects the different denominators across modules. For wives, by contrast, 250 of 1,183 women (21.1%) in the broader leave-analysis sample report taking childcare leave. Paternity leave, while formally available since 2010 and expanded in subsequent reforms, was taken by only 3 of 1,183 fathers in the leave-analysis sample during the observation period. Looking beyond this sample window, recent Japan-specific evidence suggests that rising paternal leave norms are associated with better maternal employment continuity (Nakayama and Ishikawa, 2025); the discussion returns to this point. National statistics also indicate that uptake has increased in recent years, though average leave duration

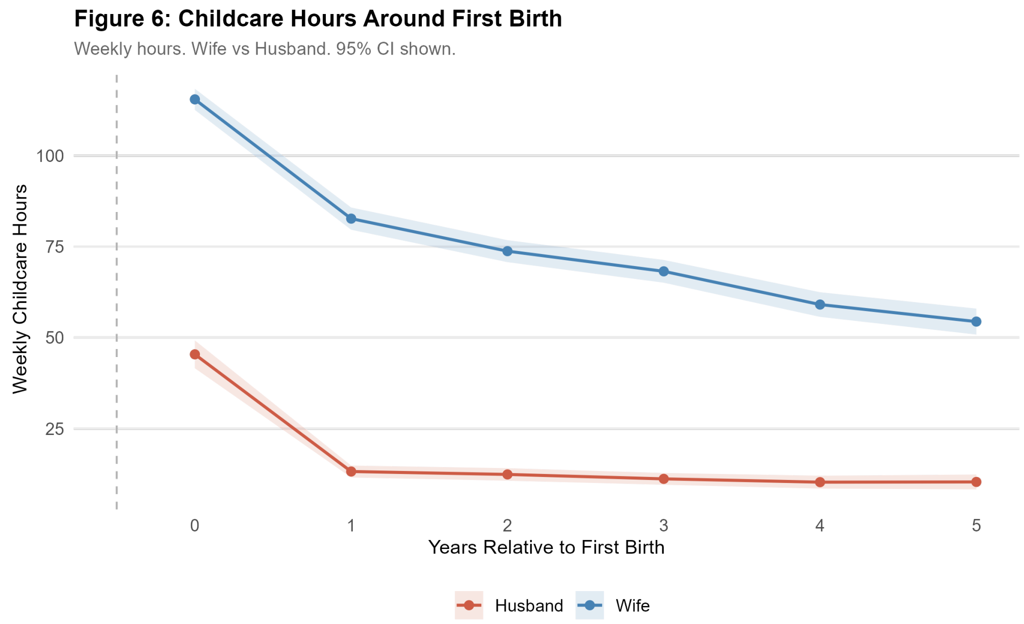

Figure 14: Mean weekly childcare hours for wives and husbands around first birth. Series are restricted to $t \geq 0$ because pre-birth survey timing can overlap pregnancy and early postpartum reporting. At $t = +1$, the wife-to-husband ratio is 6.3-to-1 (82.7 vs. 13.2 hours/week); by $t = +5$, it remains 5.3-to-1 (54.4 vs. 10.4).

among fathers remains short (OECD, 2024a; Ministry of Health, Labour and Welfare, 2019).

7.3 Interpretation

The near-zero father-hours profile in this sample is consistent with the institutional context described in Section 2. Japanese corporate culture places strong demands on male employees' time and availability, which leaves limited room for paternal caregiving. The result is a household in which the arrival of a child creates an asymmetric shock absorbed almost entirely by the mother. The extent of that adjustment then depends heavily on the structure of her job; the formal placebo event-study coefficients are reported in Appendix C.

This finding helps interpret the earlier results. When husbands work very long hours, the family's ability to redistribute childcare is correspondingly reduced, and more of the burden falls on the mother in this sample. In the regression models, husbands' overwork is imprecisely estimated and loses statistical significance once labour-market tier is included. I keep the father-side evidence as descriptive context, without elevating it to a standalone mechanism estimate. The descriptive picture established here - near-zero paternal adjustment, a persistent 5-to-6-to-1 childcare ratio, and essentially no paternity leave take-up - provides the household context against which the robustness checks in the next section are interpreted.

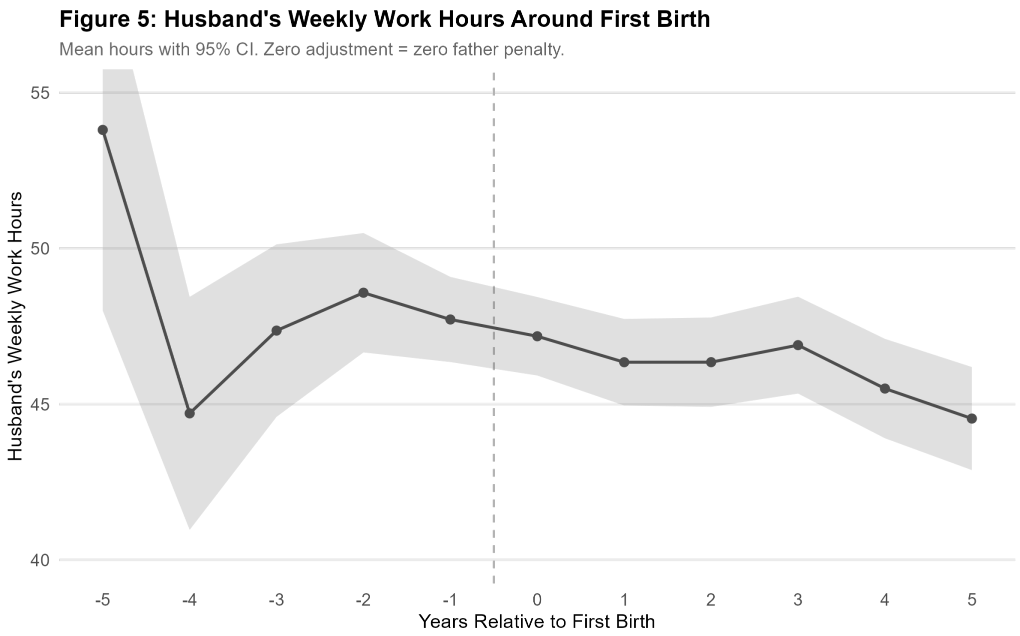

Figure 15: Father-hours benchmark around first birth. Displayed points are mean husband weekly hours with 95% confidence intervals by event time. The near-birth window is broadly flat, while far leads are more variable because support is thinner. Taken together, the profile indicates no detectable systematic paternal labour-supply adjustment around childbirth in this sample.

8 Robustness

I examine robustness and scope along several dimensions: a childless comparison group, support and pre-trend diagnostics, an industry-sector diagnostic that tests whether the main gradient is sectoral rather than contractual (Appendix D.11), second-birth dynamics, and small-sample inference for the mechanism regressions.

Table 12: Evidence-to-claim map: what each finding identifies and what it does not.

Claim	Evidence	What it identifies	What it does not identify	Level
Birth-year employment break	$t = 0$ coeff. $= -0.132$ ($p < 0.001$); near lead null; stable across cohort splits, survey sources, IPW, balanced panel, childless placebo	Sharp timing of employment decline at first birth in this panel	Causal effect of childbirth vs. anticipatory sorting; composition vs. within-person change	High
Recovery masks downgrade	Among women full-time at $t = -1$, 45.7% remain full-time at $t = +1$, 33.7% exit, and 20.6% shift to part-time; hours -10.1/wk; earnings -96.2 man-yen at $t = +5$; unconditional outcomes confirm	Persistent shift toward reduced-hours, lower-earnings re-entry	Whether downgrade is voluntary (preference) or involuntary (constrained); selection into return	High
Non-regular tier predicts exit	Childbirth margin: OR = 7.48 ($p < 0.001$), $N = 330$, 124 exits; absolute risk 13.8% vs. 64.8%; stable across missingness specs	Strong predictive gradient by contract type at $t = 0$	Whether non-regular status causes exit or proxies for unobserved worker characteristics	High
Pre-birth exit composition	$t = -2$ logit: OR $= 6.41$ ($p = 0.019$); $N_{\text{events}} = 13$; Firth/bootstrap confirm sign	Direction of pre-birth predictive association	Precise magnitude (wide CIs); causal sorting vs. selection	Moderate
Institutional interpretation	Earnings bunching at thresholds (47.8% mothers $\leq$103 man-yen; 49.7% childless married $\leq$103 man-yen) - suggests a broader secondary-earner phenomenon; training grants <3% usage with three-layer access barrier (invisibility, ineligibility, uncertainty); father hours flat; leave take-up 0.3%	Descriptive facts; threshold bunching is a pre-existing constraint on married women's earnings; training grant barriers are structurally linked to employment exit	Whether segmentation causes the penalty or merely correlates with it	Contextual / Interpretive

Notes: "High" = robust across major sensitivity checks. "Moderate" = direction is stable but precision is limited by small event counts. "Contextual / Interpretive" = institutionally grounded and not separately identified by the panel design; this label refers to the interpretive layer above the descriptive patterns and does not imply weak underlying descriptive evidence. The claim hierarchy is consistent with the evidence levels introduced in Table 1.

8.1 Childless comparison group

A within-sample childless comparison is infeasible because the harmonized classified panels are constructed from mothers only. I build a descriptive benchmark from the raw panels: married women aged 20–45 with no observed births and at least five years of panel observation. This yields N = 4,111 unique women and 50,222 person-year observations. Their average employment rate is 53.2 percent - close to the mothers' rate of 50.3 percent at $t = -1$ - with mean weekly hours of 28.5 among workers. However, the composition differs considerably: 71.9 percent of employed childless women hold non-regular positions, and 59.9 percent work in firms with fewer than 100 employees. Employment rates are stable in the 0.51–0.57 range across the 2004–2022 period, with no visible trend.

This benchmark serves two purposes. It confirms that the employment decline observed among mothers at $t = 0$ is specific to the birth event, since childless married women show no comparable cohort- or period-level drop. It also reveals that selection into motherhood is structured by job quality: the higher non-regular share among childless women (71.9 versus approximately 38 percent among employed mothers at $t = -1$) suggests that women with regular employment are more likely to appear in the mother sample, presumably because stable employment supports family formation. Under positive selection into motherhood on job quality, the observed average penalty estimates are likely conservative: the women in the analytic sample are better positioned in the labour market than the average married woman, so the population-average penalty may be understated here. The direction in which this selection affects the contract-type gradient specifically is less clear and depends on within-tier selection patterns.

I emphasize that this comparison is descriptive. The childless women differ from mothers in age, cohort, and unobserved characteristics, so the comparison does not support a difference-in-differences interpretation. See Appendix A for full details.

8.2 Attrition and support composition

Figure 16 summarizes the support profile directly and motivates the emphasis on the near-birth window in interpretation. Because the panel is unbalanced, compositional change across event time is a first-order concern. I report two diagnostics in Appendix D: support counts by event time, and baseline ($t = -1$) differences between women observed through $t = +5$ and those not observed through $t = +5$. Support thins substantially in the tails (from 186 women at $t = -5$ to 662 at $t = 0$, then 454 at $t = +5$). Observation at $t = +5$ is also weakly selective on baseline employment: women observed at $t = +5$ have lower $t = -1$ employment (47.7%) than those not observed (56.2%, $p = 0.045$), while baseline hours and income differences are not statistically significant. These diagnostics do not identify causal attrition effects, but they bound long-horizon interpretation and motivate caution for $t = +5$ coefficients.

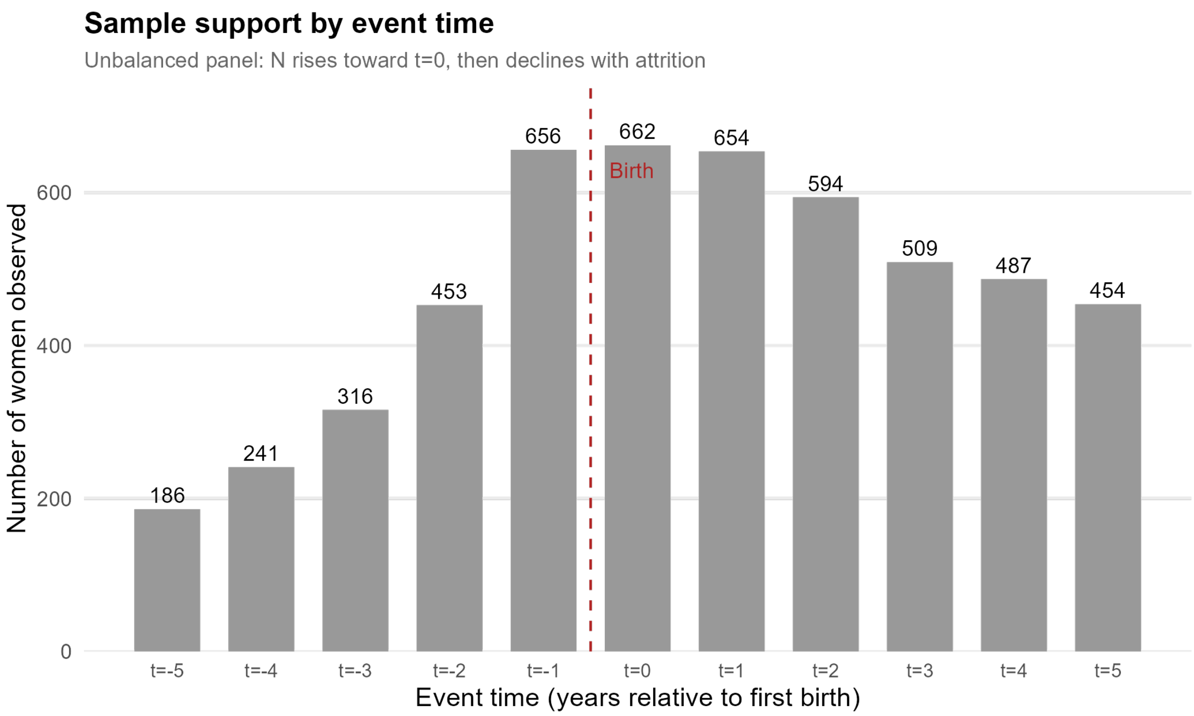

Figure 16: Sample support by event time in the unbalanced panel. The number of observed women rises toward birth and declines at longer horizons, so far leads and lags should be interpreted with greater caution.

I also re-estimate the employment event study on a balanced panel restricted to women observed continuously from $t = -2$ to $t = +1$ (N=445 women; 1,780 observations). In this sample, the key coefficients remain similar in sign and magnitude: $t = -2$ is -0.058 ($p = 0.065$), $t = 0$ is -0.128 ($p < 0.001$), and $t = +1$ is -0.121 ($p < 0.001$). This check supports the interpretation that the birth-year break is not driven by short-horizon support imbalance (Appendix D.6, Table 27).

Additional cleaner-sample checks reach the same conclusion. In an age-restricted sample (age ≥ 23), the childbirth-year drop remains large ($t = 0 = -0.123$, $p < 0.001$) and the near lead remains close to zero ($t = -2 = -0.007$, $p = 0.836$). In the balanced short-window sample ($t = -2$ to $t = +1$), the break is also preserved ($t = 0 = -0.128$, $p < 0.001$). Stricter attachment-based restrictions produce larger magnitudes. This is consistent with selection into higher-attachment subgroups: these women have more employment to lose at birth (Appendix D.6, Table 28).

8.3 Pre-trend diagnostics

The key pre-trend diagnostic in this design is whether employment is stable close to birth. In the unbalanced panel, the far leads ($t = -5, -4, -3$) show large negative coefficients that are jointly significant ($p < 0.001$). I do not interpret this pattern as evidence of a conventional pre-trend. Rather, it reflects two features of the sample: women enter the panel at different event times, so the far-lead estimates are based on a selected subsample observable early; and employment rises over the life cycle as women in their twenties enter the labour market.

The critical diagnostic is the near lead at $t = -2$, which asks whether employment is stable in the year immediately before birth. In the baseline specification, the coefficient is -0.012 ($p = 0.70$, N = 5,212). I conduct three robustness checks alongside this baseline, giving four specifications in total. In a trimmed pre-period specification that drops $t = -5, -4, -3$, the $t = -2$ coefficient remains null ($p = 0.72$, N = 4,469). In a balanced pre-support sample restricted to women observed at every pre-birth event time, the $t = -2$ coefficient remains economically small but somewhat larger in magnitude and does not cross conventional significance thresholds ($p = 0.078$, N = 1,814). This suggests some compositional difference in that stricter-support subsample and leaves the main near-lead pattern intact. Adding dataset fixed effects with a JHPS/KHPS indicator leaves the near lead unchanged ($p = 0.71$, N = 5,212).

Across all four specifications, the near lead remains economically close to zero and is statistically null in three of the four cases. Together, these checks support the interpretation that employment is stable in the year immediately before first birth. The sharp break at $t = 0$ (coefficient -0.132, $p < 0.001$) against this flat near lead supports the view that the birth-year decline is a distinct change from the earlier trajectory.

8.4 Second-birth dynamics

Birth spacing in the sample is relatively short. Among the 662 mothers, 255 (38.5 percent) have a recorded second birth within the panel window. The share with a second birth rises to 5.0 percent by $t = +1$, 19.2 percent by $t = +2$, 33.8 percent by $t = +3$, and 44.1 percent by $t = +5$. The 44.1% value is conditional on women observed at $t = +5$ (N = 454), whereas 38.5% is the full analytic-panel share (255/662); these are different conditioning sets. By $t = +3$, one-third of the observed sample has experienced a second birth within the panel window, which complicates the interpretation of medium-run coefficients after first birth.

I address this in two ways. The specification that excludes post-second-birth observations yields a $t = +5$ employment coefficient of -0.003 (N = 4,495), compared with -0.028 in the full sample (N = 5,212). This is a descriptive sensitivity, since censoring after second birth also changes sample composition at later event times. Even so, the attenuation suggests that part of the residual employment gap at $t = +5$ reflects subsequent fertility.

A stratified event study separates women by whether they eventually have a second birth within the panel window. Women without a second birth have a $t = 0$ employment coefficient of -0.124, return to baseline by $t = +3$ ($+0.002$), and modestly exceed it by $t = +5$ (+0.013). Women with a second birth show a deeper initial decline at $t = 0$ (−0.151), partial recovery by $t = +3$ (−0.072), and a persistent deficit at $t = +5$ (−0.068), with no full recovery within five years.

The gap between the two groups is present even before birth ($t = -2$: +0.042 for no second birth versus −0.089 for second birth), which indicates that women who go on to have a second child are already on a different employment trajectory at baseline. This subgroup divergence at $t = -2$ indicates that the pooled near-lead null combines offsetting trajectories across subgroups. The stratification also shows that the medium-run penalty in the pooled sample combines near-full recovery for one-child mothers with more persistent depression among those who have additional children.

8.5 Small-sample inference

The supplementary pre-birth mechanism regressions in Section 5 are estimated on 13 exit events, so they should not be treated as a central result. Even so, the coefficients are directionally stable across standard logit, Firth correction, bootstrap, and jackknife diagnostics. I report three layers of inference to assess sensitivity to this constraint.

Standard maximum likelihood yields odds ratios of 6.41 (non-regular, $p = 0.019$) and 4.14 (firms with fewer than 100 employees, $p = 0.025$). Firth's penalized likelihood estimator (Firth, 1993), which reduces finite-sample bias in rare-event logit settings, produces attenuated but still statistically significant estimates: OR = 5.32 ($p = 0.009$) and OR = 3.84 ($p = 0.021$). In addition, bootstrap inference from 2,000 nonparametric resamples yields 95 percent confidence intervals that exclude 1 for both predictors: $[1.68, 19.66]$ ($p = 0.008$) for non-regular and $[1.10, 18.34]$ ($p = 0.038$) for the sub-100-employer indicator.

The consistency across methods - MLE, penalized ML, and bootstrap - suggests that the sign of the association is not being driven solely by small-sample bias. At the same time, the confidence intervals are wide, so the exact magnitudes remain uncertain. The most defensible conclusion is directional: non-regular workers and workers in firms with fewer than 100 employees are more likely to exit on the pre-birth margin in this sample.

I further test single-observation sensitivity using leave-one-out jackknife and influence diagnostics for the two-predictor mechanism model. The non-regular odds ratio remains above one in all 185 jackknife replications (OR range: 5.69–12.82; p range: 0.016–0.030). The odds ratio on the sub-100-employer indicator also remains above one in all replications (OR range: 3.54–5.52), with significance retained in 98.9 percent of runs (p range: 0.014–0.052). Influence statistics (Cook's distance, leverage, and DFBETAs) do not indicate that a single observation overturns the core pattern. Appendix D reports these diagnostics.

Taken together, these diagnostics confirm that no single observation is driving the pre-birth mechanism result: the estimated association remains positive and of similar order of magnitude for both predictors.

See Appendix A for the childless benchmark details and placebo event study, and Appendix B for second-birth spacing and stratified event-study tables.

8.6 Summary of findings

The chapter documents four principal findings.

Aggregate break and pre-birth composition. The aggregate break in employment occurs at childbirth ($t = 0$), while the pre-birth exit margin has a distinct composition: 41.4 percent of pre-birth leavers cite marriage, and at $t = 0$ 85.0 percent cite childcare. In this sample, the near lead at $t = -2$ is flat in net terms, so the pre-birth result is best interpreted as the composition of gross exits. It does not appear as a second aggregate break.

Dual labour market predicts exit. The dual labour market is the strongest observed predictor of exit. At the childbirth margin, non-regular employment status strongly predicts non-employment (OR = 7.48, $p < 0.001$), while the sub-100-employer indicator is positive but imprecisely estimated in the same model. In the lower-power pre-birth model, both non-regular status (OR = 6.41, $p = 0.019$) and the indicator for firms with fewer than 100 employees (OR = 4.14, $p = 0.025$) are positively associated with exit. The household-level variables available in these data, such as the husband's hours and the wife's commute time, add little explanatory power once the woman's labour-market position is taken into account.

The penalty transforms over time. Although aggregate employment recovers by $t = +5$, the composition shifts persistently: among women who were full-time at $t = -1$, only 42.2 percent remain full-time at $t = +5$, a figure that includes the effect of subsequent births. Conditional hours fall by about 10 hours and conditional earnings by about 96 man-yen, with no clear sign of recovery within five years. These post-birth patterns are consistent with re-entry through lower-quality job tracks, although this design does not separately identify individual re-entry paths for all exiters.

No detectable paternal adjustment. I find no detectable systematic paternal adjustment in this sample window. Childcare-hour ratios stabilize near 5-to-1 by $t = +5$, and in the broader leave-analysis sample only 3 husbands (0.3%, N=1,183) report taking childcare leave.

9 Discussion and conclusion

9.1 Policy implications

The documented risk-stratification pattern points to several policy implications. These are implications consistent with the descriptive evidence, not estimates of the causal effects of any specific reform.

Strengthen non-regular protections. The first implication is to strengthen effective employment protections for non-regular workers. At the childbirth margin, the non-regular odds ratio is 7.48, the largest coefficient in the main risk-stratification results. Only 21.1 percent of mothers ever report taking childcare leave; for many women, the leave system is not operational because the employment relationship does not survive through childbirth. Japan's amendments to the Part-Time Workers Act and the Labour Contract Act are directionally aligned with the documented gradient, especially where they protect continuity and limit contract non-renewal around childbirth (Ministry of Health, Labour and Welfare, 2019; OECD, 2024a; Yamaguchi, 2019).

Target small-employer constraints. The firm-size gradient in exit rates is consistent with the practical difficulty of accommodating parental leave in organizations with limited staffing. In the childbirth-margin model, the conditional coefficient on the sub-100-employer indicator is positive but imprecise (OR = 1.55, $p = 0.126$), so this implication is grounded more in descriptive patterns and institutional constraints than in a precise conditional estimate. My workplace-systems data show that while 60.3 percent of mothers' employers nominally offer reduced-hours arrangements, only 10.1 percent offer work-from-home. Reducing implementation frictions for small employers - whether through administrative simplification, financial support, or dedicated guidance - is directionally indicated by the firm-size gradient in exit rates.

Reduce paternal overwork. The near-zero father adjustment observed in this sample is consistent with a work culture that makes paternal caregiving structurally difficult. Greater scope for fathers to reduce hours or take leave may be an important complement to any policy aimed at improving maternal employment continuity.

Improve post-birth mobility. High non-regular exit risk at childbirth and limited post-birth mobility back toward stronger job matches point to a bottleneck at re-entry. The descriptive evidence on training grants is consistent with this bottleneck: awareness is low, eligibility is often lost after exit, and many mothers are uncertain whether they qualify. Policies that strengthen conversion pathways for qualified non-regular workers, improve information about eligibility, and tie re-entry support more closely to actual job placement may be more effective than formal training eligibility alone.

Joint implications: segmentation and fiscal thresholds. In interpreting this re-entry pattern, I do not treat institutional segmentation and fiscal thresholds as competing explanations. They are likely complementary. Threshold bunching is a broad secondary-earner phenomenon - childless married women bunch at least as much as returning mothers - so the thresholds constrain married women's earnings independently of parenthood. What motherhood appears to add is a transition into the part-time earnings range where these pre-existing constraints are more likely to bind. Reduced-hours re-entry can reflect both demand-side constraints, such as limited access to regular-track jobs, and supply-side responses to tax and social-insurance thresholds. The policy implication is correspondingly joint: improve transition opportunities into regular employment.

9.2 External validity and broader implications

The core stratification pattern - that labour-market segmentation is associated with a larger motherhood penalty among workers with weaker contractual protections - is not unique to Japan. Dual labour markets exist throughout Southern Europe, South Korea, and parts of Latin America, wherever fixed-term contracts and firm-size-dependent regulations create tiers of employment protection (see Adda et al., 2017; Kleven et al., 2019a, 2024, for cross-country evidence on career costs). The specific institutional details differ - Japan's spousal tax thresholds and regular/non-regular divide, for example, have no direct analogue in Spain's temporary-contract system - but the broader segmentation gradient is potentially generalizable: when a large share of women hold jobs that cannot accommodate career interruption, leave expansions and childcare subsidies may underperform their potential.

These parallels are especially visible in other dual labour markets. In Spain, where approximately one quarter of the workforce holds fixed-term contracts, mothers become increasingly likely to hold temporary contracts after childbirth, and the child penalty widens from 11 percent in the first year to 28 percent after a decade (de Quinto et al., 2021). In Italy, matched employer-employee data reveal a 57 log-point earnings penalty driven partly by mothers sorting into lower-paying firms and shifting to part-time contracts - a pattern that is directionally consistent with the post-birth downgrading documented here (Casarico and Lattanzio, 2023). In Korea, the M-shaped female participation curve, the OECD's largest gender wage gap, and a non-regular share exceeding Japan's produce a similar configuration in which broad policy expansion coexists with persistent motherhood-related career discontinuity (OECD, 2024b; Schauer, 2018). The common thread across these settings is the de jure/de facto gap documented earlier in the chapter:

statutory protections designed for regular-track careers do not fully translate into effective protection for workers in precarious tiers, so aggregate policy coverage can overstate effective access for the women most at risk of exit.

The findings also speak to Goldin (2014)'s argument that persistent gender gaps depend on the structure of jobs. Goldin emphasizes temporal flexibility within occupations. My results point to a different structural margin: the regular/non-regular divide and the segmentation of employment continuity across contract tiers. In this chapter, the penalty is structured through contract tier and employment continuity. In these data, the observed constraints align more closely with institutional job features than with preference-based accounts. Recent within-firm evidence from Japan is also consistent with this interpretation: Okuyama et al. (2025) show that the initial motherhood penalty within firms is driven by overtime and bonus reductions, but over time it transforms into a promotion penalty through the job-rank hierarchy. That pattern - an immediate compensation shock transforming into a more structural career-progression penalty over time - is directionally consistent with my finding that the extensive-margin break at birth gives way to a more persistent intensive-margin penalty, though the within-firm promotion channel is not separately identifiable in the JHPS/KHPS design.

9.3 Limitations

Several limitations should be noted.

Pre-birth mechanism precision. The supplementary pre-birth mechanism regressions are estimated on 13 exit events in the complete-case sample, which limits precision and the ability to estimate richer interaction structures. Replication with a larger sample would improve precision on that margin.

Unobserved heterogeneity. I cannot rule out all forms of unobserved heterogeneity. While the selection tests address several leading alternative explanations, it remains possible that non-regular workers differ from regular workers in unobserved dimensions that independently predict exit. A design with exogenous variation in contract type would strengthen causal identification.

Marriage-related pre-birth exits. The pre-birth "marriage-stage" exits should be interpreted as marriage-related reporting patterns. In Japan, marriage and expected near-term childbirth are tightly linked, so pre-birth exits may bundle marriage transitions with anticipatory family planning. The chapter uses this stage as descriptive sequencing evidence, without isolating it as a separate causal channel.

Observation window and macro regime. My observation window of five years after first birth may not capture the full long-run trajectory, and the pooled period (2004–2022) spans major policy and macroeconomic regime changes, including childcare expansions and the COVID period. The earnings penalty shows no sign of recovery by $t = +5$, but recovery may occur at longer horizons as children enter school (cf. Angelov et al., 2016; Adda et al., 2017, who track penalties over 15–20 years). The reported estimates should be read as period-averaged patterns unless otherwise stratified (see Appendix D.10 for cohort-split and no-COVID sensitivity).

Childcare utilisation. Direct childcare-utilisation measurement is extremely limited in these panels. Although JHPS/KHPS include childcare and early-education items, many are not available consistently across waves, which prevents construction of a stable woman-year childcare panel aligned to event time. As a result, I cannot estimate dynamic effects of specific care arrangements (e.g., public/private nursery at ages 0–2 versus kindergarten pathways at ages 3–5) within the same longitudinal framework, nor can I measure whether individual women were denied childcare placement. The analysis infers childcare pressure indirectly from exit reasons, household structure, and labour-market position. Since approximately 90 percent of mothers in the sample live in nuclear households without co-resident grandparents, childcare access may be a more binding constraint than my analysis suggests.

Cross-country evidence suggests that family care structures can moderate child penalties. In China, parenthood is associated with negative labour-market effects for mothers, but those penalties are smaller where grandparental childcare is available (Meng et al., 2023; Deng et al., 2023). By contrast, evidence for Korea points to a large and persistent motherhood penalty and weaker post-birth career continuity, despite broad family-policy expansion (OECD, 2024b, 2025a). The finding that labour-market position is more visible in these data than the household characteristics measured here does not rule out the possibility that childcare access is a necessary complementary condition.

Comparison with administrative data. Recent administrative evidence from Fukai and Kondo (2025) uses municipal tax records to document that mothers' salary income falls by up to 80 percent immediately after first birth and remains at roughly 50 percent of the pre-birth level four years later. In a related study, Fukai and Kondo (2024) link childcare-admission records to parental tax data within specific municipalities and find that access to accredited childcare centres raises maternal employment by 18–40 percentage points for mothers of children under two. These studies exploit data that the JHPS/KHPS cannot match: cleaner earnings trajectories and childcare-slot assignment. The present chapter is complementary: the household panel provides contract type, firm size, and workplace-level mechanism variables that administrative tax records do not contain, at the cost of smaller samples and survey-based measurement.

This caution also applies across source panels. In the JHPS/KHPS split check, the near lead ($t = -2$) and at-birth drop ($t = 0$) are highly stable across datasets, but the long horizon ($t = +5$) differs in sign and significance. I treat the birth-year break as the most robust cross-panel result and interpret long-run recovery as panel-sensitive.

Sample scope. The analytic sample is restricted to married women around first birth. This scope matches the dominant setting of childbirth in Japan, but it narrows external validity. The conclusions apply to married first-birth trajectories in this panel context and should not be mechanically generalized to unmarried mothers.

Motherhood selection. Selection into motherhood within the panel may itself be stratified by labour-market position. In the childless benchmark, non-regular employment is more common than in the mothers' analytic sample. This pattern is consistent with regular-track attachment operating as a precondition for observed family formation in-panel, which could compress observed penalties relative to a broader at-risk population.

Japanese-language scholarship. A substantial body of research on Japanese employment and family dynamics exists in Japanese-language sources that I do not survey here. The English-language literature cited in this chapter most likely does not fully represent the state of knowledge in Japanese labour economics.

What would falsify the interpretation? Three results would substantially weaken the chapter's core claims. A matched childless placebo showing a sharp pseudo-birth break comparable to the maternal sample would weaken the event-time attribution to childbirth, but the placebo coefficient at $t = 0$ is $+0.016$ ($p = 0.608$), and the placebo post-period coefficients are small and not consistently significant, with no pattern resembling the maternal birth-year drop. A non-regular contract-tier gradient that disappeared when missingness was handled differently would weaken the predictive stratification claim, but the odds ratio is stable across complete-case (7.50), missing-indicator (7.48), and high-information (7.39) specifications. An at-birth employment break that varied substantially across early and late cohorts or across survey sources would put the generality of the result in doubt, but the $t = 0$ coefficient is virtually identical across early (-0.131), late (-0.135), no-COVID (-0.145), KHPS-only, and JHPS-only samples. None of these diagnostics overturns the chapter's core interpretation.

9.4 Concluding remarks

The chapter's findings reframe Japan's motherhood employment penalty as an institutionally patterned outcome. The central issue is whether the job itself can survive the interruption of first birth; childcare availability alone does not settle that question. Taken together, the descriptive patterns in this chapter are consistent with a segmented labour market in which continuity protections differ substantially by contract tier and employer type.

Addressing Japan's motherhood employment penalty will require more than expanding childcare or extending parental leave - policies that primarily benefit workers who already hold secure positions. It will also require addressing the dual labour market itself: the divide between those whose jobs can accommodate parenthood and those whose jobs cannot.

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Appendix A. Childless benchmark and placebo test

Because the harmonized analysis panels are constructed from mothers only, a childless comparison group is not available within the main sample. I build a descriptive benchmark from the raw panels (married women with no observed births and at least five years of observation). This group has $N_{\text{ids}} = 4{,}111$ and $N_{\text{obs}} = 50{,}222$, with mean employment 53.2 percent, mean weekly hours 28.5, 71.9 percent non-regular employment, and 59.9 percent employment in firms with fewer than 100 employees. Yearly employment rates are stable in the 0.51-0.57 range from 2004-2022. Tables A.C1-A.C3 report the childless summary statistics and annual employment rates.

Table 13: Descriptive benchmark sample of married childless women used for context only (not a causal control group). Employment rate is the share employed, and mean hours are average weekly hours across all observations.

N (obs)	N (women)	Employment rate	Mean hours/week
50,222	4,111	0.532	28.539

Table 14: Descriptive benchmark for employed observations in the childless married sample. Statistics are conditional on employment and are provided for context rather than identification.

N (obs)	N (women)	Mean hours/week	Median hours/week
26,704	3,088	28.539	26.000

Table 15: Annual employment rate in the descriptive childless benchmark sample (married women with no observed births). Rates are proportions, not percentages, and are shown to document the absence of a birth-related break in this benchmark group.

Year	N	Employment rate
2004	1,642	0.516
2005	1,642	0.543
2006	1,641	0.548
2007	2,313	0.548
2008	2,312	0.567
2009	3,625	0.533
2010	3,535	0.531
2011	3,467	0.527
2012	3,724	0.531
2013	3,611	0.518
2014	3,357	0.525
2015	3,153	0.531
2016	2,953	0.525
2017	2,684	0.534
2018	2,456	0.537
2019	2,269	0.539
2020	2,104	0.532
2021	1,916	0.523
2022	1,818	0.509

I also implement a placebo event study by assigning each childless woman a pseudo-birth year matched to mothers on dataset, age bin, and cohort bin, then estimating the same event-study specification with year fixed effects. Pseudo-event years are drawn from the mothers' first-birth-year donor pools with replacement and constrained to years in which each childless woman is actually observed, so the placebo does not create artificial missingness by construction. Match quality is strong (91.7% exact matches, 5.9% dataset-only matches, and 2.4% dataset-age matches). The placebo profile does not show a pseudo-birth discontinuity: at $t = 0$ the coefficient is $+0.016$ (SE 0.031, 95% CI $[-0.045, 0.076]$, $p = 0.608$), and the near lead at $t = -2$ is -0.003 ($p = 0.937$). Coefficients are small and not consistently significant across the window, though some post-period estimates approach conventional thresholds. I interpret this placebo as a composition diagnostic that complements, rather than replaces, the chapter's broader falsification logic: pseudo-assignment is still non-exogenous to panel support, so small non-zero coefficients can arise from lifecycle and selection drift. Table A.C4 reports the placebo event-study coefficients.

Table 16: Matched childless pseudo-event placebo: employment event-study coefficients (year fixed effects).

Event time	Estimate	SE	p-value
$t = -5$	-0.037	0.058	0.524
$t = -4$	-0.039	0.044	0.373
$t = -3$	-0.031	0.039	0.427
$t = -2$	-0.003	0.035	0.937
$t = +0$	0.016	0.031	0.608
$t = +1$	0.038	0.031	0.221
$t = +2$	0.062	0.032	0.052
$t = +3$	0.060	0.032	0.063
$t = +4$	0.056	0.034	0.099
$t = +5$	0.049	0.036	0.173

Notes: Joint Wald test for placebo far leads ($t = -5, -4, -3$): $p = 0.749$.

Appendix B. Second-birth dynamics

This appendix reports second-birth spacing, the share of women with a second birth by event time, and stratified event-study coefficients for mothers who do and do not have a second birth within the panel window. These diagnostics help interpret the muted long-run rebound in the pooled sample: in the main data, the share with a second birth rises to 19.2 percent by $t = +2$, 33.8 percent by $t = +3$, and 44.1 percent by $t = +5$ (conditional on women observed at $t = +5$). Censoring post-second-birth observations or excluding women with a second birth within three years makes the long-run coefficients less negative and close to zero by $t = +5$, while the stratified event study shows persistently more negative coefficients for the second-birth group (for example, $t = 0$: -0.151; $t = +5$: -0.068) and near-zero recovery for the no-second-birth group by $t = +3$ to $t = +5$. These patterns suggest that the muted rebound in the full sample partly reflects subsequent births, not a permanent non-recovery.

The purpose is interpretive. In this setting, subsequent fertility is an endogenous life-course process that overlaps with post-first-birth recovery time. The stratified coefficients then describe how trajectories differ across fertility paths, not how a second birth causally shifts outcomes holding all else fixed.

The key implication for the main text is scope: pooled long-horizon coefficients mix first-birth recovery with additional-child transitions. The second-birth tables and figure make that mixture explicit and show that subsequent births contribute to the medium-run persistence in the pooled sample, alongside incomplete recovery from first birth.

Table 17: Observed spacing between first and second birth in the analytic sample. Counts are the number of mothers with a recorded second birth at each spacing in years.

Years to second birth	Count
1	33
2	83
3	72
4	35
5	16
6	10
7	3
8	2
13	1

Table 18: Share of mothers with a recorded second birth by event time relative to first birth. Share is a proportion (0–1), with N denoting the number of mothers observed at each event time.

Event time	N	Share
0	662	0.000
1	654	0.050
2	594	0.192
3	509	0.338
4	487	0.407
5	454	0.441

Table 19: Employment event-study coefficients stratified by whether mothers have a second birth within the panel window. Coefficients are relative to $t = -1$ (year fixed effects); values are descriptive and reflect heterogeneous trajectories rather than causal second-birth effects.

Group	Event time	Estimate	Std. Error	CI Lower	CI Upper
No second birth	-5	-0.125	0.058	-0.239	-0.012
Second birth	-5	-0.423	0.046	-0.512	-0.333
No second birth	-4	-0.102	0.051	-0.203	-0.002
Second birth	-4	-0.361	0.047	-0.453	-0.269
No second birth	-3	-0.051	0.046	-0.142	0.040
Second birth	-3	-0.219	0.049	-0.316	-0.123
No second birth	-2	0.042	0.040	-0.037	0.121
Second birth	-2	-0.089	0.047	-0.182	0.003
No second birth	0	-0.124	0.035	-0.193	-0.054
Second birth	0	-0.151	0.044	-0.238	-0.065
No second birth	1	-0.115	0.037	-0.187	-0.043
Second birth	1	-0.149	0.045	-0.236	-0.061
No second birth	2	-0.070	0.038	-0.145	0.005
Second birth	2	-0.125	0.045	-0.214	-0.036
No second birth	3	0.002	0.040	-0.077	0.081
Second birth	3	-0.072	0.046	-0.163	0.019
No second birth	4	0.029	0.041	-0.050	0.109
Second birth	4	-0.072	0.048	-0.166	0.021
No second birth	5	0.013	0.042	-0.070	0.095
Second birth	5	-0.068	0.049	-0.164	0.029

Figure 17 is constructed from an at-risk subgroup of mothers: women who were full-time at $t = -1$ and are not full-time at $t = +1$. For each event year $t \in \{2, 3, 4, 5\}$, I compute first returns to full-time among women still at risk, then compute the cumulative first-return proportion. The resulting series is a descriptive cumulative first-return profile, not a Kaplan-Meier survival estimate, and the figure should be read as referring to the at-risk subgroup throughout.

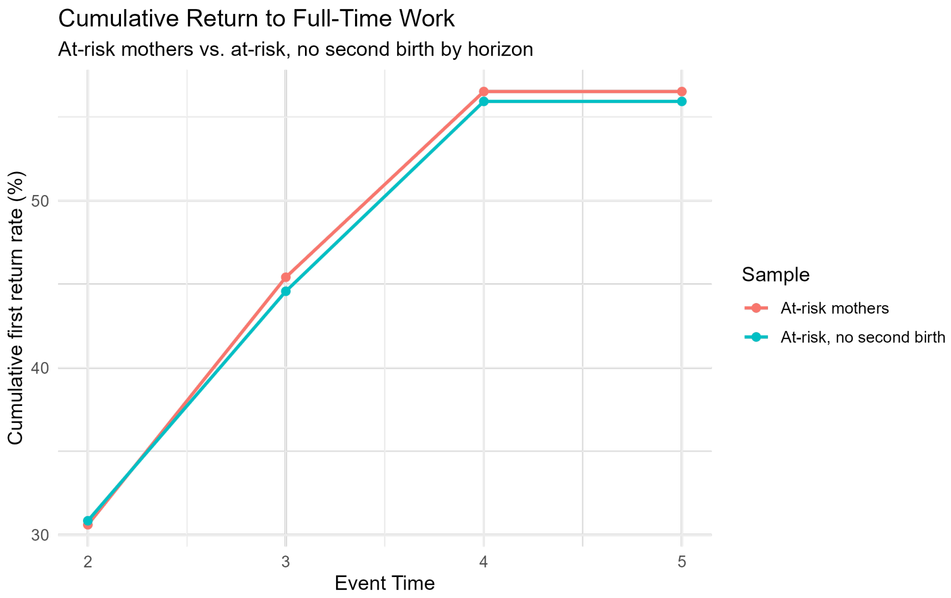

Figure 17: Appendix Figure B1: Cumulative return to full-time employment after childbirth. This curve is estimated on the at-risk subgroup only (mothers full-time at $t = -1$ and out of full-time work at $t = +1$). Points show the cumulative first-return proportion over $t = 2$ to $t = 5$, so the series is descriptive and should not be read as a Kaplan-Meier survival estimate.

Appendix C. Father placebo event study

This appendix reports the event-study coefficients for husbands' weekly hours around first birth. The estimates are small and statistically indistinguishable from zero across the event window, consistent with no systematic adjustment in fathers' labour supply.

This is a placebo-style benchmark, not a structural model of paternal labour supply. Its value is comparative: against large maternal employment and hours shifts, the husband's near-flat trajectory helps document asymmetric household adjustment around childbirth in this sample.

At the same time, a near-zero father-hours pattern does not identify an exclusion restriction for maternal outcomes. It should be read as descriptive supporting evidence for the gendered division-of-labour mechanism developed in the main text.

Table 20: Event-study coefficients: husband hours (year fixed effects).

Event time	Estimate	SE	CI Lower	CI Upper
-5	5.468	3.0004	-0.4128	11.3488
-4	-3.5484	2.0471	-7.5606	0.4639
-3	-0.978	1.5621	-4.0396	2.0837
-2	0.7091	1.1979	-1.6388	3.0570
-1	0	0	0	0
0	0.4497	0.9717	-1.4548	2.3543
1	0.2265	1.0344	-1.8009	2.2538
2	0.0181	1.0522	-2.0443	2.0804
3	0.7421	1.0791	-1.3730	2.8572
4	-0.1797	1.1171	-2.3691	2.0098
5	-0.4993	1.1313	-2.7167	1.7180

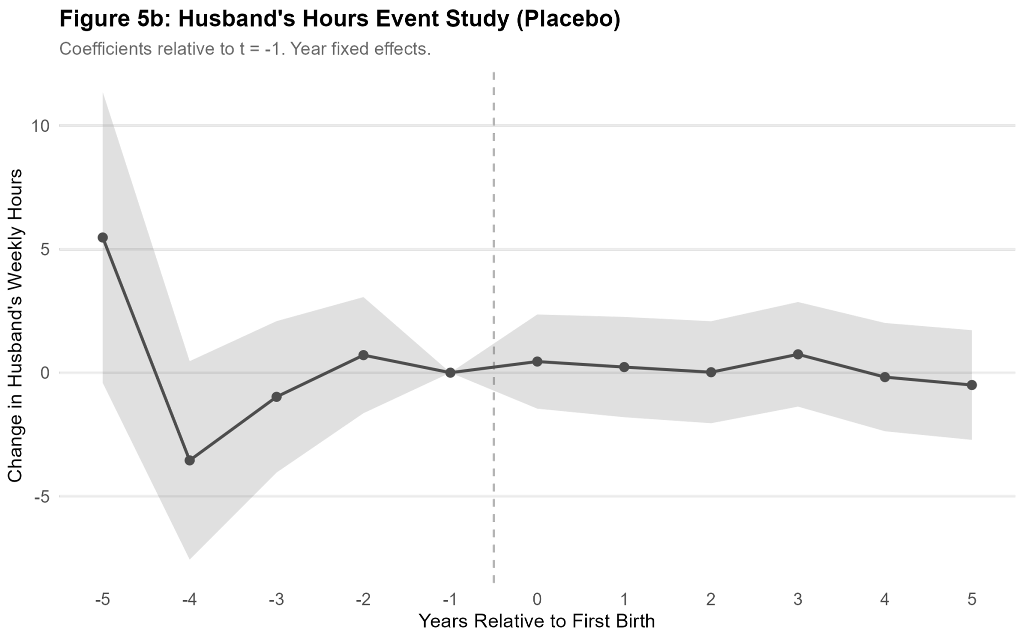

Figure 18: Appendix Figure C1: Husband weekly-hours event study (year fixed effects; robust standard errors). Coefficients are relative to $t = -1$. Estimates are close to zero in the near-birth window, while far leads are more variable because support is thinner; overall there is no stable post-birth pattern consistent with a father labour-supply penalty.

Appendix D. Identification, harmonisation, and additional sensitivity checks

This appendix consolidates diagnostics that support interpretation of the main estimates and clarifies which remaining checks are planned for the next revision cycle.

D.1 Event-time identification and individual-FE collinearity

The core design of this study uses event-time indicators around first birth with calendar-year fixed effects. In this setup, adding individual fixed effects creates an exact identification problem. The issue goes beyond a simple loss of precision. In a fully treated event-time design, event time advances one-for-one with calendar time within individuals once birth cohort is absorbed by the person fixed effect. As a result, event-time indicators and calendar-year fixed effects become collinear, and any individual-plus-year FE specification depends on arbitrary normalization choices. That setup does not yield a substantively interpretable decomposition. This is why the main text reports year-FE trajectories and uses individual-FE exercises, where shown, as transparency checks. They do not carry the chapter's primary inferential weight.

The interpretation is explicit: the reported coefficients are descriptive sample-average trajectories relative to $t = -1$, not causal within-person treatment effects. The key timing result remains that the near lead at $t = -2$ is close to zero, while the break at $t = 0$ is sharp and negative.

I also examined versions of the main event-study figures with individual and year fixed effects. In a short event window with event-time dummies, this specification is weakly identified and produces large standard errors and smooth ramp (triangular) patterns. I do not report those versions and instead rely on the year-FE specification and descriptive trajectories in the main text.

D.2 Pre-trend diagnostics in an unbalanced panel

Far leads ($t = -5, -4, -3$) are negative and jointly significant. In this application, that pattern is interpreted as composition in an unbalanced panel. It does not point to anticipatory response to childbirth. Women observed at far pre-birth horizons are a selected subgroup with different life-cycle employment profiles, and support thins markedly in those tails. The identification-relevant diagnostic is the near lead at $t = -2$, which remains small across specifications.

This interpretation is reinforced by robustness exercises that trim far leads, restrict to balanced pre-support, and add a dataset indicator. Across these alternatives, the near lead remains economically small and statistically weak, while post-birth coefficients preserve their qualitative pattern.

D.3 JHPS/KHPS pooling and harmonisation

The pooled panel is built from the official harmonisation program and variable correspondence files. Core constructs used in the chapter (employment, hours, earnings, contract type, and firm size) are mapped through that framework and checked against survey-specific coding before analysis. As a sensitivity check, the main employment specification includes a JHPS/KHPS dataset indicator; this does not materially change the near-lead or at-birth coefficients.

The practical implication is that pooling does not appear to drive the headline dynamics. Nonetheless, the estimates should be read as trajectories in the harmonised analytic sample of married first-birth women, not as nationally weighted population moments.

Table 21: Survey-split stability check for key employment event-study coefficients (year fixed effects).

Sample	Event time	Estimate	SE	p-value	N obs	N women
Pooled	$t = -2$	-0.012	0.031	0.703	5,212	662
Pooled	$t = +0$	-0.132	0.027	0.000	5,212	662
Pooled	$t = +5$	-0.028	0.032	0.384	5,212	662
KHPS	$t = -2$	0.002	0.039	0.966	3,323	394
KHPS	$t = +0$	-0.134	0.036	0.000	3,323	394
KHPS	$t = +5$	0.010	0.040	0.796	3,323	394
JHPS	$t = -2$	-0.027	0.049	0.576	1,889	268
JHPS	$t = +0$	-0.136	0.044	0.002	1,889	268
JHPS	$t = +5$	-0.126	0.057	0.027	1,889	268

Table 21 clarifies where pooling is and is not consequential. The two identification-relevant moments are stable across surveys: the near lead at $t = -2$ is close to zero and statistically insignificant in both KHPS and JHPS, and the at-birth coefficient at $t = 0$ is nearly identical in magnitude (about -13.4 percentage points in each panel). The long horizon is less stable: by $t = +5$, KHPS is near zero while JHPS remains negative. This divergence is consistent with panel-sensitive long-run composition, including different support, attrition profiles, and sample sizes at distant event times. The chapter treats the sharp at-birth decline as the most robust cross-panel finding and gives long-run recovery a more limited interpretation.

D.4 Education measurement and Newlyweds-module diagnostic

In this study I do not use education-attainment heterogeneity as a core split because JHPS/KHPS do not provide a single harmonised, respondent-level highest-attainment variable with stable coverage for the full mother event-study sample. Education-related items in the regular married-respondent instruments are not attainment stocks. In the household roster, schooling fields capture current enrolment status (in school vs not in school) and school type. In JHPS, roster schooling/work cells for "yourself" and "your spouse" are crossed out; in KHPS, roster schooling and work are combined in a status field. In both surveys, these roster items are unsuitable as highest-completed-education measures for wives. Likewise, respondent/spouse modules on current attendance, recent training actions, and time-use/education expenditures measure contemporaneous human-capital activity. They do not measure educational attainment.

The only module with an attainment-style schooling item is the Newlyweds spouse questionnaire, administered when a panel respondent newly reports marriage in-panel. This instrument is answered by the respondent's spouse and is not a standard annual respondent covariate. To provide transparency, I construct a bounded appendix diagnostic using newlyweds variable "School that R last attended", recoded into a college-plus indicator (junior college/specialized, university, graduate school).

Coverage is limited and selected in the analytic sample: of 662 mothers, 89 (13.4%) have matched non-missing newlyweds education. Within this matched subset, 60.7% are college-plus, 33.7% are non-college, and 5.6% are unclassified. Because inclusion depends on in-panel marriage timing and spouse-module availability, this diagnostic remains descriptive. The main design does not use it to identify heterogeneity.

Table 22: Coverage of newlyweds-module education in the analysis sample

Metric	Value
Analysis sample (N)	662
Matched newlyweds education (N)	89
Matched share (%)	13.4
College-plus share in matched (%)	60.7
Non-college share in matched (%)	33.7

Notes: Newlyweds education uses spouse-module item "School that R last attended". "College-plus" includes junior college/specialized, university, and graduate school categories.

D.5 Prefecture-level childcare context (external data)

To contextualise the metropolitan split in the main text, I compile prefecture-level childcare indicators from external official statistics (all 47 prefectures). The purpose is descriptive: to document the degree of geographic heterogeneity in childcare environments over time. These series are not linked to individual JHPS/KHPS respondents and are not used as identifying variation in the individual-level event-study design.

The indicators show substantial dispersion and structural transition. Day-nursery provision per 100,000 children aged 0-5 rises in median terms from 397.7 (2005) to 572.0 (2020), while cross-prefecture spread remains wide (2020: min 404.8, max 1042.8).

Nursery-utilisation medians are above full capacity in 2005 and 2014 (103.8 and 102.0), then lower in 2020 (92.2), consistent with easing average pressure alongside persistent regional variation. Public provision shares decline markedly over the same period (public day-nursery institution share median: 56.2% in 2005 to 28.1% in 2020; public day-nursery children share median: 48.8% to 26.7%). For authorised child centres, available in the later period, the 2022 median is 212.2 per 100,000 children (min 26.5, max 667.4). Table 23 reports the summary statistics. The prefecture-level series are compiled from official e-Stat tables; the harmonised extraction code and compiled panel used for these appendix summaries are documented in Gietner (2024).

Fertility, childcare demand, and childcare supply can co-evolve at the prefecture level, so these series should not be read as exogenous supply shifters.

Table 23: Prefecture-level childcare environment dispersion (all 47 prefectures). Facility indicators are reported per 100,000 children aged 0-5. Share indicators refer to public-provider institution shares or enrolled-children shares. Source: official e-Stat tables; harmonised extraction documented in Gietner (2024).

Indicator	Year	Min	P25	Median	P75	Max
Day nurseries per 100k (0-5)	2005	172.9	284.9	397.7	535.5	760.0
Day nurseries per 100k (0-5)	2014	258.9	323.6	425.5	561.9	775.6
Day nurseries per 100k (0-5)	2020	404.8	501.6	572.0	740.6	1042.8
Nursery utilisation rate (%)	2005	90.3	98.8	103.8	106.7	112.1
Nursery utilisation rate (%)	2014	88.1	96.5	102.0	106.0	115.1
Nursery utilisation rate (%)	2020	74.5	88.7	92.2	95.0	100.2
Public day-nursery share (%)	2005	18.9	41.2	56.2	66.0	82.1
Public day-nursery share (%)	2014	5.6	26.2	40.9	55.0	78.6
Public day-nursery share (%)	2020	0.4	19.6	28.1	44.8	63.6
Public day-nursery children share (%)	2005	17.0	37.0	48.8	59.5	79.0
Public day-nursery children share (%)	2014	4.9	23.7	38.2	46.3	75.4
Public day-nursery children share (%)	2020	0.6	15.0	26.7	35.7	57.9
Authorized child centers per 100k (0-5)	2022	26.5	158.7	212.2	298.0	667.4

D.6 Attrition and support composition

I report attrition/support diagnostics to make composition shifts explicit. The event-time support is strongly unbalanced: 186 women at $t = -5$, 662 at $t = 0$, and 454 at $t = +5$. I also compare baseline characteristics at $t = -1$ by whether women are observed at $t = +5$. Baseline employment differs (47.7% for those observed at $t = +5$ versus 56.2% for those not observed; $p = 0.045$), while baseline hours and income do not differ significantly ($p = 0.509$ and $p = 0.311$). These diagnostics do not convert the design into a causal panel estimator, but they quantify how much long-horizon interpretation depends on changing support.

Table 24: Attrition/support diagnostics: event-time support and baseline balance by observation at $t = +5$.

Group	N	Emp. rate	Mean hours	Mean income
Observed at $t = +5$	453	0.477	33.64	252.30
Not observed at $t = +5$	203	0.562	34.92	276.57
p-value (diff.)		0.045	0.509	0.311

Table 25: IPW diagnostics for observability at $t = +5$.

Diagnostic	Value	Note
Weight mean	1.521	Baseline sample
Weight p95 / p99	2.117 / 2.212	Tail concentration
Weight max	2.292	Extreme leverage check
Trim shares (low/high)	0.0% / 0.0%	Trim at [0.05, 0.95]
Obs-vs-full employment gap (unweighted)	-0.026	Observed minus full baseline
Obs-vs-full employment gap (IPW)	-0.006	Weighted observed minus full baseline

Table 26: IPW sensitivity for observability at $t = +5$: key employment event-study coefficients.

Specification	Event time	Estimate	SE	p-value
Unweighted	$t = -2$	-0.012	0.031	0.703
Unweighted	$t = +0$	-0.132	0.027	<0.001
Unweighted	$t = +5$	-0.028	0.032	0.384
IPW (obs at $t = +5$)	$t = -2$	-0.024	0.031	0.436
IPW (obs at $t = +5$)	$t = +0$	-0.136	0.028	<0.001
IPW (obs at $t = +5$)	$t = +5$	-0.057	0.032	0.076

The IPW model is a logit for $1\{\text{observed at } t = +5\}$ estimated on baseline covariates (employment status, weekly hours, annual income, survey indicator, and age/education fields when available in the analytic file). Balance diagnostics are reported in levels: employment is in proportions (percentage-point interpretation), hours in weekly hours, and income in annual man-yen units. In this run, estimated propensity scores already lie within the trimming bounds $[0.05, 0.95]$ (no trimming activated). IPW is constructed for observability at $t = +5$ because attrition pressure is strongest at the long horizon. It serves as a sensitivity check for late-horizon composition and does not provide a full correction for all event times. Under IPW, the long-horizon coefficient shifts from -0.028 ($p = 0.384$) to -0.057 ($p = 0.076$, 95% CI $[-0.120, 0.006]$), while the near lead and birth-year break remain unchanged in sign and significance.

As an additional support check, I re-estimate the employment event study on a balanced short window, restricting to women observed continuously from $t = -2$ through

$t = +1$. This removes short-horizon support imbalance by construction. The balanced-panel estimates are very close to the full-sample estimates in this window: the near lead remains near zero, and the childbirth-year break remains large and negative. This strengthens the interpretation that short-window panel imbalance does not drive the core $t = 0$ result.

Table 27: Balanced-panel robustness: employment event-study coefficients on the short event-time window $t = -2$ to $t = +1$. The "Full sample" column uses the full analytic sample restricted to this window; the "Balanced panel" column further restricts to women observed continuously from $t = -2$ to $t = +1$.

	Full sample			Balanced panel		
Event time	Coef.	SE	p	Coef.	SE	p
$t = -2$	-0.008	0.031	0.781	-0.058	0.031	0.065
$t = +0$	-0.131	0.028	<0.001	-0.128	0.033	<0.001
$t = +1$	-0.129	0.029	<0.001	-0.121	0.031	<0.001
N obs		2425			1780	
N women		662			445	

Notes: The balanced panel restricts to the 445 women observed at every event time from $t = -2$ to $t = +1$. Both columns are estimated on the short-window specification with year fixed effects and heteroskedasticity-robust standard errors. Reference period: $t = -1$.

I also run cleaner-sample sensitivity checks aimed at the far-lead composition concern. In an age-restricted sample (age ≥ 23), the childbirth break remains large ($t = 0 = -0.123$, $p < 0.001$) and the near lead remains close to zero ($t = -2 = -0.007$, $p = 0.836$), closely matching baseline estimates. The balanced short-window sample likewise preserves the core break ($t = 0 = -0.128$, $p < 0.001$). Stricter pre-window attachment restrictions generate larger coefficients, indicating subgroup selection toward higher-attachment women and away from a neutral support correction. The always- married pre-window specification's positive $t = -2$ coefficient also reads as composition in a selected subgroup. It does not behave like a conventional pre-trend. Table 28 reports these estimates.

Table 28: Cleaner-sample sensitivity: key employment event-study coefficients.

Specification	Event time	Estimate	SE	p-value	N obs	N women
Baseline (full analytic sample)	$t = -2$	-0.012	0.027	0.662	5212	662
Baseline (full analytic sample)	$t = 0$	-0.132	0.025	0.000	5212	662
Baseline (full analytic sample)	$t = +1$	-0.124	0.026	0.000	5212	662
Baseline (full analytic sample)	$t = +5$	-0.028	0.036	0.443	5212	662
Age-restricted (age $\geq$ 23)	$t = -2$	-0.007	0.032	0.836	3505	651
Age-restricted (age $\geq$ 23)	$t = 0$	-0.123	0.031	0.000	3505	651
Age-restricted (age $\geq$ 23)	$t = +1$	-0.115	0.041	0.005	3505	651
Age-restricted (age $\geq$ 23)	$t = +5$	-0.034	0.048	0.473	3505	651
Balanced short window ($t = -2$ to $t = +1$)	$t = -2$	-0.058	0.031	0.065	1780	445
Balanced short window ($t = -2$ to $t = +1$)	$t = 0$	-0.128	0.033	0.000	1780	445
Balanced short window ($t = -2$ to $t = +1$)	$t = +1$	-0.121	0.031	0.000	1780	445
Balanced short window ($t = -2$ to $t = +1$)	$t = +5$	–	–	–	1780	445
Pre-window attached (obs. $-3, -2, -1$; employed ≥ 2)	$t = -2$	0.063	0.027	0.018	1398	143
Pre-window attached (obs. $-3, -2, -1$; employed ≥ 2)	$t = 0$	-0.258	0.042	0.000	1398	143
Pre-window attached (obs. $-3, -2, -1$; employed ≥ 2)	$t = +1$	-0.278	0.052	0.000	1398	143
Pre-window attached (obs. $-3, -2, -1$; employed ≥ 2)	$t = +5$	-0.148	0.076	0.049	1398	143
Always-married pre-window	$t = -2$	0.137	0.026	0.000	3119	436
Always-married pre-window	$t = 0$	-0.186	0.029	0.000	3119	436
Always-married pre-window	$t = +1$	-0.187	0.029	0.000	3119	436
Always-married pre-window	$t = +5$	-0.101	0.040	0.012	3119	436

Notes: All specifications use year fixed effects and heteroskedasticity-robust standard errors. Reference period is $t = -1$. "–" indicates that the balanced short-window design does not identify $t = +5$.

D.7 Intensive-margin selection and scope

Hours, earnings, and implied hourly wage are estimated conditional on employment. These intensive-margin coefficients combine within-person adjustments with selection into who remains employed at each event time. In this study we should interpret them as evidence of post-birth labour market downgrading among employed mothers, while avoiding stronger claims that would require selection-corrected structural modelling.

I report two additional diagnostics. First, I estimate unconditional hours and earnings event studies by coding non-employment as zero outcomes. The post-birth declines remain large and precisely estimated under both heteroskedasticity-robust and woman-clustered

SEs, while the near lead remains close to zero. Second, I report Lee-style trimming sensitivity for the conditional intensive-margin outcomes. For post-birth years, the bounds remain negative for earnings from $t = +1$ onward and for hours at later horizons, though uncertainty is wider near birth. At $t = 0$ for earnings and at $t = 0/t = +1$ for hours, the Lee bounds include zero, so those early-horizon conditional effects should be treated more cautiously. These bounds are most informative in post-birth windows; far pre-birth windows require large trimming shares and should be interpreted cautiously.

Table 29: Unconditional intensive-margin event studies (hours and earnings) relative to $t = -1$.

Outcome	Event time	Estimate	SE	p-value	SE type
Annual earnings (unconditional)	$t = -2$	-6.741	7.533	0.371	Cluster (woman)
Annual earnings (unconditional)	$t = -2$	-6.741	11.481	0.557	Hetero
Annual earnings (unconditional)	$t = +0$	-55.909	6.841	<0.001	Cluster (woman)
Annual earnings (unconditional)	$t = +0$	-55.909	9.971	<0.001	Hetero
Annual earnings (unconditional)	$t = +1$	-86.845	8.076	<0.001	Cluster (woman)
Annual earnings (unconditional)	$t = +1$	-86.845	9.197	<0.001	Hetero
Annual earnings (unconditional)	$t = +5$	-52.237	11.140	<0.001	Cluster (woman)
Annual earnings (unconditional)	$t = +5$	-52.237	11.305	<0.001	Hetero
Weekly hours (unconditional)	$t = -2$	-0.575	0.940	0.541	Cluster (woman)
Weekly hours (unconditional)	$t = -2$	-0.575	1.262	0.648	Hetero
Weekly hours (unconditional)	$t = +0$	-3.930	0.798	<0.001	Cluster (woman)
Weekly hours (unconditional)	$t = +0$	-3.930	1.117	<0.001	Hetero
Weekly hours (unconditional)	$t = +1$	-6.054	0.897	<0.001	Cluster (woman)
Weekly hours (unconditional)	$t = +1$	-6.054	1.112	<0.001	Hetero
Weekly hours (unconditional)	$t = +5$	-6.034	1.120	<0.001	Cluster (woman)
Weekly hours (unconditional)	$t = +5$	-6.034	1.159	<0.001	Hetero

Notes: Outcomes are defined on the full sample rather than conditional on employment: hours are set to 0 when non-employed; earnings are set to 0 when non-employed. For employed observations with missing reported hours or earnings, outcomes remain missing. Models include calendar-year fixed effects.

Table 30: Lee-style trimming sensitivity for intensive-margin outcomes among employed women, relative to $t = -1$.

Outcome	Event	Obs. diff.	Lower	Upper	Trim	N
Annual earnings	$t = +0$	-46.86	-112.56	34.33	0.234	74
Annual earnings	$t = +1$	-124.67	-178.07	-54.86	0.195	61
Annual earnings	$t = +2$	-54.10	-84.24	-8.21	0.113	35
Annual earnings	$t = +3$	-67.81	-67.81	-67.81	0.000	0
Annual earnings	$t = +4$	-73.10	-96.60	-65.89	0.039	9
Annual earnings	$t = +5$	-74.10	-100.27	-63.19	0.056	13
Weekly hours	$t = +0$	1.47	-5.79	7.19	0.234	77
Weekly hours	$t = +1$	-3.79	-9.95	1.11	0.195	64
Weekly hours	$t = +2$	-2.81	-6.34	0.34	0.113	37
Weekly hours	$t = +3$	-5.04	-5.04	-5.04	0.000	0
Weekly hours	$t = +4$	-6.56	-7.68	-5.55	0.039	10
Weekly hours	$t = +5$	-8.81	-10.29	-7.54	0.056	13

Notes: Lee-style trimming equalizes employment shares between each event-time cell and $t = -1$ by trimming the higher-employment cell. This is a selection-robustness diagnostic for conditional outcomes, not a causal treatment-control Lee bound.

D.8 Mechanism diagnostics: pre-birth margin ($t = -2$)

The table below reports the nested pre-birth mechanism regressions (M1–M5). Because this margin contains only 13 exit events, the estimates should be read as low-powered but coefficient- stable evidence on early risk stratification. They support the chapter's interpretation without carrying the main weight of the result.

Table 31: Nested logistic regressions for pre-birth employment exit at $t = -2$. The outcome is a binary indicator equal to one if a woman exits employment between $t = -2$ and $t = -1$; the sample comprises women in the mechanism complete-case sample (13 exit events). Odds ratios are reported with log-odds standard errors in parentheses. M1 includes contract type only; M2 adds firm size; M3 adds one-way commute time (minutes); M4 adds a husband overwork indicator ($\geq$50 hours/week); M5 includes all four predictors. N varies across models because of missing covariate values.

	M1	M2	M3	M4	M5
Non-regular	7.11**	6.41**	6.00**	6.63**	6.18**
	(0.784)	(0.791)	(0.795)	(0.793)	(0.797)
Firm size below 100 employees		4.14**	3.80**	4.19**	3.94**
		(0.635)	(0.652)	(0.637)	(0.656)
Commute time			0.99		0.99
			(0.017)		(0.016)
Husband overwork				1.21	1.34
				(0.607)	(0.615)
N	185	185	169	176	160
Events	13	13	13	13	13

Notes: ** $p < 0.05$. Odds ratios above 1 indicate higher exit risk. "Non-regular" = non-regular contract status (part-time, fixed-term, temporary, or dispatch). "Firm size below 100 employees" = employer with fewer than 100 employees; the descriptive crosstabs elsewhere use the finer categories small (< 30), medium (30–99), large (100+), and government. Data source: KHPS/JHPS 2004–2022, married first-birth women.

The table 32 reports leave-one-out and influence diagnostics for the baseline two-predictor mechanism model ('non-regular' and 'firm size below 100 employees'). The jackknife ranges show stable sign and magnitude for both predictors, and influence diagnostics indicate no single case that reverses the qualitative result.

Table 32: Leave-one-out jackknife sensitivity of mechanism odds ratios (Model 2).

Predictor	Baseline OR	Baseline p	OR range	p range	Share $p < 0.05$
Non-regular	6.41	0.019	[5.69, 12.82]	[0.016, 0.030]	1.000
Firm size below 100 employees	4.14	0.025	[3.54, 5.52]	[0.014, 0.052]	0.989

D.9 Mechanism check at childbirth margin ($t = 0$)

To align mechanism evidence with the childbirth margin, I estimate a parallel model where the outcome is non-employment at $t = 0$ among women employed at $t = -1$, with predictors measured at $t = -1$. This specification targets the childcare-dominant exit margin directly and is interpreted as predictive association evidence.

Table 33: Mechanism check at childbirth margin: predictors of non-employment at $t = 0$ among women employed at $t = -1$.

Predictor	OR (logit)	SE (log-odds)	p (logit)
Non-regular ($t = -1$)	7.50	0.283	0.000
Firm size below 100 employees ($t = -1$)	1.55	0.286	0.126

Notes: Complete-case childbirth-margin logit estimated on the 282 women with non-missing contract type and firm size at $t = -1$; 100 exits occur at $t = 0$. A missing-indicator specification on the full $N = 330$ risk set is reported in Table 38.

As a small-sample robustness check, the same childbirth-margin model is re-estimated using Firth's penalized likelihood and bias-corrected bootstrap inference.

Table 34: Small-sample robustness for the childbirth-margin mechanism model.

	OR	p	95% CI	Method
Non-regular employment				
Standard logit	7.50	<0.001	–	MLE
Firth	7.31	<0.001	–	Penalized ML
Bootstrap	7.67	<0.001	[4.44, 13.89]	2,000 resamples
Firm size below 100 employees				
Standard logit	1.55	0.126	–	MLE
Firth	1.54	0.127	–	Penalized ML
Bootstrap	1.57	0.156	[0.87, 2.86]	2,000 resamples

Notes: The childbirth-margin complete-case model is estimated on 282 women with non-missing contract type and firm size at $t = -1$; 100 exits occur at $t = 0$. The non-regular coefficient remains large across all three methods, while the sub-100-employer coefficient remains positive but imprecisely estimated.

Table 35: Absolute-risk view of childbirth-margin exit: raw rates and model-implied probabilities at $t = 0$.

Profile / group	N	Exits	Exit rate
By contract type: Non-Regular	120	73	0.608
By contract type: Regular	162	27	0.167
By firm size: 100+ / government	175	53	0.303
By firm size: Below 100 employees	107	47	0.439
By contract × firm: Non-Regular × 100+ / government	66	38	0.576
By contract × firm: Non-Regular × Below 100	54	35	0.648
By contract × firm: Regular × 100+ / government	109	15	0.138
By contract × firm: Regular × Below 100	53	12	0.226
Profile			Predicted Pr(exit at $t = 0$)
Regular × 100+ / government			0.146
Regular × Below 100			0.209
Non-Regular × 100+ / government			0.562
Non-Regular × Below 100			0.665

Notes: Raw cell rates are computed on the 282 women in the childbirth-margin risk set with both contract type and firm size observed at $t = -1$. The model-implied probabilities come from the corresponding interaction-based childbirth-margin logit and closely track the raw cell means.

To increase visibility on covariate missingness, I additionally report a missingness profile for key mechanism covariates (contract type and firm size), a direct missingness-exit association check, and side-by-side coefficient sensitivity across complete-case, missing-indicator, and high-information samples.

Table 36: Missingness profile for mechanism covariates in the childbirth-margin risk set (employed at $t = -1$, observed at $t = 0$).

Covariate missingness	N	Share of risk set	Exit rate at $t = 0$
Contract type missing	36	0.109	0.444
Firm size missing	16	0.048	0.750
Either missing	48	0.145	0.500

Notes: The risk-set size is $N = 330$ women. Exit is defined as non-employment at $t = 0$ among women employed at $t = -1$.

Table 37: Association between covariate missingness and childbirth-margin exit.

Missingness indicator	OR	p-value
Contract type missing	1.243	0.554
Firm size missing	5.254	0.005

Notes: Logit model with exit at $t = 0$ as the dependent variable. Odds ratios above 1 indicate higher exit odds for observations with missing covariates.

Table 38: Childbirth-margin mechanism sensitivity to missing-data handling.

Specification	N	Exits	OR Non-regular	p	OR Firm size < 100	p
Complete-case	282	100	7.505	0.000	1.548	0.126
Missing-indicator	330	124	7.476	0.000	1.289	0.345
High-information	237	83	7.393	0.000	1.860	0.047

Notes: The high-information sample requires non-missing contract type, firm size, commute indicator, and husband's overwork indicator at $t = -1$. The missing-indicator specification includes missing dummies for contract type and firm size. The binary firm-size predictor equals one for employers with fewer than 100 employees.

These diagnostics clarify the missingness pattern in the childbirth-margin risk set. Covariate missingness is non-trivial (14.5% missing on either contract type or firm size). Contract-type missingness is not strongly associated with exit (OR = 1.24, $p = 0.554$), while firm-size missingness is strongly associated with exit (OR = 5.25, $p = 0.005$; exit rate 75% among missing). The key non-regular coefficient remains essentially unchanged across complete-case, missing-indicator, and high-information specifications, while estimates on the sub-100-employer indicator vary in precision and are not statistically distinguishable from zero under bootstrap inference. I treat the contract-tier gradient as robust and the binary firm-size gradient as supportive but sample-sensitive.

A natural follow-up concern is whether contract-type missingness is itself systematically related to baseline characteristics. If women with missing contract information differ on observable dimensions, the complete-case estimates could reflect selection into observability and give a less clear view of the underlying labour-market gradient. I report a balance diagnostic comparing baseline hours, income, and survey source between women with observed and missing contract type at $t = -1$ (Table 39), together with a logit for the probability of contract-type observability.

Table 39: Balance diagnostic: baseline characteristics by contract-type observability at $t = -1$ in the childbirth-margin risk set ($N = 330$).

Panel A: Means by observability group

Variable	Observed ($N = 294$)	Missing ($N = 36$)	p-value
Exit at $t = 0$	0.367	0.444	0.389
Hours/week (at $t = -1$)	35.626	21.472	<0.001
Annual income (man-yen, at $t = -1$)	280.918	137.750	0.002
JHPS share	0.449	0.389	0.612

Panel B: Logit for 1{contract type observed}

Predictor	OR	SE (log-odds)	p-value
Intercept	1.234	0.379	0.579
Hours/week at $t = -1$	1.037	0.014	0.009
Annual income (man-yen) at $t = -1$	1.004	0.002	0.016
JHPS indicator	1.146	0.384	0.722

McFadden pseudo-$R^2 = 0.1364$

Notes: The risk set is women employed at $t = -1$ and observed at $t = 0$ ($N = 330$). Panel A reports means and two-sample t-test p-values (proportion test for JHPS share). Panel B reports logit odds ratios for the probability that contract type is non-missing at $t = -1$. Missingness is predictably related to baseline hours and income, but not to survey source, and the explanatory power of these observables remains limited in absolute terms.

The balance diagnostic reveals that contract-type missingness is not random: women with missing contract type at $t = -1$ work significantly fewer hours (21.5 vs 35.6 hours/week, $p < 0.001$) and earn significantly less (137.8 vs 280.9 man-yen, $p = 0.002$) than women with observed contract type. The logit for observability confirms this (McFadden pseudo-$R^2 = 0.14$). However, this pattern is consistent with contract-type missingness being concentrated in marginal employment where formal classification is ambiguous or unreported. Because these women resemble the non-regular profile on observable dimensions, the complete-case non-regular gradient is likely conservative and does not appear to be inflated by selection into observability. The non-regular OR is virtually identical across complete-case (7.50), missing-indicator (7.48), and high-information (7.39) specifications, confirming that the core stratification result is not driven by which women have observed contract type. The exit-rate difference between groups (44.4% vs 36.7%) is not statistically significant ($p = 0.389$), and survey source (JHPS vs KHPS) does not predict observability ($p = 0.722$).

A coding concern is that some women on leave may report zero market hours at $t = 0$ and be classified as non-employed. To assess this, I run a leave-recoding sensitivity: women coded as non-working at $t = 0$ are reclassified as employed if they report leave take-up near birth. The adjusted estimates attenuate the childbirth coefficient but preserve the core result: the birth-year break remains large, negative, and statistically significant. I treat leave-related coding as affecting level magnitude, not the qualitative timing pattern.

Table 40: Leave-as-employment sensitivity at childbirth ($t = 0$).

Specification	Coef. at $t = 0$	SE	p-value	Reclassified at $t = 0$
Original (hours > 0)	-0.1319	0.0275	<0.001	0
Leave-adjusted	-0.0988	0.0276	<0.001	22

Notes: This sensitivity recodes women as employed at $t = 0$ if they are coded non-working but report leave take-up near birth. The coefficient remains strongly negative after reclassification.

D.10 Policy-period cohort split

The observation window (2004-2022) spans substantial policy changes in Japan's work - family infrastructure, including the 2010 introduction of paternity leave, the 2012 revised Child Care and Family Care Leave Act, the acceleration of childcare expansion under the Abe government's "Womenomics" agenda from 2013, and the 2019 free early childhood education reform. If these reforms materially altered the motherhood employment penalty, pooling across the full period could obscure regime-specific effects.

To assess sensitivity, I split the sample by birth-report year: an early cohort (first birth reported in 2004-2012, before the acceleration of Womenomics-era reforms) and a late cohort (2013-2022). I additionally re-estimate the baseline specification excluding pandemic-era births (2020-2022) to rule out COVID-related distortions.

Table 41 reports the key event-study coefficients ($t = -2$, $t = 0$, $t = +5$) for each subsample. The near-lead coefficient ($t = -2$) is statistically indistinguishable from zero in all four samples. The childbirth-year break ($t = 0$) is negative in every split: -0.131 ($p < 0.001$) in the early cohort, -0.135 ($p = 0.002$) in the late cohort, and -0.145 ($p < 0.001$) in the no-COVID sample. The long-horizon coefficient ($t = +5$) is small and insignificant in all splits.

Two implications follow. First, the birth-year motherhood employment penalty remains negative in both the pre-2013 and post-2013 policy periods in this sample, despite substantial expansions in childcare capacity and leave generosity over the period. This is consistent with the interpretation that structural labour-market segmentation is the main constraint on employment continuity through childbirth. Policy generosity alone does not appear to resolve it. Second, excluding pandemic-era births does not alter the core estimates, indicating that COVID-related labour market disruptions do not drive the pooled results.

Table 41: Cohort-split sensitivity: key employment event-study coefficients by birth-year period (year fixed effects, heteroskedasticity-robust standard errors).

Sample	Event time	Estimate	SE	p-value	N obs	N women
Pooled	$t = -2$	-0.012	0.031	0.703	5,212	662
	$t = +0$	-0.132	0.027	<0.001	5,212	662
	$t = +5$	-0.028	0.032	0.384	5,212	662
Early (≤2012)	$t = -2$	0.020	0.041	0.627	3,062	390
	$t = +0$	-0.131	0.036	<0.001	3,062	390
	$t = +5$	-0.020	0.045	0.658	3,062	390
Late (≥2013)	$t = -2$	-0.063	0.047	0.186	2,150	272
	$t = +0$	-0.135	0.044	0.002	2,150	272
	$t = +5$	-0.025	0.056	0.657	2,150	272
No-COVID (≤2019)	$t = -2$	-0.004	0.032	0.906	4,822	577
	$t = +0$	-0.145	0.029	<0.001	4,822	577
	$t = +5$	-0.028	0.034	0.399	4,822	577

Notes: Each panel re-estimates the baseline employment event study (is_working $\sim i(\text{event\_time}, \text{ref} = -1)$ | calendar_year) on the indicated subsample. "Early" includes mothers whose first birth was reported in 2004–2012; "Late" in 2013–2022. "No-COVID" excludes birth years 2020–2022. All specifications use year fixed effects and heteroskedasticity-robust standard errors.

To probe whether the late cohort is masking distinct short-run dynamics in the pandemic years, I also compare births in 2013–2019 with those in 2020–2022 over the shorter horizon that the latter period can support. The birth-year employment break remains negative in both groups: -0.184 for the late pre-COVID cohort ($p < 0.001$) and -0.148 for the COVID-era cohort ($p = 0.168$). The COVID-era estimate is imprecise because it is based on only 85 women, so this split works best as a coarse sensitivity check. It does not support a precise cohort decomposition. Even so, the estimates do not suggest a qualitative break in the sign or timing of the motherhood penalty in the pandemic tail. The near lead remains close to zero in both groups, and the broader no-COVID specification is slightly larger in magnitude than the pooled estimate (-0.145 versus -0.132), which points in the same direction. The pooled results are therefore unlikely to be materially driven by COVID-era disruptions, and the no-COVID specification supports the chapter's period-averaged interpretation.

Table 42: Focused late-cohort sensitivity: short-horizon employment event-study coefficients for 2013–2019 versus 2020–2022 births (year fixed effects, heteroskedasticity-robust standard errors).

Sample	Event time	Estimate	SE	p-value	N obs	N women
Late pre-COVID (2013–2019)	$t = -2$	-0.051	0.055	0.351	1,760	187
	$t = +0$	-0.184	0.053	<0.001	1,760	187
	$t = +1$	-0.158	0.056	0.005	1,760	187
COVID-era (2020–2022)	$t = -2$	0.009	0.112	0.937	390	85
	$t = +0$	-0.148	0.107	0.168	390	85
	$t = +1$	-0.254	0.155	0.103	390	85

Notes: This focused check is designed to answer a narrower question than the main cohort split: whether the pooled late cohort masks a sharp break between late pre-COVID births (2013–2019) and the short COVID-era tail (2020–2022). Because the COVID-era window is short and recent, the table reports only the near-lead and short-horizon coefficients ($t = -2$, $t = 0$, $t = +1$). These estimates should be read as a coarse sensitivity diagnostic rather than as a precise cohort decomposition.

Figure 19: Earnings distribution by event time for employed mothers ($t = +3, +4, +5$), with reference lines at 103 and 130 man-yen. The threshold concentration pattern is visible at each post-birth horizon. For the direct comparison with childless married employed women, see Figure 11 in the main text. Taken together, the figures indicate a broader secondary-earner phenomenon that extends beyond motherhood-specific responses.

The same threshold pattern appears separately at $t = +3$, $t = +4$, and $t = +5$ (Figure 19). As noted in the main text, childless married employed women show at least as much concentration below these thresholds (49.7% at or below 103 man-yen, 60.0% at or below 130 man-yen). This supports an interpretation in which the bunching reflects broader secondary-earner constraints, including but extending beyond motherhood-specific behavioural responses.

D.11 Industry-sector diagnostic

A natural concern is whether the motherhood employment penalty varies by pre-birth industry sector, and specifically whether women in sectors with stronger institutional protections (e.g., government, finance/insurance, utilities, and education) experience smaller employment breaks than those in other sectors.

To investigate, I assign each woman to one of three groups based on her last observed industry in the pre-birth window ($t = -5$ to $t = -1$): "stable sector" (government, finance/insurance, utilities, education), "other sector" (all remaining industries), and "unknown / no pre-birth industry" (women with no industry data in the pre-birth window, predominantly not employed before birth). This assignment avoids conditioning on employment at any single event time, which would create a mechanical negative pre-trend.

I report two diagnostics. First, a full-sample event-study stratification retains all 662 mothers across the three groups (Table 43). The "stable sector" ($N = 86$) and "other sector" ($N = 305$) subsamples both show significant employment declines at $t = 0$, but both also exhibit significant negative coefficients at $t = -2$, indicating that sector assignment based on pre-birth employment history induces composition differences that appear as pre-trends. The "unknown" group ($N = 271$) shows positive coefficients at $t = 0$ and $t = +5$, reflecting selection: these women were predominantly not employed before birth, and some enter employment over the panel window. These patterns confirm that sector-stratified event-study *levels* in this sample are not cleanly interpretable as sector-specific penalties.

Second, to avoid these composition artefacts, I estimate a conditional transition diagnostic restricted to the 328 women employed at $t = -1$ with an observed sector assignment (Table 44). Exit rates at birth are similar across sectors: 37.1% in the other-sector group versus 39.0% in the stable-sector group, a difference of 1.9 percentage points ($p = 0.764$). Employment rates at $t = +5$ are 67.5% in the other-sector group versus 57.4% in the stable-sector group, a difference of -10.1 percentage points ($p = 0.192$), suggestive of weaker long-run recovery in the stable sector but statistically imprecise.

Two implications follow. First, among women employed before birth, pre-birth sector type does not significantly predict exit probability at childbirth in this sample. This is consistent with an interpretation in which the employment break at birth operates primarily through job-type segmentation, especially regular versus non-regular status, with industry affiliation carrying less signal. Second, the imprecise long-horizon difference leaves open the possibility that stable-sector women who exit face different return barriers, though the small cell sizes ($N = 54$ and $N = 160$ at $t = +5$) preclude firm conclusions. This is consistent with the main-text interpretation that contract type is the primary observed predictor of childbirth-margin exit in this sample, while industry affiliation appears less informative.

Table 43: Industry stratification: key employment event-study coefficients by pre-birth sector assignment.

Sample	Event time	Coefficient (SE)	N obs	N women
All women	$t = -2$	-0.012 (0.031)	5,212	662
	$t = +0$	-0.132 (0.027)	5,212	662
	$t = +5$	-0.028 (0.032)	5,212	662
Stable sector	$t = -2$	-0.219 (0.072)	682	86
	$t = +0$	-0.316 (0.067)	682	86
	$t = +5$	-0.357 (0.075)	682	86
Other sector	$t = -2$	-0.083 (0.036)	2,492	305
	$t = +0$	-0.291 (0.036)	2,492	305
	$t = +5$	-0.199 (0.042)	2,492	305
Unknown / no pre-birth industry	$t = -2$	0.003 (0.010)	2,038	271
	$t = +0$	0.139 (0.023)	2,038	271
	$t = +5$	0.357 (0.038)	2,038	271

Notes: Sector is assigned using the most recent observed pre-birth industry in event times $[-5, -1]$ among years with employment observed. "Stable sector" includes government, finance/insurance, utilities, and education. Women with no observed pre-birth industry are retained as a separate "Unknown" group. This is a descriptive sensitivity check.

Table 44: Industry and birth-transition outcomes among women employed at $t = -1$.

Sector group	N women	Exit rate at $t = 0$	N at $t = +5$	Employed rate at $t = +5$
Other sector	251	0.371	160	0.675
Stable sector	77	0.390	54	0.574

Difference (Stable − Other), linear probability model with heteroskedasticity-robust SEs:

Exit at $t = 0$ (1 = not working): 0.019 (SE 0.064, $p = 0.764$, $N = 328$)

Employed at $t = +5$ (1 = working): -0.101 (SE 0.077, $p = 0.192$, $N = 214$)

Notes: Sector is assigned using the most recent observed pre-birth industry in event times $[-5, -1]$ among employed observations. "Stable sector" includes government, finance/insurance, utilities, and education. This table is a conditional transition diagnostic and not a full-sample sector event study.

D.12 Standard-error sensitivity (heteroskedastic vs clustered)

The table 45 compares key employment event-study coefficients using heteroskedasticity-robust and woman-level clustered standard errors. The point estimates are unchanged by construction, and inference is substantively identical across SE choices: the near lead at $t = -2$ remains null, the birth-year drop at $t = 0$ remains strongly significant, and the long-horizon coefficient at $t = +5$ remains statistically indistinguishable from zero.

Table 45: Sensitivity of key employment event-study coefficients to SE choice (year fixed effects).

Event time	Heteroskedastic	p	Clustered	p
$t = -2$	-0.012 (0.031)	0.703	-0.012 (0.024)	0.620
$t = +0$	-0.132 (0.027)	<0.001	-0.132 (0.020)	<0.001
$t = +5$	-0.028 (0.032)	0.384	-0.028 (0.032)	0.387

D.13 Pre-pregnancy job-type stability ($t = -5$ to $t = -3$)

To address the concern that contract type in the pre-birth mechanism models may reflect immediate anticipatory switching, I report a stability diagnostic in the pre-pregnancy window. Specifically, I track broad job type (Regular vs Non-Regular) from $t = -5$ to $t = -3$ on common support and report persistence rates and transition counts. This is a descriptive check. It does not remove long-run sorting into job types, but it does provide suggestive evidence against abrupt compositional reclassification immediately before the mechanism margin. In the common-support sample observed at both $t = -5$ and $t = -3$ ($N = 32$), overall job-type stability is 81.2% (Regular→Regular: 77.8%; Non-Regular→Non-Regular: 85.7%). Some switching remains, but the pattern is more consistent with persistent segmentation than with abrupt pre-birth reclassification. Given the small common-support sample, this stability estimate is best treated as indicative.

Table 46: Pre-birth job-type stability check ($t = -5$ to $t = -3$).

Metric	Value	Interpretation	Window
Women observed at both $t = -5$ and $t = -3$	32	Common-support sample	Pre-pregnancy
Overall job-type stability	81.2%	Same broad type at $t = -5$ and $t = -3$	Pre-pregnancy
Regular → Regular	77.8%	Persistence among regular workers	Pre-pregnancy
Non-Regular → Non-Regular	85.7%	Persistence among non-regular workers	Pre-pregnancy

D.14 Core robustness summary and denominator map

To make inference priorities and sample definitions fully transparent, I report (i) one consolidated robustness table for the key employment coefficients at $t = -2$, $t = 0$, and $t = +5$, and (ii) a denominator map for the main samples used in the chapter. The robustness table focuses on the same estimand across alternative sample/weighting/SE choices; the denominator map is a reading guide so each reported percentage is tied to the correct risk set.

Table 47: Master robustness table: key employment coefficients at $t = -2$, $t = 0$, and $t = +5$.

Specification	$t = -2$	$t = 0$	$t = +5$
Balanced short window ($t = -2$ to $t = +1$)	-0.058 (0.065)	-0.128 (<0.001)	–
Baseline (year-FE robust)	-0.012 (0.702)	-0.132 (<0.001)	-0.028 (0.384)
Clustered SE (woman)	-0.012 (0.620)	-0.132 (<0.001)	-0.028 (0.387)
Early cohort (≤2012)	0.020 (0.627)	-0.131 (<0.001)	-0.020 (0.658)
IPW (obs at $t = +5$)	-0.024 (0.436)	-0.136 (<0.001)	-0.057 (0.076)
JHPS only	-0.027 (0.576)	-0.136 (0.002)	-0.126 (0.027)
KHPS only	0.002 (0.966)	-0.134 (<0.001)	0.010 (0.796)
Late cohort (≥2013)	-0.063 (0.186)	-0.135 (0.002)	-0.025 (0.657)
Leave-adjusted coding	–	-0.099 (<0.001)	–
No-COVID sample	-0.004 (0.906)	-0.145 (<0.001)	-0.028 (0.399)
Second-birth censored	-0.011 (0.724)	-0.135 (<0.001)	-0.003 (0.935)

Notes: Cells report coefficient and p-value in parentheses. Leave-adjusted coding is defined for the childbirth margin ($t = 0$) only.

Table 48: Denominator map for main reported quantities

Quantity	Denominator	Definition
Main employment event study	662 women	Analytic first-birth sample in the event-study panel
Childbirth-margin mechanism ($t = 0$)	330 women	Women employed at $t = -1$ and observed at $t = 0$
Pre-birth mechanism ($t = -2$)	185 women	Women employed at $t = -2$ and observed at $t = -1$
Father leave (partner-linked analytic)	661 partners	Linked partner records in the analytic panel
Leave take-up tabulation (broader module)	1,183 couples	First-birth-linked leave-analysis module sample
Newlyweds education coverage	89 women	Mothers with non-missing Newlyweds-module education

Notes: Denominators differ by design because each statistic is defined on its own valid risk set or module coverage. Reported percentages should be interpreted within row, not compared mechanically across rows.

Appendix E. Extended institutional context

This appendix does not introduce new evidence. Its purpose is to restate, in compressed form, the institutional features from Section 2 that matter most for interpretation of the event-study design.

E.1 Dual labour market structure

Section 2.1 gives the main discussion. The key point for the chapter is that continuity through childbirth depends on contract status and employer capacity: regular-track and larger-firm jobs provide more scope for continued attachment than non-regular jobs and smaller employers.

E.2 Work-family policy and eligibility

Section 2.5 explains the policy architecture in detail. For interpretation here, the relevant point is the gap between formal eligibility and effective access: leave rights are strongest where employment continuity is already most secure, and much weaker where contracts are short, renewal is uncertain, or employer capacity is limited.

E.3 Long-hours norms and gendered care

Section 2.3 develops the household-side argument. The main implication is that paternal labour supply remains close to flat while mothers provide roughly five-to-six times more childcare across the post-birth window, leaving limited room for within-household reallocation once a child arrives.

E.4 Fiscal incentives and secondary earners

Section 2.5 and Section 6 discuss the threshold evidence. The relevant point for identification is that bunching around the 103 and 130 man-yen thresholds extends beyond mothers. In this setting, motherhood appears to shift women into an already constrained secondary-earner earnings range, with the constraint predating the birth transition itself.

E.5 Implications for identification

These institutional features are important for interpreting the event-study results. The sharp exit at birth and the durable shift into reduced-hours work - often on non-regular contracts - are consistent with a system in which job status and firm-level constraints shape who can remain employed. The mechanisms do not require strong changes in preferences; instead, they reflect the structure of available jobs and the rules governing leave eligibility and earnings incentives.